

Network Autoregressive Models for Functional Panel Data

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Abstract

This study proposes a novel functional vector autoregressive framework for analyzing network interactions of functional outcomes in panel data settings. In this framework, an individual's outcome function is influenced by the outcomes of others through a simultaneous equation system. To estimate the functional parameters of interest, we need to cope with the endogeneity issue arising from these simultaneous interactions among outcome functions. We address this issue by developing a novel functional moment-based estimator. We establish the consistency, convergence rate, and pointwise asymptotic normality of the proposed estimator. Additionally, we discuss the estimation of marginal effects and impulse response analysis. As an empirical illustration, we analyze the demand for a bike-sharing service in the U.S. The results reveal statistically significant spatial interactions in bike availability across stations, with interaction patterns varying over the time of day.

Keywords: functional data analysis, panel data, endogeneity, network autoregressive models, bike-sharing systems

1 Introduction

The availability of functional data has been rapidly expanding across all fields of research, leading to a growing need for statistical tools that appropriately account for the unique characteristics of each type of functional data. In the analysis of socioeconomic data, there are at least two key aspects that should be addressed. The first is that an individual's decision or behavioral pattern may influence that of others through social networks – *interactions* between individuals. The second is that individuals are intrinsically heterogeneous, even after controlling for observable characteristics – *unobserved heterogeneity* of individuals. Therefore, analyzing socioeconomic functional data requires functional models that jointly capture both of these aspects, which is the aim of this study.

More specifically, to account for the interactions among units, we extend the network (or spatial) autoregressive (NAR) modeling approach to a functional response model. To address the unobserved individual heterogeneity, we introduce the functional fixed effects approach, given the availability of panel data. When the response variable is a scalar rather than a function, there already exists a vast body of studies investigating fixed-effect NAR models for panel data, such as

$$Y_{it} = \alpha_0 \sum_{j=1}^n w_{i,j} Y_{jt} + X_{it}^\top \beta_0 + f_{0i} + \varepsilon_{it}, \quad i = 1, \dots, n, \quad t = 1, \dots, T \quad (1.1)$$

and its variants (e.g., [Yu et al., 2008](#); [Lee and Yu, 2010, 2014](#); [Kuersteiner and Prucha, 2020](#), among others). Here, Y_{it} is a scalar outcome, $w_{i,j}$ denotes a known weight term measuring the social or geographical proximity between units i and j , X_{it} is a vector of covariates, f_{0i} represents a fixed effect specific to each i , and ε_{it} denotes an error term. The term $\sum_{j=1}^n w_{i,j} Y_{jt}$ captures the local trend of the outcome variable in the neighborhood of i . Model (1.1) is typically applied in fields such as health, real estate, transportation, education, and municipal data. However, with the increasing availability of functional data in these fields, such as real-time activity recognition, real-time population mobility and congestion patterns, and regional wealth distributions, scalar models like (1.1) may fail to appropriately capture the complex nature of these interactions. For an illustrative example, we refer the reader to the empirical analysis in Subsection 7.3.

The above discussion motivates us to extend (1.1) to the following model: for $s \in [0, 1]$,

$$Y_{it}(s) = \alpha_0(s) \sum_{j=1}^n w_{i,j} A(Y_{jt}, s) + X_{it}^\top \beta_0(s) + f_{0i}(s) + \varepsilon_{it}(s), \quad (1.2)$$

where Y_{it} represents the outcome function of interest, which may or may not be a smooth function of s , α_0 is the interaction effect function, β_0 is a vector of functional coefficients, f_{0i} is the fixed effect function, and ε_{it} is the functional error term with mean zero at each s . Here, $A(\cdot, s)$ denotes a known functional, whose functional form may differ according to the research interest.

Since the response variable is a function, we can consider various types of interaction patterns. The most typical form of interaction would be the "concurrent" interaction, where only the responses of others at the same evaluation point s are influential. In this case, $A(\cdot, s)$ is a point-evaluation functional at s : $A(Y_{jt}, s) = Y_{jt}(s)$. When s represents time, then the past outcome should affect the future outcome (but the converse should not), which motivates us to employ $A(Y_{jt}, s) = \int_0^1 Y_{jt}(u)\nu(u, s)du$, where $\nu(u, s)$ is a user-chosen function that is non-negative, increasing in u up to s , and $\nu(u, s) = 0$ for $u > s$. For other examples, if others' responses at all evaluation points are equivalently influential, we may use $A(Y_{jt}, s) = \int_0^1 Y_{jt}(u)du$. In this case, $A(Y_{jt}, s)$ is independent of s . These examples can be represented as an integral operator $A(Y_{jt}, s) = \int_0^1 Y_{jt}(u)\nu(u, s)du$ with some kernel weight function $\nu(u, s)$. For example, in the case of the point-evaluation functional, we can set $\nu(u, s) = \delta(u - s)$, where δ denotes the Dirac delta function.

Here, we provide three typical empirical topics to which model (1.2) would be nicely applied.

Example 1.1 (Health data analysis). In the health literature, researchers have increasingly focused on real-time activity data collected through wearable devices or smartphone apps (see, e.g., [Di et al. \(2024\)](#) for a review). As a typical example, $Y_{it}(s)$ represents the activity level of individual i , measured by an accelerometer at time s on day t . Now, suppose the dataset consists of an elderly population, where some people in the same neighborhood frequently engage in fitness activities such as running or Tai Chi. If we apply our model to their activity-level data, with $w_{i,j}$ representing neighborhood membership and $A(Y_{jt}, s) = Y_{jt}(s)$, we may observe a significantly positive interaction effect.

Example 1.2 (Demographic data analysis). Demographic analysis is a major application of functional data analysis (FDA). For instance, functional analysis of regional age distributions (i.e., *population pyramids*) has been studied extensively (e.g., [Delicado, 2011](#); [Hron et al., 2016](#); [Hoshino, 2024](#)). Among these studies, [Hoshino \(2024\)](#) considered a functional spatial autoregressive model in which $Y_i(s)$ represents the s -th age quantile of city i , allowing interactions with the age quantiles of neighboring cities. Another common application in FDA is the analysis of mortality and fertility rates (e.g., [Hyndman and Ullah, 2007](#); [Chen and Müller,](#)

2012). In such a setting, the outcome function $Y_{it}(s)$ may represent the average number of births of women aged s in city i in year t . When one uses $A(Y_{jt}, s) = \int_0^1 Y_{jt}(u)\nu(u, s)du$ such that $\nu(u, s) \neq 0$ for u in the neighborhood of s , the presence of regional interactions in fertility within similar age group would be unsurprising.

Example 1.3 (Transportation data analysis). Functional data analysis of transportation data, such as traffic flows and demand for transportation services, has been gaining significant attention (see, e.g., [Ma et al. \(2024\)](#) for recent advancements). In the empirical application of this study, we apply our model to analyze the bike use data in a U.S. bike-sharing system. In our empirical analysis, $Y_{it}(s)$ represents the availability of bikes at station i at time s during week t . We find statistically significant positive or negative spatial interactions among bike availabilities of nearby stations, depending on the time of day. Further details are provided in Section 7.

In model (1.2), the parameters of our primary interest to be estimated are α_0 and β_0 . With the total time periods T possibly large or small, we apply a first-differencing transformation to eliminate the individual fixed effects from the model. For the transformed model, rather than estimating $\alpha_0(s)$ and $\beta_0(s)$ pointwise at many different s -values separately, we approximate them using orthonormal basis expansions and estimate their entire functional forms jointly in a single estimation. Our proposed estimator is based on the generalized method of moments (GMM). Specifically, we first derive a set of moment conditions at each s , in a similar manner to [Lin and Lee \(2010\)](#), and then integrate these conditions numerically over $s \in [0, 1]$. These integrated moment functions define our GMM objective function, and the resulting estimator is referred to as the integrated-GMM estimator. Once $\alpha_0(s)$ and $\beta_0(s)$ are estimated, if necessary, we can estimate the fixed effects $f_{01}(s), \dots, f_{0n}(s)$ simply by taking the individual-level mean of the residuals.

Note that in model (1.2), the outcome functions appear on both the left- and right-hand sides, implying that it is formulated as a system of simultaneous functional equations. Depending on the true values of the functional parameters, the model may exhibit an explosive network interaction process, leading to inconsistency of the proposed estimator. Thus, we first derive a condition for the *completeness* of our model – a condition ensuring that the model is uniquely characterized by a stable solution. We consequently show that the magnitude of network interactions must reside within a certain range.

Then, under the uniqueness condition on network interactions, along with some regularity conditions, we derive the convergence rates of the integrated-GMM estimators for α_0 and β_0 . In addition, we prove that the estimators are asymptotically normal at each evaluation point

s. These theoretical results are numerically corroborated through a series of Monte Carlo experiments.

As an empirical illustration, we apply our method to demand analysis for a bike-sharing system in the San Francisco Bay Area, U.S. Using publicly available data from *Bay Area Bike Share*, we study spatial interactions in bike availability across 70 stations from May 2014 to August 2015. Our results reveal significantly positive spatial interactions in bike availability during the morning hours, while negative interactions emerge in the early evening. Importantly, when we estimate a non-functional model (1.1) by averaging bike availability over time, the estimated α_0 is negative and close to zero. This highlights that ignoring the functional nature of the data may lead to misleading policy implications. Furthermore, we conduct an impulse response analysis to demonstrate how a reduction in bike availability at a given station propagates over time to nearby stations. These findings underscore the importance of spatial interactions in shared mobility services and demonstrate the practical applicability of our method.

Our paper relates to a broad range of theoretical and empirical literature. From a theoretical perspective, our study contributes to both the FDA literature and the network/spatial interactions literature by proposing a new model that connects the two. In this sense, one of the most closely related studies to ours is [Zhu et al. \(2022\)](#). They proposed a functional NAR model similar to (1.2) but not in panel data settings. In contrast to [Zhu et al. \(2022\)](#), our GMM estimator requires neither parametric assumptions nor IID conditions for the disturbance term. This weak requirement arises from the fact that we treat the individual effects as parameters, whereas [Zhu et al. \(2022\)](#) perform functional principal component analysis to control them based on some homogeneity condition. Moreover, they considered only a concurrent interaction case (i.e., $A(Y_{jt}, s) = Y_{jt}(s)$). As described earlier, the variable s typically represents a time on some scale.¹ The concurrent interaction rules out interactions even with immediate past outcomes and allows only strictly simultaneous interactions, which should limit the interpretability of the model. Computationally, our estimator can recover the full functional forms of the functional parameters in a single step, while the estimator in [Zhu et al. \(2022\)](#) must be repeatedly applied at each evaluation point s .

On the empirical side, demand forecasting for bike-sharing systems has been an active topic in the data science literature (e.g., [Faghih-Imani and Eluru, 2016](#); [Lin et al., 2018](#); [Eren and Uz, 2020](#); [Torti et al., 2021](#), among others). Among these studies, [Faghih-Imani and](#)

¹In this case, both s and T represent time, but they capture different time scales. Typically, s is used to describe short-term cycles, such as intraday stock price movements, or daily variation in temperature. In contrast, T captures more macro-level trends and thus corresponds to a longer time scale.

Eluru (2016) is most closely related to our study in that they employed a spatial panel model similar to (1.1) to analyze the spatial and temporal interaction structure for the bike-sharing system in New York City, CitiBike. In their approach, however, the data are not treated as functional, and thus the model parameters are not allowed to vary over time. By contrast, Torti *et al.* (2021) analyzed the flow of bikes in the bike-sharing system in Milan, BikeMi, through a functional linear model with functional coefficients; however, they did not account for the spatial interactions of mobility. Thus, our empirical study can be viewed as combining the strengths of these two papers.

Paper organization The rest of the paper is organized as follows. In Section 2, we present our model and discuss its uniqueness condition. Section 3 introduces our integrated-GMM estimator and investigates its asymptotic properties. In Section 4, we discuss additional topics related to our model, including the estimation of marginal effects and network impulse response analysis. Section 5 conducts a set of Monte Carlo simulations to numerically demonstrate the properties of our estimator. Section 6 discusses several extensions that should be useful in empirical applications. Section 7 presents our empirical analysis on the U.S. bike-sharing data, and Section 8 concludes. Proofs of technical results, additional empirical material, and the programming codes used in the numerical studies are provided in the online supplementary material.

Notation For a function h defined on $[0, 1]$ and $p \in [1, \infty)$, the L^p norm of h is written as $\|h\|_{L^p} := (\int_0^1 |h(s)|^p ds)^{1/p}$, and $L^p(0, 1)$ denotes the set of h 's such that $\|h\|_{L^p} < \infty$. For a random variable X , the L^p norm of X is written as $\|X\|_p := (\mathbb{E}|X|^p)^{1/p}$. For a matrix M , $\|M\|$, $\|M\|_1$, and $\|M\|_\infty$ denote the Frobenius norm, the maximum absolute column sum, and the maximum absolute row sum of M , respectively. If M is a symmetric matrix, we use $\lambda_{\max}(M)$ and $\lambda_{\min}(M)$ to denote its largest and smallest eigenvalues, respectively. For a positive integer Z , we denote $[Z] := \{1, \dots, Z\}$. We use I_Z to denote an identity matrix of dimension Z . Finally, $X \lesssim Y$ if $X = O(Y)$ almost surely, and $X \lesssim_P Y$ if $X = O_P(Y)$.

2 Functional Network Autoregressive Model

2.1 The model

Suppose that we have balanced panel data of size (n, T) : $\{(Y_{it}, X_{it}, w_{i,1}, \dots, w_{i,n}) : i \in [n], t \in [T]\}$. The number of time periods $T \geq 2$ can be either fixed or tending to infinity jointly with

the sample size n . Here, $Y_{it} : [0, 1] \rightarrow \mathbb{R}$ denotes a random outcome function of interest with the common support $[0, 1]$, $X_{it} = (X_{it}^1, \dots, X_{it}^{d_x})^\top$ denotes a vector of covariates, and $w_{i,j} \in \mathbb{R}$ is the (i, j) -th element of an $n \times n$ time-invariant interaction matrix $W_n = (w_{i,j})$. Throughout, we assume that the functional outcome $Y_{it}(s)$ is fully observed over $s \in [0, 1]$. In our empirical application, $Y_{it}(s)$ represents the number of available bikes at station i , where s denotes the time of day. The data record the timing of all pickups and returns at each station, which implies the complete observability of Y_{it} as a function of s . As in the case of bike availability, we do not rule out situations in which $Y_{it}(s)$ is discontinuous in s . The covariates X_{it} do not include a constant term and vary with both i and t . The value of each $w_{i,j}$ is pre-determined non-randomly. In social network analysis, it is common to set $w_{i,j} = c_{i,j} \mathbf{1}\{i \text{ and } j \text{ are peers}\}$, where $c_{i,j}$ is some normalizing constant. Similarly, if each i represents a spatial unit, one may use $w_{i,j} = c_{i,j} \mathbf{1}\{\Delta(i, j) \leq \bar{\Delta}\}$, where $\Delta(i, j)$ is the distance between i and j , and $\bar{\Delta}$ is a given threshold. As is the convention, we set $w_{i,i} = 0$ for all i for normalization.

As shown in (1.2), our working model is given as follows: for $s \in [0, 1]$,

$$Y_{it}(s) = \alpha_0(s)A(\bar{Y}_{it}, s) + X_{it}^\top \beta_0(s) + f_{0i}(s) + \varepsilon_{it}(s), \quad i \in [n], \quad t \in [T]$$

where $\bar{Y}_{it} = \sum_{j=1}^n w_{i,j} Y_{jt}$. Considering analytical transparency and interpretational simplicity in empirical applications, we assume $A(\cdot, \cdot)$ is linear in its first argument so that we have $\sum_{j=1}^n w_{i,j} A(Y_{jt}, \cdot) = A(\bar{Y}_{it}, \cdot)$. While this linear specification provides a reasonable starting point, more complicated forms are studied in Subsection 6.2.

The parameters of primary interest are the interaction effect function $\alpha_0(s)$ and the coefficient functions $\beta_0(s) = (\beta_{01}(s), \dots, \beta_{0d_x}(s))^\top$. The functional individual effects $f_{01}(s), \dots, f_{0n}(s)$ are treated as nuisance parameters. Restricting the support of s to be a unit interval is a normalization, which is harmless as long as the response functions have the identical interval support. For simplicity, we do not explicitly assume that X_{it} is a function of s , which can be relaxed easily at the expense of more complicated notation and proofs. In the rest of this section, we assume that $Y_{it} \in L^2(0, 1)$, $\varepsilon_{it} \in L^2(0, 1)$, and that α_0 and β_0 are continuous functions. To derive the asymptotic properties of our estimator, we impose stronger regularity conditions later.

2.2 Completeness

We discuss the *completeness* of our model (in the sense of Tamer (2003); Lewbel (2007)). That is, since the model is a system of simultaneous functional equations, it may not have a unique interior solution in general, depending on the parameter values. Thus, we need to

impose some conditions on the model to ensure that the outcome functions follow a unique data-generating process and prevent explosive behavior.

Let $Y_t(s) = (Y_{1t}(s), \dots, Y_{nt}(s))^\top$, $\mathbf{A}(Y_t, s) = (A(Y_{1t}, s), \dots, A(Y_{nt}, s))^\top$, $X_t = (X_{1t}, \dots, X_{nt})^\top$, $F_0(s) = (f_{01}(s), \dots, f_{0n}(s))^\top$, and $\mathcal{E}_t(s) = (\varepsilon_{1t}(s), \dots, \varepsilon_{nt}(s))^\top$. Then, we can re-write (1.2) in matrix form as

$$Y_t(s) = \alpha_0(s)W_n\mathbf{A}(Y_t, s) + X_t\beta_0(s) + F_0(s) + \mathcal{E}_t(s), \quad t \in [T].$$

This expression clearly indicates that our model is characterized as T independent systems of functional equations of size n . We introduce the following assumption to ensure that the model has a unique stable solution.

Assumption 2.1 (Completeness). (i) $\bar{\alpha}_0 \lesssim 1$ and $\|W_n\|_\infty \lesssim 1$ such that $\bar{\alpha}_0\|W_n\|_\infty < 1$, where $\bar{\alpha}_0 := \max_{s \in [0,1]} |\alpha_0(s)|$. (ii) For any $h \in L^2(0, 1)$, $\|A(h, \cdot)\|_{L^2} \leq \|h\|_{L^2}$.

Assumption 2.1(i) requires that the magnitude of the network interaction is not too strong. With Assumption 2.1(ii), we have for any $h, h' \in L^2(0, 1)$ that $\|A(h - h', \cdot)\|_{L^2} \leq \|h - h'\|_{L^2}$, which indicates the non-expansive property of the operator A . This assumption still accommodates many empirically interesting interaction patterns. For example, in the case of the point-evaluation functional $A(h, s) = h(s)$, it trivially satisfies $\|A(h, \cdot)\|_{L^2} = \|h\|_{L^2}$. For another example, suppose $A(h, s) = \int_0^1 h(u)\nu(u, s)du$ for some continuous ν . Since

$$\|A(h, \cdot)\|_{L^2}^2 \leq \int_0^1 \int_0^1 |h(u)\nu(u, s)|^2 dud s \leq \bar{\nu}^2 \|h\|_{L^2}^2,$$

where $\bar{\nu} := \max_{u,s \in [0,1]^2} |\nu(u, s)|$, Assumption 2.1(ii) is implied if $\bar{\nu} \leq 1$ holds.

Now, denote $\mathcal{H}_{n,p} := \{H = (h_1, \dots, h_n) : h_i \in L^p(0, 1), i \in [n]\}$, and define a linear operator $\mathcal{A}$ as

$$(\mathcal{A}H)(s) := \alpha_0(s)W_n\mathbf{A}(H, s), \quad H \in \mathcal{H}_{n,p}.$$

Then, we can write our model symbolically as follows:

$$Y_t = \mathcal{A}Y_t + X_t\beta_0 + F_0 + \mathcal{E}_t.$$

Further, denoting Id to be the identity operator, if the inverse operator $(\text{Id} - \mathcal{A})^{-1}$ exists, the

solution Y_t of the system can be uniquely determined up to an equivalence class in $\mathcal{H}_{n,2}$ as

$$Y_t = (\text{Id} - \mathcal{A})^{-1}[X_t\beta_0 + F_0 + \mathcal{E}_t].$$

The next proposition states that Assumption 2.1 is sufficient for the existence of $(\text{Id} - \mathcal{A})^{-1}$.

Proposition 2.1. Suppose that Assumption 2.1 holds. Then, $(\text{Id} - \mathcal{A})^{-1}$ exists, and for each $t \in [T]$, Y_t is the only solution of (1.2) in the Banach space $(\mathcal{H}_{n,2}, \|\cdot\|_{\infty,2})$, where $\|H\|_{\infty,p} := \max_{1 \leq i \leq n} \|h_i\|_{L^p}$.

The proof of Proposition 2.1 is given in Appendix B. Note that the explicit form of the inverse operator $(\text{Id} - \mathcal{A})^{-1}$ cannot be derived in general, except for some simple cases such as $(\mathcal{A}Y_t)(s) = \alpha_0(s)W_n Y_t(s)$. In this case, $(\text{Id} - \mathcal{A})^{-1}$ is obtained as $(I_n - \alpha_0(\cdot)W_n)^{-1}$. However, in practice, we can approximate it with arbitrary precision by truncating the Neumann series expansion $(\text{Id} - \mathcal{A})^{-1} = \sum_{\ell=0}^{\infty} \mathcal{A}^\ell$ at a sufficiently large order (see, e.g., Kress, 2014). See Remark 1 in Zhu *et al.* (2022) for a related discussion.

Finally, we note that Assumption 2.1 is only one type of sufficient condition for the existence of $(\text{Id} - \mathcal{A})^{-1}$. Condition (i) can be relaxed by allowing $\alpha_0(s)$ to exhibit locally explosive network dependence for some $s \in [0, 1]$, provided that the operator $\mathcal{A}$ satisfies a condition more restrictive than condition (ii); see Subsection B.1.2 in Appendix B for details. Accordingly, unlike standard non-functional NAR models, the completeness of our model can still be satisfied even when the network effect $\alpha_0(s)$ takes locally extreme values. That said, we adopt the current form of Assumption 2.1 for technical convenience in the subsequent analysis.

3 Estimation and Asymptotic Theory

3.1 Integrated-GMM estimation

To estimate the unknown functional parameters α_0 and β_0 , there are broadly two approaches. The first is a "local" approach that estimates the values of these functions at specific s -values, repeating the estimation across different points to recover the full functional forms. The second is a "global" approach that estimates the entire functional forms in a single step using a series approximation method. Although both approaches are theoretically valid, the local approach typically requires more computation time and often leads to larger variance (but smaller bias) because it does not exploit information from nearby evaluation points. This study adopts the global approach.

Let $\{\phi_k : k = 1, 2, \dots\}$ be a series of orthonormal basis functions. We throughout assume that ϕ_k 's are continuous on $[0, 1]$. Then, if the functions α_0 and β_0 are sufficiently smooth, we can approximate

$$\alpha_0(s) \approx \sum_{k=1}^K \phi_k(s) \theta_{0\alpha,k}, \quad \text{and} \quad \beta_{0j}(s) \approx \sum_{k=1}^K \phi_k(s) \theta_{0j,k}, \quad j \in [d_x],$$

uniformly in $s \in [0, 1]$, for some coefficient vectors $\theta_{0\alpha} = (\theta_{0\alpha,1}, \dots, \theta_{0\alpha,K})^\top$ and $\theta_{0j} = (\theta_{0j,1}, \dots, \theta_{0j,K})^\top$, $j \in [d_x]$. Here, $K \equiv K_{nT}$ is a sequence of positive integers tending to infinity as nT increases. For simplicity of presentation, we use the same basis function ϕ_k and the same basis order K to approximate both α_0 and β_0 . Define $\theta_0 = (\theta_{0\alpha}^\top, \theta_{01}^\top, \dots, \theta_{0d_x}^\top)^\top$, $\phi^K(s) = (\phi_1(s), \dots, \phi_K(s))^\top$,

$$R_{it}(s) := (A(\bar{Y}_{it}, s), X_{it}^\top)^\top, \quad H_{it}(s) := R_{it}(s) \otimes \phi^K(s), \quad \text{and} \quad H_t(s) = (H_{1t}(s), \dots, H_{nt}(s))^\top.$$

Then, we can further re-write the model in (1.2) as

$$Y_t(s) = H_t(s) \theta_0 + F_0(s) + V_t(s) + \mathcal{E}_t(s), \quad t \in [T].$$

Here, $V_t(s) = (v_{1t}(s), \dots, v_{nt}(s))^\top$ is an $n \times 1$ vector of series approximation errors:

$$v_{it}(s) := A(\bar{Y}_{it}, s) \{ \alpha_0(s) - \phi^K(s)^\top \theta_{0\alpha} \} + \sum_{j=1}^{d_x} X_{it}^j \{ \beta_{0j}(s) - \phi^K(s)^\top \theta_{0j} \}.$$

Under the assumptions we will introduce, this approximation error diminishes to zero at a certain rate as K goes to infinity. How to choose an appropriate K will be discussed in Remark 3.2.

Further, let

$$N := n(T - 1),$$

$$\mathbf{Y}(s) = \begin{pmatrix} Y_1(s) \\ \vdots \\ Y_T(s) \end{pmatrix}, \quad \mathbf{H}(s) = \begin{pmatrix} H_1(s) \\ \vdots \\ H_T(s) \end{pmatrix}, \quad \mathbf{V}(s) = \begin{pmatrix} V_1(s) \\ \vdots \\ V_T(s) \end{pmatrix}, \quad \mathcal{E}(s) = \begin{pmatrix} \mathcal{E}_1(s) \\ \vdots \\ \mathcal{E}_T(s) \end{pmatrix},$$

and $\mathbf{D}_{N \times nT} = (d_{ij})$ be the one-period lag operator, whose (i, j) -th element is defined as

$$d_{ij} = \begin{cases} -1 & \text{if } i = j \\ 1 & \text{if } n + i = j \\ 0 & \text{otherwise} \end{cases}$$

Then, we can remove the unknown fixed effects from the model in the following manner:

$$\mathbf{D}\mathbf{Y}(s) = \mathbf{D}\mathbf{H}(s)\theta_0 + \mathbf{D}\mathbf{V}(s) + \mathbf{D}\mathbf{\mathcal{E}}(s).$$

We estimate θ_0 based on this expression. In order to consistently estimate θ_0 , we need to address the endogeneity issue caused by the simultaneous interactions of the response functions; that is, since $A(\bar{Y}_{it}, s)$ is correlated with the error term $\varepsilon_{it}(s)$ in general, simply regressing $\mathbf{D}\mathbf{Y}(s)$ on $\mathbf{D}\mathbf{H}(s)$ does not yield a consistent estimate of θ_0 . To tackle this issue, we employ an instrumental variable (IV) approach.

Suppose we have a $d_q \times 1$ vector of IVs $Q_{it} = (Q_{it}^1, \dots, Q_{it}^{d_q})^\top$ for $A(\bar{Y}_{it}, s)$. Define

$$B_{it} := (Q_{it}^\top, X_{it}^\top)^\top, \quad Z_{it}(s) := B_{it} \otimes \phi^K(s), \quad Z_t(s) = (Z_{1t}(s), \dots, Z_{nt}(s))^\top, \quad \text{and} \quad \mathbf{Z}(s) = \begin{pmatrix} Z_1(s) \\ \vdots \\ Z_T(s) \end{pmatrix}.$$

Then, we have the *linear* moment conditions

$$\mathbb{E}[\mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{\mathcal{E}}(s)] = \mathbf{0}_{(d_q + d_x)K}.$$

Noting that $W_n Y_t = W_n \mathcal{A}Y_t + W_n X_t \beta_0 + W_n F_0 + W_n \mathcal{E}_t$, we can find that the network lagged covariates $\bar{X}_{it} := \sum_{j=1}^n w_{i,j} X_{jt}$ (and also their lags) are valid candidates for Q_{it} . However, when X_{it} is locally homogeneous, $\bar{X}_{it}$ and X_{it} become strongly correlated, which leads to a weak IV problem. Moreover, even when they are not strongly correlated, if the impact of X_{it} on Y_{it} is small (i.e., $\beta_0 \approx 0$), a weak IV problem may still occur.

Although one can estimate θ_0 based on the linear moment conditions only, which results in a two-stage least squares (2SLS) type estimator, we can utilize additionally the *quadratic* moment conditions to improve the efficiency of estimation (see, e.g., [Lin and Lee, 2010](#)). That is, under the independence assumption on the error terms $\{\varepsilon_{it}(s)\}_{i \in [n], t \in [T]}$ (Assumption 3.3(i) below), for any $N \times N$ matrices $P_m := I_{T-1} \otimes P_{m,1}$, where $P_{m,1}$ ($m = 1, \dots, M$) is an $n \times n$

matrix whose diagonal elements are all zero, we have²

$$\mathbb{E}[\boldsymbol{\mathcal{E}}(s)^\top \boldsymbol{D}^\top P_m \boldsymbol{D} \boldsymbol{\mathcal{E}}(s)] = 0, \quad m \in [M].$$

Some examples of $P_{m,1}$ include $P_{m,1} = W_n$ and $P_{m,1} = W_n^\top W_n - \text{diag}(W_n^\top W_n)$. The use of quadratic moment conditions is particularly important when the weakness of the IV $\boldsymbol{Z}(s)$ is a concern. As discussed in [Yang et al. \(2025\)](#) and also demonstrated in our numerical results, quadratic moment conditions can help mitigate weak identification when $\boldsymbol{Z}(s)$ are not strong IVs.

Combining the linear and quadratic moment conditions, we can construct our estimator based on the following $d_g := (d_q + d_x)K + M$ moment conditions: for $s \in [0, 1]$,

$$\frac{1}{N} \begin{pmatrix} \mathbb{E}[\boldsymbol{Z}(s)^\top \boldsymbol{D}^\top \boldsymbol{D} \boldsymbol{\mathcal{E}}(s)] \\ \mathbb{E}[\boldsymbol{\mathcal{E}}(s)^\top \boldsymbol{D}^\top P_1 \boldsymbol{D} \boldsymbol{\mathcal{E}}(s)] \\ \vdots \\ \mathbb{E}[\boldsymbol{\mathcal{E}}(s)^\top \boldsymbol{D}^\top P_M \boldsymbol{D} \boldsymbol{\mathcal{E}}(s)] \end{pmatrix} = \mathbf{0}_{d_g}.$$

As the empirical counterpart of these moment conditions, given a candidate value θ for θ_0 , we define

$$g_{nT}(s; \theta) := \frac{1}{N} \begin{pmatrix} \boldsymbol{Z}(s)^\top \boldsymbol{D}^\top \boldsymbol{D} \boldsymbol{E}(s; \theta) \\ \boldsymbol{E}(s; \theta)^\top \boldsymbol{D}^\top P_1 \boldsymbol{D} \boldsymbol{E}(s; \theta) \\ \vdots \\ \boldsymbol{E}(s; \theta)^\top \boldsymbol{D}^\top P_M \boldsymbol{D} \boldsymbol{E}(s; \theta) \end{pmatrix},$$

where $\boldsymbol{E}(s; \theta) := \boldsymbol{Y}(s) - \boldsymbol{H}(s)\theta$. Since we have a set of moment conditions that holds continuously over $s \in [0, 1]$, to construct our GMM estimator, we numerically integrate these conditions. Specifically, for a set of pre-specified $L \equiv L_{nT} \geq K$ grid points in $[0, 1]$, denoted by $0 \leq s_1 \leq \dots \leq s_L \leq 1$, we define

$$\bar{g}_{nT}(\theta) := \frac{1}{L} \sum_{l=1}^L g_{nT}(s_l; \theta).$$

² Throughout the paper, we assume that M is fixed. However, as pointed out by a referee, it would be interesting to consider an increasing number of quadratic moment conditions by setting, for example, $P_{m,1} = W_n^m - \text{diag}(W_n^m)$. The quadratic moments constructed in this way are strongly correlated with each other, and the contribution of higher-order quadratic moments may be marginal. This setting corresponds to GMM with many potentially weak moments (e.g., [Han and Phillips, 2006](#)), for which a separate and careful discussion is required. Thus, we leave this issue for future research.

Remark 3.1 below provides practical guidance on how to choose the grid $\{s_l\}$. Situations where the response functions are not fully observable are discussed in Subsection 6.3.

Now, we are ready to introduce our estimator:

$$\hat{\theta}_{nT} = (\hat{\theta}_{nT,\alpha}^\top, \hat{\theta}_{nT,1}^\top, \dots, \hat{\theta}_{nT,d_x}^\top)^\top := \underset{\theta \in \Theta_K}{\operatorname{argmin}} \mathcal{Q}_{nT}(\theta),$$

$$\text{where } \mathcal{Q}_{nT}(\theta) := \bar{g}_{nT}(\theta)^\top \Omega_{nT} \bar{g}_{nT}(\theta),$$

Ω_{nT} is a $d_g \times d_g$ positive definite symmetric weight matrix. For each $K \geq 1$, $\Theta_K \subset \mathbb{R}^{(d_x+1)K}$ is a compact parameter space containing θ_0 in its interior. Moreover, we assume that $\phi^K(s)^\top \theta_\alpha$ and $\phi^K(s)^\top \theta_j$ are bounded for any $\theta = (\theta_\alpha^\top, \theta_1^\top, \dots, \theta_{d_x}^\top)^\top \in \Theta_K$, uniformly in $s \in [0, 1]$. For one example of the weight matrix, we can use $\Omega_{nT} = I_{d_g}$. Another example is the 2SLS-type weight matrix:

$$\Omega_{nT} = \begin{pmatrix} \left(\sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbf{Z}(s_l) / (NL) \right)^{-1} & \mathbf{0}_{(d_q+d_x)K \times M} \\ \mathbf{0}_{M \times (d_q+d_x)K} & I_M \end{pmatrix}. \quad (3.1)$$

Once $\hat{\theta}_{nT}$ is obtained, the estimators of $\alpha_0(s)$ and $\beta_0(s)$ are given as

$$\hat{\alpha}_{nT}(s) := \phi^K(s)^\top \hat{\theta}_{nT,\alpha}, \text{ and } \hat{\beta}_{nT,j}(s) := \phi^K(s)^\top \hat{\theta}_{nT,j}, \quad j \in [d_x]$$

which we refer to as the integrated-GMM estimators. Additionally, if one is interested in the estimation of individual fixed effect functions, the following estimator can be used:

$$\hat{f}_{ni}(s) := \frac{1}{T} \sum_{t=1}^T \left(Y_{it}(s) - \hat{\alpha}_{nT}(s) A(\bar{Y}_{it}, s) - X_{it}^\top \hat{\beta}_{nT}(s) \right), \quad i \in [n]$$

where $\hat{\beta}_{nT}(s) = (\hat{\beta}_{nT,1}(s), \dots, \hat{\beta}_{nT,d_x}(s))^\top$. Consistent estimation of $f_{0i}(s)$ by $\hat{f}_{ni}(s)$ requires T to increase to infinity, while $\alpha_0(s)$ and $\beta_0(s)$ can be consistently estimated even when T is fixed. Unlike $\hat{\alpha}_{nT}(s)$ and $\hat{\beta}_{nT}(s)$, the estimator $\hat{f}_{ni}(s)$ is not necessarily continuous, as we do not preclude cases where $Y_{it}(s)$ and $A(\bar{Y}_{it}, s)$ are discontinuous in s .

3.2 Asymptotic theory

To derive the asymptotic properties of our estimator, we first need to specify the structure of our sampling space. Following Jenish and Prucha (2012), let $\mathcal{D} \subset \mathbb{R}^d$ be a possibly uneven lattice, and $\mathcal{D}_n \subset \mathcal{D}$ be the set of observation locations. Once the observation locations are

determined for a given sample of size n , we assume that they do not vary over time t . For spatial data, $\mathcal{D}$ would be defined by a geographical space with $d = 2$.³

We first derive the rates of convergence of our estimator under the following set of assumptions.

Assumption 3.1 (Sampling space). (i) The maximum coordinate difference (i.e., the Chebyshev distance) between any two observations $i, j \in \mathcal{D}$, which we denote as $\Delta(i, j)$, is at least (without loss of generality) 1; and (ii) a threshold distance $\bar{\Delta}$ exists such that $w_{i,j} = 0$ if $\Delta(i, j) > \bar{\Delta}$.

Assumption 3.2 (Observables). (i) $\{(X_{it}, Q_{it})\}_{i \in [n], t \in [T]}$ are non-stochastic and uniformly bounded; and (ii) for all $s \in [0, 1]$, $i \in [n]$, and $t \in [T]$, $\|Y_{it}(s)\|_p \lesssim 1$ for some $p > 4$.

Assumption 3.3 (Error term). (i) $\{\varepsilon_{it}\}_{i \in [n], t \in [T]}$ are independent; (ii) for all $s \in [0, 1]$, $i \in [n]$, and $t \in [T]$, $\mathbb{E}[\varepsilon_{it}(s)] = 0$, $\|\varepsilon_{it}(s)\|_2 > 0$, and $\|\varepsilon_{it}(s)\|_4 \lesssim 1$; and (iii) for all $i \in [n]$ and $t \in [T]$, $\sum_{k=1}^K \left(L^{-2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_{it}(s_l, s_{l'}) \phi_k(s_l) \phi_k(s_{l'}) \right) \lesssim 1$ uniformly in K , where $\Gamma_{it}(s_l, s_{l'}) := \text{Cov}(\varepsilon_{it}(s_l), \varepsilon_{it}(s_{l'}))$.

Assumption 3.4 (Interaction operator). There exists a function ω_p satisfying $|A(h, s)|^p \leq \int_0^1 |h(u)|^p \omega_p(u, s) du$ for any given $1 \leq p < \infty$, such that $\int_0^1 \omega_p(u, s) du \leq 1$ for all $s \in [0, 1]$.

Assumption 3.5 (Weight matrices). (i) For all $m \in [M]$, $P_{m,1}$ is symmetric, $\text{diag}(P_{m,1}) = \mathbf{0}_n$, and $\|P_{m,1}\|_1 (= \|P_{m,1}\|_\infty) \lesssim 1$. In addition, writing $P_{m,1} = (p_{m,i,j})$, a threshold distance $\bar{\Delta}_m$ exists such that $p_{m,i,j} = 0$ if $\Delta(i, j) > \bar{\Delta}_m$; and (ii) the GMM weight matrix, including the special case of the 2SLS-type weight (3.1), satisfies $0 < \lambda_{\min}(\Omega_{nT}) \leq \lambda_{\max}(\Omega_{nT}) \lesssim 1$ for all sufficiently large nT .

Assumption 3.6 (Identification). For all sufficiently large nT , $0 < \lambda_{\min}(\Pi_{nT}^\top \Pi_{nT}) \leq \lambda_{\max}(\Pi_{nT}^\top \Pi_{nT}) \lesssim 1$ uniformly in K , where $\Pi_{nT} := (NL)^{-1} \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\mathbf{H}(s_l)]$.

Assumption 3.7 (Series approximation). $\{\phi_k : k = 1, 2, \dots\}$ is a series of continuous orthonormal basis functions satisfying $\sup_{s \in [0,1]} |\alpha_0(s) - \phi^K(s)^\top \theta_{0\alpha}| \lesssim K^{-\pi}$ and $\max_{j \in [d_x]} \sup_{s \in [0,1]} |\beta_{0j}(s) - \phi^K(s)^\top \theta_{0j}| \lesssim K^{-\pi}$.

Assumptions 3.1(i) and (ii) together imply that the number of interacting partners for each unit is bounded (i.e., the network must be sparse). These assumptions play a crucial role in characterizing the stochastic process of the outcome functions. In Assumption 3.2, part

³Note that $\mathcal{D}$ does not necessarily have to be exactly observable to us. For example, $\mathcal{D}$ is possibly a complex space of general social and economic characteristics. In this case, we can consider it to be an embedding of individuals in a latent space, rather than their physical locations.

(i) assumes that the covariates are non-stochastic and bounded. This type of assumption is frequently utilized in the spatial and network literature and can be interpreted as viewing the analysis conditional on the realized values of the covariates. Meanwhile, part (ii) is introduced to ensure some convergence results for the quadratic moments.

Assumption 3.3(i) allows the error terms to be fully heteroskedastic. Part (ii) should be standard. Part (iii) is a high-level condition, which plays an important role to obtain the parametric convergence rate for the GMM estimator. This assumption essentially requires the covariance operator Γ_{it} to be *trace class* (see, e.g., [Hsing and Eubank, 2015](#)). More specifically, if Γ_{it} belongs to $L^2([0, 1]^2)$, it admits the following series expansion:

$$\Gamma_{it}(s, s') = \sum_{k_1, k_2=1}^{\infty} \kappa_{it, k_1, k_2} \phi_{k_1}(s) \phi_{k_2}(s')$$

for some sequence of constants $\{\kappa_{it, k_1, k_2}\}$. By the orthonormality of ϕ_k ,

$$\begin{aligned} \int_0^1 \int_0^1 \Gamma_{it}(s, s') \phi_k(s) \phi_k(s') ds ds' &= \sum_{k_1, k_2=1}^{\infty} \kappa_{it, k_1, k_2} \left(\int_0^1 \phi_k(s) \phi_{k_1}(s) ds \right) \left(\int_0^1 \phi_k(s') \phi_{k_2}(s') ds' \right) \\ &= \kappa_{it, k, k}. \end{aligned}$$

Since $L^{-2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_{it}(s_l, s_{l'}) \phi_k(s_l) \phi_k(s_{l'})$ can be seen as a numerical approximation of the left-hand side of the above expression, Assumption 3.3(iii) essentially requires that $\sum_{k=1}^K \kappa_{it, k, k} \lesssim 1$ uniformly in K . This type of assumption is commonly made in the FDA literature.

Assumption 3.4 is not restrictive in most empirically relevant situations. For example, in the case where $A(h, s) = h(s)$, we can set $\omega_p(u, s) = \delta(u - s)$ for any p . For another example, when $A(h, s) = \int_0^1 h(u) \nu(u, s) du$ for some kernel $\nu(u, s)$, since $|A(h, s)|^p \leq \int_0^1 |h(u)|^p |\nu(u, s)|^p du$, we can set $|\nu(u, s)|^p = \omega_p(u, s)$ in this case.

In Assumption 3.5, we assume that the weight matrices in the quadratic moments are symmetric. Note that this assumption does not lose any generality because $A^\top P_{m,1} A = A^\top P_{m,1}^\top A$ for any $n \times 1$ vector A . If $P_{m,1}$ is not symmetric in practice, we can always symmetrize it as $(P_{m,1} + P_{m,1}^\top)/2$. The assumption for the existence of a threshold distance $\bar{\Delta}_m$ may be non-standard, but it simplifies the proof. Since $P_{m,1}$'s are usually created from the interaction matrix W_n and its powers, this assumption is consistent with Assumption 3.1(ii).

Assumption 3.6 is a regularity condition to ensure the identifiability of θ_0 . Assumption 3.7 is standard in the series estimation literature. As shown in Theorem 3.2, we require $\pi > 3/2$ in order to establish the asymptotic normality of the estimator. This condition is satisfied, for example, when spline basis functions are used and the functions α_0 and β_{0j} are twice

continuously differentiable (see, e.g., [Chen, 2007](#); [Belloni et al., 2015](#)).

Remark 3.1 (Choice of the locations and the number of grid points). The choice of the grid $\{s_l\}$ determines the locations of s at which the continuum of moment conditions is evaluated, which affects both the identifiability of parameters and the performance of our estimator. As long as all of the above conditions are satisfied, any set of grid points can in principle be employed. For example, when the covariance operator Γ_{it} is indeed a trace-class operator, Assumption 3.3(iii) is more likely to be satisfied if the grid points are placed quasi-uniformly over $[0, 1]$ with a sufficiently large L . Further, to satisfy Assumption 3.6, it is generally necessary that the collection of basis vectors $\{\phi^K(s_1), \dots, \phi^K(s_L)\}$ contains at least K linearly independent columns. In particular, when the 2SLS-type weight matrix Ω_{nT} in (3.1) is used, the nonsingularity of $L^{-1} \sum_{l=1}^L \phi^K(s_l) \phi^K(s_l)^\top$ is required to satisfy Assumption 3.5(ii). Thus, in applications, it is advisable to choose grid points with equal spacing and to take L sufficiently large. Our numerical simulation results suggest that setting $L = 2K$ is sufficient to produce accurate estimates (see Appendix D).

Theorem 3.1 (Rates of convergence). Suppose that Assumptions 2.1, 3.1 – 3.7 hold. In addition, assume that $K/\sqrt{nT} \rightarrow 0$ and $K^{1/2-\pi} \rightarrow 0$ as $nT \rightarrow \infty$. Then,

- (i) $\|\hat{\theta}_{nT} - \theta_0\| \lesssim_p 1/\sqrt{nT} + K^{-\pi}$
- (ii) $\|\hat{\alpha}_{nT} - \alpha_0\|_{L^2} \lesssim_p 1/\sqrt{nT} + K^{-\pi}$, and $\sup_{s \in [0,1]} |\hat{\alpha}_{nT}(s) - \alpha_0(s)| \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi}$
- (iii) $\|\hat{\beta}_{nT,j} - \beta_{0j}\|_{L^2} \lesssim_p 1/\sqrt{nT} + K^{-\pi}$, and $\sup_{s \in [0,1]} |\hat{\beta}_{nT,j}(s) - \beta_{0j}(s)| \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi}$ for all $j \in [d_x]$.

The proof of Theorem 3.1 is given in Appendix B. Result (i) shows that, if the functional parameters are sufficiently smooth so that $K^{-\pi} \lesssim (nT)^{-1/2}$, the series coefficient estimator is consistent and converges at the parametric rate. The same convergence rate applies to the L^2 -convergence rate of the functional estimators, as shown in (ii) and (iii). Although these results may appear somewhat unexpected since the dimension of θ_0 diverges as $K \rightarrow \infty$, parametric convergence rates are commonly observed in dense functional data settings (e.g., [Huang et al., 2004](#); [Li and Hsing, 2010](#); [Hu et al., 2021](#)), our setup corresponding to an extreme case in which the entire functions are observed. The key condition underlying this rate is Assumption 3.3(iii), which ensures that, once K is sufficiently large, adding further basis terms does not inflate the variance of the series coefficient estimator. The uniform convergence rate for these estimators is $K^{1/2}$ slower than the L^2 -convergence rate.⁴

⁴Note that the uniform convergence results obtained here are not necessarily the sharpest, and the theoretically optimal convergence rates under our setup are also unknown. These points are left for future research.

We next present the limiting distribution of our estimators. To this end, we introduce the following notations and additional assumption:

$$\begin{aligned}
J_{nT}(s; \theta) &:= \frac{\partial g_{nT}(s; \theta)}{\partial \theta^\top}, \quad \bar{J}_{nT}(\theta) := \frac{1}{L} \sum_{l=1}^L J_{nT}(s_l; \theta), \quad \bar{J}_{nT} := \mathbb{E}[\bar{J}_{nT}(\theta_0)] \\
g_{1,nT}(s) &:= \frac{1}{N} \begin{pmatrix} \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\varepsilon}(s) \\ \boldsymbol{\varepsilon}(s)^\top \mathbf{D}^\top P_1 \mathbf{D} \boldsymbol{\varepsilon}(s) \\ \vdots \\ \boldsymbol{\varepsilon}(s)^\top \mathbf{D}^\top P_M \mathbf{D} \boldsymbol{\varepsilon}(s) \end{pmatrix}, \quad \bar{g}_{1,nT} := \frac{1}{L} \sum_{l=1}^L g_{1,nT}(s_l) \\
\mathcal{V}_{nT} &:= N \mathbb{E}[\bar{g}_{1,nT} \bar{g}_{1,nT}^\top] \\
\Sigma_{nT} &:= \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \bar{J}_{nT}^\top \Omega_{nT} \mathcal{V}_{nT} \Omega_{nT} \bar{J}_{nT} \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1}.
\end{aligned}$$

More explicit forms of the matrices $J_{nT}(s; \theta)$ and $\mathcal{V}_{nT}$ can be found in (A.2) and in (A.3) in Appendix A, respectively. Further, let $\mathbb{S}_\alpha$ and $\mathbb{S}_j$ be the $K \times (d_x + 1)K$ selection matrices such that $\theta_{0\alpha} = \mathbb{S}_\alpha \theta_0$ and $\theta_{0j} = \mathbb{S}_j \theta_0$ hold.

Assumption 3.8 (Misc.). For all sufficiently large nT , (i) $\lambda_{\max} \left((NL)^{-1} \sum_{l=1}^L \mathbb{E}[\mathbf{H}(s_l)^\top \mathbf{H}(s_l)] \right) \lesssim 1$; (ii) $0 < \lambda_{\min} \left(\bar{J}_{nT}^\top \bar{J}_{nT} \right) \leq \lambda_{\max} \left(\bar{J}_{nT}^\top \bar{J}_{nT} \right) \lesssim 1$; and (iii) $0 < \lambda_{\min}(\mathcal{V}_{nT}) \leq \lambda_{\max}(\mathcal{V}_{nT}) \lesssim 1$.

Theorem 3.2 (Asymptotic normality). Suppose that Assumptions 2.1, 3.1 – 3.8 hold. In addition, assume that $K/\sqrt{nT} \rightarrow 0$, $K^2/(\sqrt{nT} \|\phi^K(s)\|^2) \rightarrow 0$, and $\sqrt{nT} K^{-\pi} \rightarrow 0$ as $nT \rightarrow \infty$. Then, for any given $s \in [0, 1]$,

$$\text{(i)} \quad \frac{\sqrt{N} (\hat{\alpha}_{nT}(s) - \alpha_0(s))}{\sigma_{nT,\alpha}(s)} \xrightarrow{d} \mathcal{N}(0, 1), \quad \text{(ii)} \quad \frac{\sqrt{N} (\hat{\beta}_{nT,j}(s) - \beta_{0j}(s))}{\sigma_{nT,j}(s)} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $[\sigma_{nT,\alpha}(s)]^2 := \phi^K(s)^\top \mathbb{S}_\alpha \Sigma_{nT} \mathbb{S}_\alpha^\top \phi^K(s)$, $[\sigma_{nT,j}(s)]^2 := \phi^K(s)^\top \mathbb{S}_j \Sigma_{nT} \mathbb{S}_j^\top \phi^K(s)$, and recall that $N = n(T - 1)$.

Corollary 3.1. Suppose that the assumptions in Theorem 3.2 hold. If $K^3/(nT) \rightarrow 0$ and $K^{3/2-\pi} \rightarrow 0$ as $nT \rightarrow \infty$ additionally hold, we have

$$\text{(i)} \quad \frac{\sqrt{N} (\hat{\alpha}_{nT}(s) - \alpha_0(s))}{\hat{\sigma}_{nT,\alpha}(s)} \xrightarrow{d} \mathcal{N}(0, 1), \quad \text{(ii)} \quad \frac{\sqrt{N} (\hat{\beta}_{nT,j}(s) - \beta_{0j}(s))}{\hat{\sigma}_{nT,j}(s)} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $\hat{\sigma}_{nT,\alpha}(s)$ and $\hat{\sigma}_{nT,j}(s)$ are standard error estimators, whose definitions are provided in (C.1) in Appendix C.

Theorem 3.2 and Corollary 3.1 establish the pointwise asymptotic normality of the integrated-GMM estimators. We impose additional undersmoothing conditions to ensure that the bias terms vanish sufficiently quickly. See also Proposition C.1 in Appendix C, where additional asymptotic results for consistent variance estimation are established.

For the interpretation of these results, there are several points that should be noted. First, the standard deviations $\sigma_{nT,\alpha}(s)$ and $\sigma_{nT,j}(s)$ depend on the choice of the grid $\{s_l\}$ and its size L , as shown in their definitions. In this sense, one could consider a data-driven procedure for selecting the grid so as to improve the efficiency. We leave the investigation of such procedures for future research. Second, a straightforward calculation shows that the order of the standard deviations and their estimators is governed by $\|\phi^K(s)\|$, which is typically of order $\sqrt{K}$ for $s \in (0, 1)$. Therefore, the theorem does not necessarily assert the $\sqrt{N}$ -asymptotic normality for our functional estimator (similar to Theorem 2 of Newey (1997), for example). Third, the asymptotic normality results hold for any evaluation point $s \in [0, 1]$, including the boundary points $s = 0$ and $s = 1$. Nevertheless, both the required conditions on K and the structure of the standard deviations depend on s through the behavior of $\phi^K(s)$. For example, consider $\phi^K(0) = (1, 0, \dots, 0)^\top$, as is typical in practice. In this case, $\|\phi^K(0)\|^2 = 1$. Since Theorem 3.2 requires $K^2/(\sqrt{nT}\|\phi^K(s)\|^2) \rightarrow 0$, this implies that achieving the normality at the boundary needs a more restrictive condition on K than interior points.

Remark 3.2 (Choice of K). Suppose that K is proportional to $(nT)^c$ for some $0 < c < \infty$, and that $c_1 K \leq \|\phi^K(s)\|^2 \leq c_2 K$ for some $0 < c_1 \leq c_2 < \infty$. Then, to obtain the asymptotic normality results in Corollary 3.1, we require $K^3/(nT) \rightarrow 0$ and $\sqrt{nT}K^{-\pi} \rightarrow 0$ simultaneously. These conditions can be reduced to the following restriction on c : $1/(2\pi) < c < 1/3$. Thus, when the functional coefficients are believed to be sufficiently smooth, setting, for example, $K = \lfloor c_K(nT)^{1/5} \rfloor$ for some constant $c_K > 0$ would be a reasonable choice.

As a more data-driven approach, we also consider a cross-validation procedure in our numerical studies. Specifically, we split the data into a training sample consisting of periods $t = 1, \dots, T_{\text{train}}$ and a validation sample consisting of $t = T_{\text{train}} + 1, \dots, T$. We then estimate α_0 and β_0 using the training sample to predict the outcome difference $Y_{i,t+1} - Y_{it}$ for $t = T_{\text{train}} + 1, \dots, T - 1$, and the value of K is chosen to minimize the average mean squared prediction error (AMSPE):

$$\text{AMSPE} := \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \left(\frac{1}{n(T - T_{\text{train}} - 1)} \sum_{i=1}^n \sum_{t=T_{\text{train}}+1}^{T-1} \left\{ [\hat{Y}_{i,t+1}(s) - \hat{Y}_{it}(s)] - [Y_{i,t+1}(s) - Y_{it}(s)] \right\}^2 \right),$$

where $\mathcal{S}$ is a given set of evaluation points, and $|\mathcal{S}|$ denotes its cardinality. In our simulation analysis, this procedure performs reasonably well. For further details, see Appendix D.

4 Network Multiplier Effects: Marginal Effects and Impulse Responses

Once the model is estimated, as a next step, one might be interested in computing the marginal effects of covariates on the outcome. In a standard linear regression model without network interaction, the estimated coefficients directly represent the marginal effects of their corresponding covariates. However, in the presence of intricate functional interaction, this is no longer the case.

As shown in Section 2, under Assumption 2.1, we have the following moving-average type representation:

$$Y_t = \sum_{\ell=0}^{\infty} \mathcal{A}^{\ell} X_t \beta_0 + \sum_{\ell=0}^{\infty} \mathcal{A}^{\ell} F_0 + \sum_{\ell=0}^{\infty} \mathcal{A}^{\ell} \mathcal{E}_t, \quad t \in [T].$$

This expression indicates that the marginal effect of increasing X_{it}^j by one unit on Y_t is given by $\partial Y_t / (\partial X_{it}^j) = \lim_{c \rightarrow 0} \sum_{\ell=0}^{\infty} [\mathcal{A}^{\ell}(X_t^j + \mathbf{e}_i c) \beta_{0j} - \mathcal{A}^{\ell} X_t^j \beta_{0j}] / c = \sum_{\ell=0}^{\infty} \mathcal{A}^{\ell} \mathbf{e}_i \beta_{0j}$ by the linearity of $\mathcal{A}^{\ell}$, where X_t^j is the j -th column of X_t , and $\mathbf{e}_i$ denotes the i -th column of I_n . Alternatively, a little more informative expression can be obtained as follows: letting $\gamma(h, s) := \alpha_0(s) A(h, s)$,

$$\begin{aligned} M(i, j, s) &:= \partial Y_t(s) / (\partial X_{it}^j) = \mathbf{e}_i \beta_{0j}(s) + W_n \mathbf{e}_i \gamma(\beta_{0j}, s) + W_n^2 \mathbf{e}_i \gamma^2(\beta_{0j}, s) + \cdots \\ &= \sum_{\ell=0}^{\infty} W_n^{\ell} \mathbf{e}_i \gamma^{\ell}(\beta_{0j}, s), \end{aligned}$$

where $\gamma^0(\beta_{0j}, s) = \beta_{0j}(s)$, and $\gamma^{\ell}(\beta_{0j}, s) = \gamma(\gamma^{\ell-1}(\beta_{0j}, \cdot), s)$ for $\ell \geq 1$. From this, we can clearly see that the marginal effects $M(i, j, s)$ of increasing X_{it}^j consist of the direct effect on unit i , the indirect effect on i 's immediate neighbors, the second-order indirect effect on i 's neighbors' neighbors, and so forth, highlighting the presence of the *network multiplier* effect. More specifically, recall that when W_n represents a (weighted) adjacency matrix, the (i, j) -th element of W_n^{ℓ} corresponds to the number of (weighted) *walks* between i and j of length ℓ . Thus, the k -th element of $M(i, j, s)$ is interpreted as the weighted sum of the number of walks from i to k , where the contribution of each length- ℓ walk to the sum decays exponentially at $\gamma^{\ell}(\beta_{0j}, s)$.

To estimate the marginal effects, not just replacing the unknown parameters with their estimators, the infinite sum generally needs to be approximated by a truncated sum: for some positive integer S ,

$$\widehat{M}_{nT}^S(i, j, s) := \sum_{\ell=0}^S W_n^\ell \mathbf{e}_i \widehat{\gamma}_{nT}^\ell(\widehat{\beta}_{nT,j}, s),$$

where $\widehat{\gamma}_{nT}(h, s) := \widehat{\alpha}_{nT}(s)A(h, s)$. Meanwhile, in the special case of concurrent interaction such that $\gamma(h, s) = \alpha_0(s)h(s)$, $\gamma^2(h, s) = (\alpha_0(s))^2 h(s)$, $\dots$, it is easy to see that $M(i, j, s) = \sum_{\ell=0}^{\infty} (\alpha_0(s)W_n)^\ell \mathbf{e}_i \beta_{0j}(s) = (I_n - \alpha_0(s)W_n)^{-1} \mathbf{e}_i \beta_{0j}(s)$ holds. Thus, in this case, we can estimate $M(i, j, s)$ directly as $(I_n - \widehat{\alpha}_{nT}(s)W_n)^{-1} \mathbf{e}_i \widehat{\beta}_{nT,j}(s)$, without computing the infinite sum.

In the above discussion, we have demonstrated how the impacts of shifting one's covariate propagate to others. Similarly, just like the impulse response analysis in time-series vector autoregression, we can consider network impulse responses when an external shock occurs at a given unit. In particular, in a similar spirit to [Koop *et al.* \(1996\)](#), we define

$$I(i, \eta, s) := \mathbb{E}[Y_t(s) \mid \varepsilon_{it} = \eta] - \mathbb{E}[Y_t(s)],$$

where η is a given "function" representing the external shock. By a similar calculation as above, we obtain

$$I(i, \eta, s) = \sum_{\ell=0}^{\infty} W_n^\ell \mathbf{e}_i \gamma^\ell(\eta, s).$$

When plotting each element of $W_n^\ell \mathbf{e}_i \gamma^\ell(\eta, s)$ against $\ell = 0, 1, 2, \dots$, it can be interpreted as a network version of the impulse response function (as a function of ℓ), similarly to [Denbee *et al.* \(2021\)](#).

When assuming a concurrent interaction model, the impulse responses at s take the following form: $I(i, \eta, s) = (I_n - \alpha_0(s)W_n)^{-1} \mathbf{e}_i \eta(s)$. Thus, if there is no exogenous shock at s , i.e., if $\eta(s) = 0$, the expected outcome at s remains unaffected. This implies, for instance, that a travel demand shock that occurred five minutes ago has no impact on current mobility availability, which is unrealistic. On the other hand, if the interaction structure is given by $A(h, s) = \int_0^1 h(u) \nu(u, s) du$ with $\nu(s', s) \neq 0$ for $s' < s$, then a shock occurring at s' can transmit to the outcome at s , leading to nonzero impulse responses at s even when $\eta(s) = 0$.

The estimation of $I(i, \eta, s)$ can be performed in the same manner as above. For some positive integer S , we estimate $I(i, \eta, s)$ by $\widehat{I}_{nT}^S(i, \eta, s) := \sum_{\ell=0}^S W_n^\ell \mathbf{e}_i \widehat{\gamma}_{nT}^\ell(\eta, s)$. The next proposition

provides the convergence rate of $\widehat{M}_{nT}^S(i, j, s)$ and that of $\widehat{I}_{nT}^S(i, \eta, s)$.

Proposition 4.1. Suppose that the assumptions in Theorem 3.1 hold. In addition, assume that $\bar{\alpha}_0 < 1$. Then, uniformly in $s \in [0, 1]$,

$$\begin{aligned} \text{(i)} \quad & \max_{i \in [n]} \left\| \widehat{M}_{nT}^S(i, j, s) - M(i, j, s) \right\|_{\infty} \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi} + \bar{\alpha}_0^{S+1}, \\ \text{(ii)} \quad & \max_{i \in [n]} \left\| \widehat{I}_{nT}^S(i, \eta, s) - I(i, \eta, s) \right\|_{\infty} \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi} + \bar{\alpha}_0^{S+1}. \end{aligned}$$

This proposition indicates that the uniform convergence rates for the marginal effect and the impulse response estimators depend on the uniform convergence rate of the integrated-GMM estimator and the summation order S . Since the approximation error from truncating the infinite sum decreases geometrically as S increases, if $\bar{\alpha}_0$ is close to zero, setting $S = 4$ or 5 would be sufficient. Although $\bar{\alpha}_0$ is unknown in practice, it can be naturally estimated by $\widehat{\alpha}_{nT} := \max_{s \in [0, 1]} |\widehat{\alpha}_{nT}(s)|$. Then, if the value of $\widehat{\alpha}_{nT}$ is relatively large, it would be preferable to use a larger truncation order, say $S = 9$ or 10.

5 Monte Carlo Simulation

In this section, we conduct a series of Monte Carlo experiments to evaluate the finite-sample performance of the integrated-GMM estimator. To save space, we only summarize the main findings from the Monte Carlo simulations here; a complete description of the simulation design and full numerical results are provided in Appendix D. We compare the performance of three estimators: **GMM 1**: the integrated-GMM estimator using the weight matrix given in (3.1), **GMM 2**: the integrated-GMM estimator using the identity weight matrix, and (integrated) **2SLS**: GMM 1 estimator without utilizing the quadratic moment conditions.

Overall, all estimators perform well in terms of bias; however, GMM 1 consistently achieves the lowest RMSE (root-mean-squared error) across almost all settings, highlighting the efficiency gains from incorporating quadratic moment conditions. Interestingly, GMM 2 underperforms compared to 2SLS in many cases, indicating that the choice of the GMM weight matrix is equally (or potentially more) crucial to the inclusion of additional moment conditions. Meanwhile, as expected, 2SLS is more sensitive to the strength of IV than the others, with its RMSE for estimating α_0 decreasing more drastically as the IV strength increases. For the estimation of β_0 , the strength of IV has almost no impact, which is intuitive. Overall, the choice of L seems to have only minor impacts on performance. Lastly, when we increase the sample size from $nT = 240$ to $nT = 960$, the RMSE values are roughly halved, demonstrating

$\sqrt{nT}$ -consistency of the estimators, which numerically corroborates our theoretical result in Theorem 3.1(ii).

6 Extensions

In this section, we present several extensions of the current framework that are useful in empirical applications and can be implemented with relatively minor modifications to our method.

6.1 Functional NAR models with two-way fixed effects

So far, we have assumed that the error terms are independent over time, which may be restrictive in practice. A practical approach to accommodating time effects is to extend the present model (1.2) to the following two-way fixed effects model: for $s \in [0, 1]$,

$$Y_{it}(s) = \alpha_0(s) \sum_{j=1}^n w_{i,j} A(Y_{jt}, s) + X_{it}^\top \beta_0(s) + f_{0i}(s) + c_{0t}(s) + \varepsilon_{it}(s). \quad (6.1)$$

To facilitate the analysis of this model, we introduce the following additional conditions.

Assumption 6.1. (i) For all $i \in [n]$, $\sum_{j=1}^n w_{i,j} = 1$; and (ii) $\text{diag}(\mathcal{R}_n^\top P_{m,1} \mathcal{R}_n) = \mathbf{0}_n$.

Assumption 6.1(i) rules out isolated units. In condition (ii), we assume that $P_{m,1}$ satisfies Assumption 3.5(i), while the restriction $\text{diag}(P_{m,1}) = \mathbf{0}_n$ is now no longer necessary.

Under Assumption 6.1(i), to eliminate the time effects $c_{0t}(s)$ from the model, we can consider de-meaning by the neighborhood average rather than the conventional de-meaning by the global average. Specifically, letting $\mathcal{R}_n := I_n - W_n$ and $\mathcal{R}_{N \times N} := I_{T-1} \otimes \mathcal{R}_n$, we obtain

$$\mathcal{R}DY(s) = \mathcal{R}DH(s)\theta_0 + \mathcal{R}DV(s) + \mathcal{R}D\mathcal{E}(s).$$

It is straightforward to verify that the linear moment conditions

$$\mathbb{E}[\mathbf{Z}(s)^\top D^\top \mathcal{R}^\top \mathcal{R}D\mathcal{E}(s)] = \mathbf{0}_{(d_q + d_x)K}$$

hold. In addition, by Assumption 6.1(ii), the quadratic moment condition

$$\mathbb{E}[\mathcal{E}(s)^\top D^\top \mathcal{R}^\top P_m \mathcal{R}D\mathcal{E}(s)] = 0$$

also holds. Based on these moment conditions, we can construct a GMM estimator for α_0 and β_0 as in Subsection 3.1. With an abuse of notation, let $\hat{\alpha}_{nT}(s)$ and $\hat{\beta}_{nT,j}(s)$ be the resulting estimators of $\alpha_0(s)$ and $\beta_{0j}(s)$, respectively.

Theorem 6.1 (Asymptotic normality of the two-way fixed effects estimator). Suppose that the assumptions in Corollary 3.1 and Assumption 6.1 are satisfied. Then, we have

$$(i) \quad \frac{\sqrt{N}(\hat{\alpha}_{nT}(s) - \alpha_0(s))}{\hat{\sigma}_{nT,\alpha}(s)} \xrightarrow{d} \mathcal{N}(0, 1), \quad (ii) \quad \frac{\sqrt{N}(\hat{\beta}_{nT,j}(s) - \beta_{0j}(s))}{\hat{\sigma}_{nT,j}(s)} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $\hat{\sigma}_{nT,\alpha}(s)$ and $\hat{\sigma}_{nT,j}(s)$ are standard error estimators in this context, whose definitions are given in Appendix F.

The proof of Theorem 6.1 is outlined in Appendix F. It is worth noting that constructing a weight matrix $P_{m,1}$ satisfying $\text{diag}(\mathcal{R}_n^\top P_{m,1} \mathcal{R}_n) = \mathbf{0}_n$ is relatively straightforward. For an arbitrarily chosen $\tilde{P}_{m,1} = (\tilde{p}_{m,i,j})$, let $\mathcal{L} = \text{diag}(\ell_{1,1}, \dots, \ell_{n,n})$ be a diagonal matrix and set $P_{m,1} = \tilde{P}_{m,1} + \mathcal{L}$. Then, for each $i \in [n]$,

$$(\mathcal{R}_n^\top P_{m,1} \mathcal{R}_n)_{ii} = \sum_{j=1}^n \sum_{k=1}^n r_{j,i} \tilde{p}_{m,j,k} r_{k,i} + \sum_{j=1}^n \ell_{j,j} r_{j,i}^2,$$

where $r_{j,i}$ denotes the (j, i) -th element of $\mathcal{R}_n$. Hence, the constraint $\text{diag}(\mathcal{R}_n^\top P_{m,1} \mathcal{R}_n) = \mathbf{0}_n$ is equivalent to the linear system

$$\mathbf{y} = \mathbf{C}\boldsymbol{\ell},$$

where $\mathbf{y} = -\text{diag}(\mathcal{R}_n^\top \tilde{P}_{m,1} \mathcal{R}_n)$, $\mathbf{C} = (c_{i,j})$, $c_{i,j} = r_{j,i}^2$, and $\boldsymbol{\ell} = (\ell_{1,1}, \dots, \ell_{n,n})^\top$. Therefore, one may choose $\boldsymbol{\ell} = \mathbf{C}^- \mathbf{y}$, where $\mathbf{C}^-$ is a generalized inverse of $\mathbf{C}$.

6.2 More general interaction structures

As a straightforward extension of our current NAR model, one may consider a nonlinear NAR model:

$$\text{Nonlinear transformation : } Y_{it}(s) = \alpha_0(s) \sum_{j=1}^n w_{i,j} A(m_s(Y_{jt}), s) + \mu_{it}(s), \quad (6.2)$$

where $\mu_{it}(s) := X_{it}^\top \beta_0(s) + f_{0i}(s) + \varepsilon_{it}(s)$, and m_s is a known transformation (up to parameters). For example, consider $m_s(Y_{jt}) = b_1(s)Y_{jt} + b_2(s)Y_{jt}^2 + \dots + b_P(s)Y_{jt}^P$. Then, setting $\alpha_{0p}(s) :=$

$\alpha_0(s)b_p(s)$, we can write the model as

$$Y_{it}(s) = \sum_{p=1}^P \alpha_{0p}(s) A(\bar{Y}_{it}^{(p)}, s) + \mu_{it}(s), \quad (6.3)$$

where $\bar{Y}_{it}^{(p)} := \sum_{j=1}^n w_{i,j} Y_{jt}^p$.

Another potentially useful extension is a multiple-network model:

$$\textbf{Multiple networks : } Y_{it}(s) = \sum_{p=1}^P \alpha_{0p}(s) \sum_{j=1}^n w_{i,j}^{(p)} A(Y_{jt}, s) + \mu_{it}(s),$$

where $W_n^{(p)} = (w_{i,j}^{(p)})$, $p \in [P]$, are different interaction matrices. In this case, if we define $\bar{Y}_{it}^{(p)} := \sum_{j=1}^n w_{i,j}^{(p)} Y_{jt}$, the model can again be expressed as (6.3).

For both cases, as long as valid IVs Q_{it} for the multiple endogenous components $A(\bar{Y}_{it}^{(p)}, s)$, $p \in [P]$, are available, essentially the same estimation and inference procedure as in Section 3 can be applied, while we need to impose more restrictive conditions to ensure completeness. Additionally, models with multiple kernels $\nu^{(p)}$, $p \in [P]$, or heterogeneous interaction effects of the form $\alpha_0(s) = \sum_{p=1}^P x_{it}^{(p)} \alpha_{0p}(s)$ can also be analyzed in the same way.

6.3 Incompletely observed response function

The integrated-GMM estimator is often infeasible because the response functions are typically observed only at a finite set of points in $[0, 1]$. Even in such cases, we can approximate the entire functional form of Y_{it} using a linear interpolation method.

Suppose that for each (i, t) , Y_{it} is observed at L_{it} distinct points $0 \leq s_{it,1} \leq s_{it,2} \leq \dots \leq s_{it,L_{it}} \leq 1$. Then, for each given $s \in [s_{it,l}, s_{it,l+1}]$, define

$$Y_{it}^{\text{int}}(s) := Y_{it}(s_{it,l}) + \frac{Y_{it}(s_{it,l+1}) - Y_{it}(s_{it,l})}{s_{it,l+1} - s_{it,l}} (s - s_{it,l}).$$

When $s < s_{it,1}$ (resp. $s > s_{it,L_{it}}$), we set $Y_{it}^{\text{int}}(s) := Y_{it}(s_{it,1})$ (resp. $Y_{it}^{\text{int}}(s) := Y_{it}(s_{it,L_{it}})$).

Other than linear interpolation, one may also use a kernel method, as in [Zhu et al. \(2022\)](#), to obtain $Y_{it}^{\text{int}}(s)$. Then, using $Y_{it}^{\text{int}}(s)$ in place of $Y_{it}(s)$, we can write

$$Y_{it}^{\text{int}}(s) = \alpha_0(s) A(\bar{Y}_{it}^{\text{int}}, s) + X_{it}^\top \beta_0(s) + f_{0i}(s) + \varepsilon_{it}(s) + u_{it}(s),$$

where $u_{it}(s)$ is the interpolation error: $u_{it}(s) := Y_{it}^{\text{int}}(s) - Y_{it}(s) + \alpha_0(s) A(\bar{Y}_{it} - \bar{Y}_{it}^{\text{int}}, s)$. Thus, if $u_{it}(s)$ converges to zero sufficiently quickly for all $s \in [0, 1]$, $i \in [n]$, and $t \in [T]$, we can

apply the same estimation and inference strategy as above.

7 Analyzing the Demand of Bike-Sharing System

As an empirical application, we analyze spatial interactions in the demand for a bike-sharing system in the U.S. Demand analysis of shared mobility has been a highly active research topic in recent years across various areas, including transportation research, marketing, economics, and environmental studies. In particular, bike-sharing systems have attracted increasing attention. For a comprehensive review of this literature, see [Eren and Uz \(2020\)](#).

7.1 Data

The dataset used in this analysis comes from the *Bay Area Bike Share* in San Francisco, which was established in August 2013 and is now known as *Bay Wheels*. The dataset is publicly available on the *Kaggle* website.⁵ It contains detailed information about the system from August 2013 to August 2015, including station locations, the number of available bicycles at each station over time, and all trip-level data during this time period. The trip data include details such as start and end times and stations, as well as the user type (subscriber or casual user). In this dataset, there are 70 bike stations in total; for a map of all 70 station locations, see Figure E.1 in Appendix E.

Since the initial installation of stations in August 2013, the 70th station (*Ryland Park* station) was added in April 2014. Accordingly, we use data from May 2014 to August 2015 for this analysis, which represents the largest balanced panel dataset that can be extracted from the raw data.

One concern in the analysis is that shared mobility services often relocate bikes across stations in order to rebalance bike availability. To mitigate the effects of rebalancing operations in demand analysis, several approaches have been considered in the literature. For example, using the number of trips, pickups, or returns at each station as the outcome variable is a common strategy (e.g., [El-Assi et al., 2017](#); [Sun et al., 2018](#)). However, this approach is not straightforwardly applicable in the present functional data framework. Another approach is to detect potential rebalancing operations using heuristic thresholding rules (e.g., [Faghih-Imani et al., 2014, 2017](#)). Although it is generally impossible to access formal records of relocation operations, as noted by [Gammelli et al. \(2022\)](#), the majority of bike-sharing systems are rebalanced during nighttime (this is called *static rebalancing*). Then, if we observe sudden jumps

⁵<https://www.kaggle.com/datasets/benhamner/sf-bay-area-bike-share>

or drops in bike counts during nighttime hours, they are likely due to relocation operations conducted by the service provider. We adopt this approach in our analysis. Specifically, we first identify instances where the number of available bicycles jumps up/down by more than or equal to 10 all at once. We then examine the distribution of such events across hours and days, as shown in Figure E.2 in Appendix E. This figure indicates that sudden drops or increases in bike availability tend to occur between midnight and early morning, particularly on Sundays. Based on this, we exclude these time periods from our analysis.

Another concern is the enormous size of the dataset. Because the original data are recorded in minutes every day, using the raw data directly can lead to a memory problem. Moreover, daily data tend to fluctuate and to be noisy due to random events.

To address the aforementioned issues, we first rounded the trip data to 15-minute intervals and then averaged over Monday through Friday at each interval, discarding data from Saturdays and Sundays. Furthermore, to avoid potential bike relocation events in weekdays, we restrict the analysis to the time period from 6 AM to 9 PM. Consequently, our final dataset is a weekly-level panel with $n = 70$ stations and $T = 69$ weeks. The outcome of interest is the number of bicycles at each station for $s \in [0, 1]$, where $s = 0$ corresponds to 6 AM and $s = 1$ corresponds to 9 PM.

Figure 7.1 presents the trajectories of average bike availability for all 70 stations during the first week in our panel. It clearly shows that most of the variation in bike availability occurs between 6 AM and 9 PM.

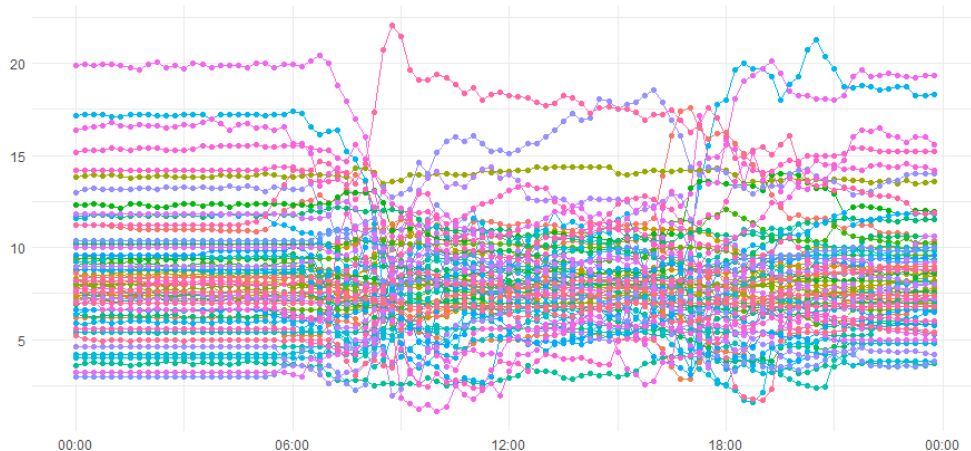


Figure 7.1: Availability of bikes at each station (averaged over 5-9 May, 2014)

7.2 Empirical results

Based on the dataset constructed as described above, we estimate model (1.2), where

$$\begin{aligned}
Y_{it}(s) &= \text{number of available bikes at each station} \\
A(Y_{jt}, s) &= \text{average of } Y_{jt} \text{ in the past one hour} \\
X_{it} &= [\text{ratio of round trips, ratio of subscribers (departing from station } i), \text{ratio of} \\
&\quad \text{subscribers (arriving at station } i), \text{rainy day dummy, month dummies}]^\top \\
w_{ij} &= \frac{\tilde{w}_{ij}}{\sum_{j \neq i} \tilde{w}_{ij}}, \text{ where } \tilde{w}_{ij} = \frac{\mathbf{1}\{\text{dist}(i, j) \leq 1\text{km}\}}{\text{dist}(i, j)}
\end{aligned}$$

Here, $\text{dist}(i, j)$ denotes the Euclidean distance between stations i and j . The estimation procedure is the same as the GMM 1 estimator in Section 5. The number of basis terms is selected via the cross-validation procedure described in Appendix D, using the last 11 weeks as the validation sample. As a result, we set $K = 8$ and $L = 3K$. The rainy day dummy and month dummy variables are not used as IVs. All integrals are approximated by finite summations over grid points at 15-minute intervals.

The estimation result for the interaction effect function α_0 is presented in Figure 7.2. In the figure, the shaded area depicts the (pointwise) 95% confidence interval. From the figure, we observe that positive spatial interaction in bicycle availability exists during the morning hours. It is plausible that as bike-sharing becomes more popular particularly among commuters, it encourages further use of the service, thereby reinforcing demand during the morning. Meanwhile, interestingly, negative interaction appears around 5–7 PM. In the evening, main users may include not only returning commuters but also individuals going out for dining, shopping, concerts, etc. As a result, bicycles might accumulate at certain popular stations while nearby less-popular stations experience lower availability, leading to the negative interaction. Alternatively, as kindly suggested by a referee, the negative correlation may also be driven by rebalancing operations conducted by the service provider, which artificially accumulate bikes at popular origin stations. However, it should be noted that autoregressive models of the current type are generally able to identify the presence of autocorrelation but are not designed to distinguish its underlying mechanisms. Investigating the detailed mechanisms generating such autocorrelation would require a different empirical framework.

To save space, the estimation results for $\beta_0(s)$ are presented in Figure E.4 in Appendix E, excluding the coefficients for the month dummies. Among the key covariates, we observe that only the ratio of arriving subscribers has a statistically significant positive impact on bike

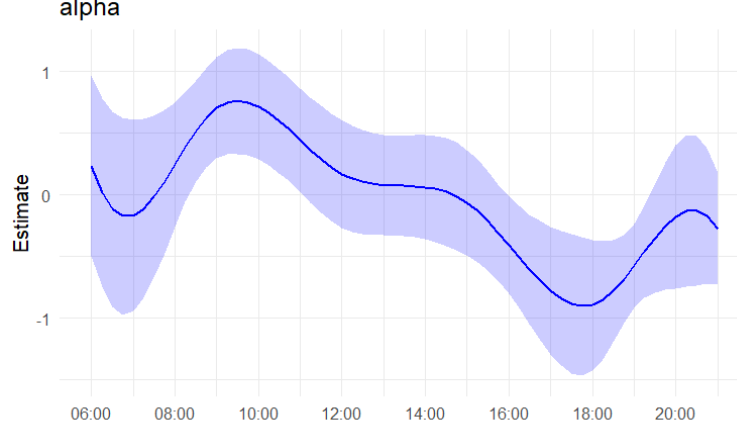


Figure 7.2: Estimated $\alpha_0(s)$

availability. This result is intuitive, as stations with a higher number of regular users arriving are expected to hold a richer stock of bikes. For other variables, for instance, the rainy day dummy has a positive effect on bike availability, which is consistent with previous studies, though the effect is not statistically significant. One possible explanation is that the rainy "day" dummy does not capture detailed temporal variations (i.e., it is not a function of s), and since our dataset is averaged over weeks, these may have diluted its impact.

Lastly, we conduct an impulse response analysis. The figures summarizing the results are presented in Figure 7.3. For illustration, we arbitrarily select the *Embarcadero at Folsom* station as the target station receiving an external shock. Specifically, we consider a hypothetical scenario in which the bike stock at this station is reduced by 2 at the peak of 9 AM (panel (a)). Panels (b) and (c) illustrate how the shock propagates to its two nearest stations, *Spear at Folsom* and *Temporary Transbay Terminal*. These figures indicate that the external shock spills over to these stations with a slight time delay, peaking just before 10 AM. Since the magnitude of both the external shock and spatial interaction is moderate in this analysis, the impulse responses for both stations are relatively mild.

7.3 Robustness check and comparison with a non-functional method

For robustness checks, we also estimated the model using $K = 7$ and $K = 9$. The corresponding estimates of α_0 are presented in Figure E.3 in Appendix E. In these figures, we can observe that the estimated interaction function exhibits the same pattern, with positive interaction in the morning hours and negative interaction at night.

Finally, for comparison, we also estimate a non-functional panel spatial autoregressive model with individual fixed effects in Lee and Yu (2010). The dependent variable is the average

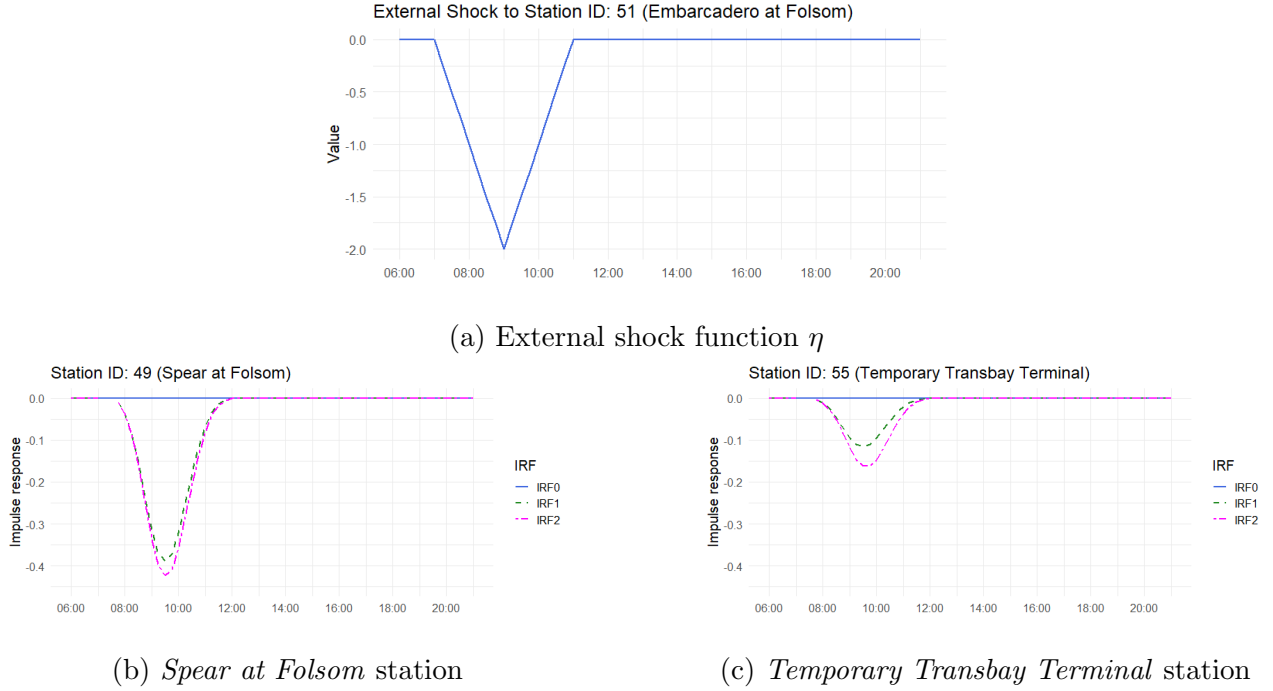


Figure 7.3: Impulse responses

$$\text{IRF: } \text{IRF0} = W_n^0 e_i \hat{\gamma}_{nT}^0(\eta, s), \text{IRF1} = \text{IRF0} + W_n^1 e_i \hat{\gamma}_{nT}^1(\eta, s), \text{ and } \text{IRF2} = \text{IRF1} + W_n^2 e_i \hat{\gamma}_{nT}^2(\eta, s).$$

weekly bike availability at each station, obtained by averaging $Y_{it}(s)$ over $s \in [6 \text{ AM}, 9 \text{ PM}]$. To save space, the result is reported in Table E.1 in Appendix E. Notably, the estimated spatial autoregressive parameter is slightly negative and statistically significant, possibly capturing the negative interaction during nighttime, as observed in Figure 7.2. However, as the figure also shows, positive interaction occurs during the morning hours. This illustrates that ignoring the functional nature of time-varying bike availability may result in a misinterpretation of the magnitude and direction of spatial interactions that change dynamically over time.

8 Conclusion

In this paper, we proposed a novel functional regression framework to analyze spatial and network interactions in functional panel data settings. By extending the standard NAR model to accommodate functional outcomes and individual fixed effects, we developed an integrated-GMM estimator that can estimate the functional parameters potentially more efficiently than 2SLS-based estimators. Under certain conditions, we established the theoretical properties of our estimator, including consistency, convergence rates, and asymptotic normality, and confirmed its finite-sample performance through Monte Carlo simulations. As an empirical

application, we analyzed the demand for a bike-sharing system in the San Francisco Bay Area and discovered significant spatial interactions in bike availability that vary over the time of day. This complex interaction pattern cannot be observed if the functional nature of the response variable is ignored, and thus our finding highlights the importance of accounting for functional spatial dependencies in modeling the demand for shared mobility services.

There are various avenues to extend our methodological and theoretical contributions. For example, following [Beyaztas *et al.* \(2024\)](#) and [Hoshino \(2024\)](#), one may instead consider the specification $\int_0^1 Y_{jt}(u)\alpha_0(u, s)du$ as an alternative to $\alpha_0(s)\int_0^1 Y_{jt}(u)\nu(u, s)du$, which corresponds to the standard functional linear form in the FDA literature. When m_s in (6.2) is unknown and estimated nonparametrically, this formulation leads to an ill-posed estimation problem (e.g., [Newey, 2013](#); [Hoshino, 2022](#); [Yang *et al.*, 2025](#)), which substantially complicates the asymptotic analysis. Also, one might want to consider a dynamic interaction structure in which the outcomes in the previous time period affect the current outcome through $A(Y_{j,t-1}, s)$. Such a model can be viewed as a functional extension of [Zhu *et al.* \(2017\)](#). The choice between our simultaneous model and the dynamic alternative should be decided by the data type. For instance, in the analysis of local interactions of mobility usage, where demand propagation occurs almost instantaneously, the simultaneous model may be more appropriate. In summary, many different types of functional interaction structures, as well as their combinations, can be considered. We leave these extensions to future research.

Acknowledgments

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Supplementary Appendix for "Network Autoregressive Models for Functional Panel Data"

Appendix A collects preparatory materials used to establish our theoretical results. Appendix B contains the proofs of Theorems 3.1 and 3.2, Corollary 3.1, and Proposition 4.1. Appendix C develops additional asymptotic results for consistent variance estimation. Appendices D and E contain additional materials for the Monte Carlo simulation results and the empirical analysis, respectively. Appendix F provides the proof of Theorem 6.1 on the asymptotic normality of the two-way fixed effects estimator.

A Preparation

The following definition is from [Jenish and Prucha \(2012\)](#).

Definition A.1 (Near-epoch dependence). Let $\mathbf{x} = \{x_{n,i} : i \in \mathcal{D}_n; n \geq 1\}$ and $\mathbf{e} = \{e_{n,i} : i \in \mathcal{D}_n; n \geq 1\}$ be triangular arrays of random fields, where x and e are real-valued and general (possibly infinite-dimensional) random variables, respectively. Then, the random field $\mathbf{x}$ is said to be L^p -near-epoch dependent (NED) on $\mathbf{e}$ if

$$\|x_{n,i} - \mathbb{E}[x_{n,i} \mid \mathcal{F}_{n,i}(\delta)]\|_p \leq c_{n,i}\rho(\delta)$$

for an array of finite positive constants $\{c_{n,i} : i \in \mathcal{D}_n; n \geq 1\}$ and some function $\rho(\delta) \geq 0$ with $\rho(\delta) \rightarrow 0$ as $\delta \rightarrow \infty$, where $\mathcal{F}_{n,i}(\delta)$ is the σ -field generated by $\{e_{n,j} : \Delta(i, j) \leq \delta\}$. The $c_{n,i}$'s and $\rho(\delta)$ are called the NED scaling factors and NED coefficient, respectively. The $\mathbf{x}$ is said to be uniformly L^p -NED on $\mathbf{e}$ if $c_{n,i}$ is uniformly bounded. If $\rho(\delta) \lesssim \varrho^\delta$ for some $0 < \varrho < 1$, then it is called geometrically L^p -NED.

In the following, for a general $\theta = (\theta_\alpha^\top, \theta_1^\top, \dots, \theta_{d_x}^\top)^\top \in \Theta_K$, we denote

$$\begin{aligned}\alpha(s; \theta) &:= \phi^K(s)^\top \theta_\alpha \\ \beta_j(s; \theta) &:= \phi^K(s)^\top \theta_j, \quad j \in [d_x]\end{aligned}$$

Recall that we have assumed that $\alpha(s; \theta)$ and $\beta_j(s; \theta)$ are uniformly bounded on $[0, 1]$ for all $\theta \in \Theta_K$ and K . For a given θ , the residual vector can be written as $\mathbf{E}(s; \theta) = (E_1(s; \theta)^\top, \dots, E_T(s; \theta)^\top)^\top$,

where

$$E_t(s; \theta) = (e_{1t}(s; \theta), \dots, e_{nt}(s; \theta))^\top$$

$$e_{it}(s; \theta) = Y_{it}(s) - \alpha(s; \theta)A(\bar{Y}_{it}, s) - \sum_{j=1}^{d_x} X_{it}^j \beta_j(s; \theta).$$

Under Assumptions 2.1(i), 3.2, and 3.4, we have

$$\|e_{it}(s; \theta)\|_p \leq \|Y_{it}(s)\|_p + |\alpha(s; \theta)| \sum_{j=1}^n |w_{i,j}| \|A(Y_{jt}, s)\|_p + \sum_{j=1}^{d_x} |X_{it}^j| \cdot |\beta_j(s; \theta)| \quad (\text{A.1})$$

$$\lesssim 1$$

for $p > 4$, uniformly in $s \in [0, 1]$, $\theta \in \Theta_K$, and (i, t) .

Finally, for ease of reference, we provide a list of some basic facts below:

$$\begin{aligned} \mathbf{D}\mathbf{E}(s; \theta) &= \mathbf{D}\mathbf{H}(s)(\theta_0 - \theta) + \mathbf{D}\mathbf{V}(s) + \mathbf{D}\mathbf{E}(s) \\ \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{E}(s; \theta) &= \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{H}(s)(\theta_0 - \theta) + \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{V}(s) + \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{E}(s) \\ \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{E}(s; \theta_0) &= \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{V}(s) + \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{E}(s) \\ \mathbf{E}(s; \theta)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{E}(s; \theta) &= (\theta_0 - \theta)^\top \mathbf{H}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{H}(s)(\theta_0 - \theta) + \mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{V}(s) \\ &\quad + \mathbf{E}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{E}(s) + 2\mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{H}(s)(\theta_0 - \theta) \\ &\quad + 2\mathbf{E}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{H}(s)(\theta_0 - \theta) + 2\mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{E}(s) \\ \mathbf{E}(s; \theta_0)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{E}(s; \theta_0) &= \mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{V}(s) + \mathbf{E}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{E}(s) + 2\mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\mathbf{E}(s) \end{aligned}$$

Empirical moment function:

$$g_{nT}(s; \theta) := \frac{1}{N} \begin{pmatrix} \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{E}(s; \theta) \\ \mathbf{E}(s; \theta)^\top \mathbf{D}^\top P_1 \mathbf{D}\mathbf{E}(s; \theta) \\ \vdots \\ \mathbf{E}(s; \theta)^\top \mathbf{D}^\top P_M \mathbf{D}\mathbf{E}(s; \theta) \end{pmatrix}$$

Jacobian of $g_{nT}(s; \theta)$:

$$J_{nT}(s; \theta) := \frac{\partial g_{nT}(s; \theta)}{\partial \theta^\top} = -\frac{1}{N} \begin{pmatrix} \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}\mathbf{H}(s) \\ 2\mathbf{E}(s; \theta)^\top \mathbf{D}^\top P_1 \mathbf{D}\mathbf{H}(s) \\ \vdots \\ 2\mathbf{E}(s; \theta)^\top \mathbf{D}^\top P_M \mathbf{D}\mathbf{H}(s) \end{pmatrix} \quad (\text{A.2})$$

Decompose $\bar{g}_{nT}(\theta_0) = \bar{g}_{1,nT} + \bar{g}_{2,nT}$ with

$$\begin{aligned}\bar{g}_{1,nT} &:= \frac{1}{NL} \sum_{l=1}^L \begin{pmatrix} \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) \\ \boldsymbol{\mathcal{E}}(s_l)^\top \mathbf{D}^\top P_1 \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) \\ \vdots \\ \boldsymbol{\mathcal{E}}(s_l)^\top \mathbf{D}^\top P_M \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) \end{pmatrix} \\ \bar{g}_{2,nT} &:= \frac{1}{NL} \sum_{l=1}^L \begin{pmatrix} \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbf{V}(s_l) \\ \mathbf{V}(s_l)^\top \mathbf{D}^\top P_1 \mathbf{D} \mathbf{V}(s_l) + 2\mathbf{V}(s_l)^\top \mathbf{D}^\top P_1 \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) \\ \vdots \\ \mathbf{V}(s_l)^\top \mathbf{D}^\top P_M \mathbf{D} \mathbf{V}(s_l) + 2\mathbf{V}(s_l)^\top \mathbf{D}^\top P_M \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) \end{pmatrix}\end{aligned}$$

The variance-covariance matrix of $\sqrt{N}\bar{g}_{1,nT}$:

$$\mathcal{V}_{nT} := N\mathbb{E}[\bar{g}_{1,nT}\bar{g}_{1,nT}^\top] = \begin{pmatrix} \mathcal{V}_{z,nT} & \mathbf{0}_{(d_q+d_x)K \times 1} & \cdots & \mathbf{0}_{(d_q+d_x)K \times 1} \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \mathcal{V}_{11,nT} & \cdots & \mathcal{V}_{1M,nT} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \mathcal{V}_{M1,nT} & \cdots & \mathcal{V}_{MM,nT} \end{pmatrix} \quad (\text{A.3})$$

Here, for convenience, we introduce the following relabeling: $(it) = (11) \iff i = 1$, $(it) = (21) \iff i = 2, \dots$, $(it) = (nT) \iff i = \mathbf{n}$ (where $\mathbf{n} = nT$). Then, we can write

$$\begin{aligned}\mathcal{V}_{z,nT} &:= \frac{N}{(NL)^2} \sum_{l=1}^L \sum_{l'=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\boldsymbol{\mathcal{E}}(s_l)\boldsymbol{\mathcal{E}}(s_{l'})^\top] \mathbf{D}^\top \mathbf{D} \mathbf{Z}(s_{l'}) \\ &= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{i=1}^{\mathbf{n}} z_i^\dagger(s_l) z_i^\dagger(s_{l'})^\top \Gamma_i(s_l, s_{l'}),\end{aligned} \quad (\text{A.4})$$

$z_i^\dagger(s)$ denotes the i -th column of $\underbrace{\mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D}}_{(d_q+d_x)K \times nT}$. Furthermore, let $\underbrace{\mathbf{D}^\top P_m \mathbf{D}}_{nT \times nT} =: \tilde{P}_m = (\tilde{p}_{m,i,j})$, and

$$\underbrace{\mathbf{D}_1}_{(T-1) \times T} := \begin{pmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & 1 \end{pmatrix}$$

such that $\mathbf{D} = D_1 \otimes I_n$. Then, we have

$$\begin{aligned} \text{diag}(\tilde{P}_m) &= \text{diag}((D_1^\top \otimes I_n)(I_{T-1} \otimes P_{m,1})(D_1 \otimes I_n)) \\ &= \text{diag}((D_1^\top D_1) \otimes P_{m,1}) = \mathbf{0}_{nT}. \end{aligned}$$

Since $\tilde{P}_m$ is symmetric,

$$\begin{aligned} \mathcal{V}_{ab,nT} &:= \frac{N}{(NL)^2} \sum_{l=1}^L \sum_{l'=1}^L \mathbb{E}[\boldsymbol{\mathcal{E}}(s_l)^\top \mathbf{D}^\top P_a \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) \boldsymbol{\mathcal{E}}(s_{l'})^\top \mathbf{D}^\top P_b \mathbf{D} \boldsymbol{\mathcal{E}}(s_{l'})] \\ &= \frac{1}{N} \sum_{1 \leq i,j,k,l \leq n} \tilde{p}_{a,i,j} \tilde{p}_{b,k,l} \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \mathbb{E}[\varepsilon_i(s_l) \varepsilon_j(s_l) \varepsilon_k(s_{l'}) \varepsilon_l(s_{l'})] \\ &= \frac{1}{N} \sum_{1 \leq i,k \leq n} \tilde{p}_{a,i,k} \tilde{p}_{b,i,k} \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_k(s_l, s_{l'}) \\ &\quad + \frac{1}{N} \sum_{1 \leq i,k \leq n} \tilde{p}_{a,i,k} \tilde{p}_{b,k,i} \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_k(s_l, s_{l'}) \\ &= \frac{2}{N} \sum_{1 \leq i,k \leq n} \tilde{p}_{a,i,k} \tilde{p}_{b,i,k} \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_k(s_l, s_{l'}). \end{aligned} \tag{A.5}$$

Note that the cross terms between the linear and quadratic moments are zero:

$$\begin{aligned} &\frac{N}{(NL)^2} \sum_{l=1}^L \sum_{l'=1}^L \mathbb{E}[\mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) \boldsymbol{\mathcal{E}}(s_{l'})^\top \mathbf{D}^\top P_m \mathbf{D} \boldsymbol{\mathcal{E}}(s_{l'})] \\ &= \frac{N}{(NL)^2} \sum_{l=1}^L \sum_{l'=1}^L \sum_{1 \leq i,j,k \leq n} z_i^\dagger(s_l) \tilde{p}_{m,j,k} \mathbb{E}[\varepsilon_i(s_l) \varepsilon_j(s_{l'}) \varepsilon_k(s_{l'})] = \mathbf{0}_{(d_q+d_x)K}. \end{aligned}$$

B Proofs

B.1 Results for Proposition 2.1

B.1.1 Proof of Proposition 2.1

Under Assumption 2.1, we have

$$\begin{aligned}
\|\{\mathcal{A}H\}_i\|_{L^2} &= \left\| \alpha_0(\cdot) \sum_{j=1}^n w_{i,j} A(h_j, \cdot) \right\|_{L^2} \leq \sum_{j=1}^n |w_{i,j}| \|\alpha_0(\cdot) A(h_j, \cdot)\|_{L^2} \\
&= \sum_{j=1}^n |w_{i,j}| \left(\int_0^1 |\alpha_0(s) A(h_j, s)|^2 ds \right)^{1/2} \\
&\leq \bar{\alpha}_0 \sum_{j=1}^n |w_{i,j}| \|A(h_j, \cdot)\|_{L^2} \\
&\leq \bar{\alpha}_0 \|W_n\|_\infty \max_{1 \leq j \leq n} \|h_j\|_{L^2} < \|H\|_{\infty, 2} < \infty
\end{aligned} \tag{B.1}$$

for any $H \in \mathcal{H}_{n,2}$. This implies that $\mathcal{A}H \in \mathcal{H}_{n,2}$. As is well known, if the operator norm of $\mathcal{A}$ is smaller than one, $(\text{Id} - \mathcal{A})^{-1}$ exists (e.g., Theorem 2.14, Kress (2014)). It is immediate from (B.1) that $\|\mathcal{A}H\|_{\infty, 2} < 1$ follows for any H such that $\|H\|_{\infty, 2} = 1$, which yields the desired result. ■

B.1.2 Alternative sufficient condition for the completeness

As an alternative to Assumption 2.1, suppose that

$$\begin{aligned}
\text{(i)} \quad & \|\alpha_0\|_{L^2} \|W_n\|_\infty < 1, \\
\text{(ii)} \quad & A(h, s) = \int_0^1 h(u) \nu(u, s) du, \text{ and } \sup_{(u,s) \in [0,1]^2} |\nu(u, s)| \leq 1.
\end{aligned} \tag{B.2}$$

In contrast to the condition $\bar{\alpha}_0 \|W_n\|_\infty < 1$ introduced in the main text, the network effect function $\alpha_0(s)$ does not need to be uniformly bounded in a small range to satisfy condition (i) in (B.2). Under these conditions, we obtain

$$\begin{aligned}
\int_0^1 |\alpha_0(s) A(h, s)|^2 ds &= \int_0^1 |\alpha_0(s)|^2 \left| \int_0^1 h(u) \nu(u, s) du \right|^2 ds \\
&\leq \int_0^1 |\alpha_0(s)|^2 \int_0^1 |h(u)|^2 |\nu(u, s)|^2 du ds \\
&\leq \|\alpha_0\|_{L^2}^2 \|h\|_{L^2}^2.
\end{aligned}$$

Hence, similarly to (B.1), we have

$$\begin{aligned}\|\{\mathcal{A}H\}_i\|_{L^2} &\leq \sum_{j=1}^n |w_{i,j}| \left(\int_0^1 |\alpha_0(s)A(h_j, s)|^2 ds \right)^{1/2} \\ &\leq \|\alpha_0\|_{L^2} \|W_n\|_\infty \max_{1 \leq j \leq n} \|h_j\|_{L^2} < \|H\|_{\infty, 2}.\end{aligned}$$

B.2 Proof of Theorem 3.1

B.2.1 Lemmas

Lemma B.1. Suppose that Assumptions 2.1(i), 3.2, 3.3(ii), 3.4, 3.5(i), and 3.7 hold. Then, $\mathbb{E}[g_{nT}(s; \theta_0)] \lesssim \mathbf{1}_{d_g} K^{-\pi}$ element-wisely.

Proof. Observe that

$$\mathbb{E}[g_{nT}(s; \theta_0)] = \frac{1}{N} \begin{pmatrix} \mathbf{Z}(s)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\mathbf{V}(s)] \\ \mathbb{E}[\mathbf{V}(s)^\top \mathbf{D}^\top P_1 \mathbf{D} \mathbf{V}(s)] + 2\mathbb{E}[\mathbf{V}(s)^\top \mathbf{D}^\top P_1 \mathbf{D} \boldsymbol{\varepsilon}(s)] \\ \vdots \\ \mathbb{E}[\mathbf{V}(s)^\top \mathbf{D}^\top P_M \mathbf{D} \mathbf{V}(s)] + 2\mathbb{E}[\mathbf{V}(s)^\top \mathbf{D}^\top P_M \mathbf{D} \boldsymbol{\varepsilon}(s)] \end{pmatrix}$$

By Assumptions 3.2(ii) and 3.4,

$$\begin{aligned}|A(\mathbb{E}[Y_{jt}], s)| &\leq \int_0^1 |\mathbb{E}[Y_{jt}(u)]| \omega_1(u, s) du \\ &\leq \int_0^1 \mathbb{E}|Y_{jt}(u)| \omega_1(u, s) du \lesssim 1\end{aligned}$$

uniformly in $s \in [0, 1]$, implying that $\sup_{s \in [0, 1]} |A(\mathbb{E}[Y_{jt}], s)| \lesssim 1$. Then, we have

$$\begin{aligned}|\mathbb{E}[v_{it}(s)]| &\leq \sum_{j=1}^n |w_{i,j}| \cdot |A(\mathbb{E}[Y_{jt}], s)| \cdot |\alpha_0(s) - \alpha(s; \theta_0)| + \sum_{j=1}^{d_x} |X_{it}^j| \cdot |\beta_{0j}(s) - \beta_j(s; \theta_0)| \\ &\lesssim K^{-\pi}\end{aligned}$$

uniformly in $s \in [0, 1]$ and (i, t) under Assumption 3.7. This implies that the first $(d_q + d_x)K$ elements of $\mathbb{E}[g_{nT}(s; \theta_0)]$ are of order $K^{-\pi}$.

Next, by Cauchy-Schwarz inequality and the fact that $\lambda_{\max}(\mathbf{D}^\top P_m \mathbf{D}) \lesssim 1$ under Assump-

tion 3.5(i), we obtain

$$\begin{aligned}
\mathbb{E} |\mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{V}(s)| &\leq \sqrt{\mathbb{E} \|\mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D}\|^2} \sqrt{\mathbb{E} \|\mathbf{V}(s)\|^2} \\
&= \sqrt{\text{trace}\{\mathbf{D}^\top P_m \mathbf{D} \mathbf{D}^\top P_m \mathbf{D} \mathbb{E}[\mathbf{V}(s) \mathbf{V}(s)^\top]\}} \sqrt{\mathbb{E} \|\mathbf{V}(s)\|^2} \\
&\lesssim \sum_{t=1}^T \sum_{i=1}^n \mathbb{E} |v_{it}(s)|^2.
\end{aligned}$$

Similarly as above, by the c_r inequality,

$$\begin{aligned}
\mathbb{E} |v_{it}(s)|^2 &\leq 2 \mathbb{E} \left| \sum_{j=1}^n w_{i,j} A(Y_{jt}, s) [\alpha_0(s) - \alpha(s; \theta_0)] \right|^2 + 2 \left| \sum_{j=1}^{d_x} X_{it}^j (\beta_{0j}(s) - \beta_j(s; \theta_0)) \right|^2 \\
&\leq 2 \sum_{j=1}^n \sum_{j'=1}^n w_{i,j} w_{i,j'} \mathbb{E} [A(Y_{jt}, s) A(Y_{j't}, s)] [\alpha_0(s) - \alpha(s; \theta_0)]^2 + cK^{-2\pi}.
\end{aligned}$$

By Cauchy-Schwarz inequality,

$$|\mathbb{E}[A(Y_{jt}, s) A(Y_{j't}, s)]| \leq \|A(Y_{jt}, s)\|_2 \|A(Y_{j't}, s)\|_2.$$

Further, Assumptions 3.2(ii) and 3.4 imply that $\mathbb{E}|A(Y_{jt}, s)|^2 \leq \int_0^1 \mathbb{E}|Y_{jt}(u)|^2 \omega_2(u, s) du \lesssim 1$ uniformly in $s \in [0, 1]$ and (j, t) . Thus, $|\mathbb{E}[A(Y_{jt}, s) A(Y_{j't}, s)]|$ is uniformly bounded, and we have

$$\|v_{it}(s)\|_2 \lesssim K^{-\pi} \quad (\text{B.3})$$

uniformly in $s \in [0, 1]$ and (i, t) .

Lastly, by Cauchy-Schwarz and Minkowski's inequalities,

$$\begin{aligned}
\mathbb{E}[\mathbf{V}(s)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s)] &= \sum_{t \in [T-1]} \sum_{1 \leq i, j \leq n} p_{m,i,j} \mathbb{E}[(v_{i,t+1}(s) - v_{it}(s))(\varepsilon_{j,t+1}(s) - \varepsilon_{jt}(s))] \\
&\leq \sum_{t \in [T-1]} \sum_{1 \leq i, j \leq n} |p_{m,i,j}| \mathbb{E}|(v_{i,t+1}(s) - v_{it}(s))(\varepsilon_{j,t+1}(s) - \varepsilon_{jt}(s))| \\
&\leq \sum_{t \in [T-1]} \sum_{1 \leq i, j \leq n} |p_{m,i,j}| \cdot \|v_{i,t+1}(s) - v_{it}(s)\|_2 \|\varepsilon_{j,t+1}(s) - \varepsilon_{jt}(s)\|_2 \\
&\leq \sum_{t \in [T-1]} \sum_{1 \leq i, j \leq n} |p_{m,i,j}| \cdot \{\|v_{i,t+1}(s)\|_2 + \|v_{it}(s)\|_2\} \{\|\varepsilon_{j,t+1}(s)\|_2 + \|\varepsilon_{jt}(s)\|_2\} \\
&\lesssim NK^{-\pi}
\end{aligned}$$

where the last inequality follows from (B.3) and Assumptions 3.3(ii) and 3.5(i). Combining these results gives the desired result. $\square$

Denote the population GMM objective function as follows:

$$\mathcal{Q}_{nT}^*(\theta) := \mathbb{E}[\bar{g}_{nT}(\theta)]^\top \Omega_{nT} \mathbb{E}[\bar{g}_{nT}(\theta)]$$

Lemma B.2. Suppose that Assumptions 2.1(i), 3.2, 3.3(ii), 3.4, 3.5, 3.6, and 3.7 hold. In addition, assume that $K^{(1-2\pi)/2} \rightarrow 0$ as $nT \rightarrow \infty$. Then, for any $\theta \in \Theta_K$ and $e > 0$ such that $\|\theta - \theta_0\| \geq e$, there exists a constant $c_e > 0$ such that $\mathcal{Q}_{nT}^*(\theta) - \mathcal{Q}_{nT}^*(\theta_0) > c_e$ for all sufficiently large nT .

Proof. Decompose

$$\begin{aligned} \mathcal{Q}_{nT}^*(\theta) - \mathcal{Q}_{nT}^*(\theta_0) &= \underbrace{(\mathbb{E}[\bar{g}_{nT}(\theta)] - \mathbb{E}[\bar{g}_{nT}(\theta_0)])^\top \Omega_{nT} (\mathbb{E}[\bar{g}_{nT}(\theta)] - \mathbb{E}[\bar{g}_{nT}(\theta_0)])}_{=: A_{nT}(\theta)} \\ &\quad + 2 (\mathbb{E}[\bar{g}_{nT}(\theta)] - \mathbb{E}[\bar{g}_{nT}(\theta_0)])^\top \Omega_{nT} \mathbb{E}[\bar{g}_{nT}(\theta_0)] \end{aligned}$$

In view of

$$\begin{aligned} &\mathbb{E}[\bar{g}_{nT}(\theta)] - \mathbb{E}[\bar{g}_{nT}(\theta_0)] \\ &= \frac{1}{NL} \sum_{l=1}^L \begin{pmatrix} \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\mathbf{H}(s_l)] \\ \mathbb{E}[(\mathbf{H}(s_l)(\theta_0 - \theta) + 2\mathbf{V}(s_l) + 2\mathbf{E}(s_l))^\top \mathbf{D}^\top P_1 \mathbf{D} \mathbf{H}(s_l)] \\ \vdots \\ \mathbb{E}[(\mathbf{H}(s_l)(\theta_0 - \theta) + 2\mathbf{V}(s_l) + 2\mathbf{E}(s_l))^\top \mathbf{D}^\top P_M \mathbf{D} \mathbf{H}(s_l)] \end{pmatrix} (\theta_0 - \theta), \end{aligned}$$

we can find that $A_{nT}(\theta)$ is bounded below from $\lambda_{\min}(\Omega_{nT}) \lambda_{\min}(\Pi_{nT}^\top \Pi_{nT}) \|\theta_0 - \theta\|^2 \geq c_1 e^2$ for some $c_1 > 0$ for all sufficiently large nT , under Assumptions 3.5(ii) and 3.6. Further, Cauchy-Schwarz inequality and Lemma B.1 give that

$$\begin{aligned} \left| (\mathbb{E}[\bar{g}_{nT}(\theta)] - \mathbb{E}[\bar{g}_{nT}(\theta_0)])^\top \Omega_{nT} \mathbb{E}[\bar{g}_{nT}(\theta_0)] \right| &\leq (A_{nT}(\theta))^{1/2} (\mathbb{E}[\bar{g}_{nT}(\theta_0)]^\top \Omega_{nT} \mathbb{E}[\bar{g}_{nT}(\theta_0)])^{1/2} \\ &\leq c_2 (A_{nT}(\theta))^{1/2} K^{(1-2\pi)/2} \end{aligned}$$

Hence, since $(A_{nT}(\theta))^{1/2}$ is bounded below from zero and $K^{(1-2\pi)/2} \rightarrow 0$, we have

$$\begin{aligned}\mathcal{Q}_{nT}^*(\theta) - \mathcal{Q}_{nT}^*(\theta_0) &\geq A_{nT}(\theta) - 2c_2(A_{nT}(\theta))^{1/2}K^{(1-2\pi)/2} \\ &= (A_{nT}(\theta))^{1/2}((A_{nT}(\theta))^{1/2} - 2c_2K^{(1-2\pi)/2}) > 0\end{aligned}$$

for all sufficiently large nT . This completes the proof. $\square$

Lemma B.3. Suppose that Assumptions 2.1, 3.1, 3.2, and 3.4 hold. Then, for any given $s \in [0, 1]$ and all $t \in [T]$, $\{Y_{it}(s) : i \in \mathcal{D}_n; n \geq 1\}$ is uniformly and geometrically L^2 -NED on $\{\varepsilon_{it} : i \in \mathcal{D}_n; n \geq 1\}$.

Proof. We prove the lemma in a similar manner to Jenish (2012) and Hoshino (2022). Recall that Y_t is uniquely determined in $\mathcal{H}_{n,2}$ as $Y_t = (\text{Id} - \mathcal{A})^{-1}[X_t\beta_0 + F_0] + (\text{Id} - \mathcal{A})^{-1}\mathcal{E}_t$ under Assumption 2.1 for all $t \in [T]$. We denote

$$[\xi_{1t}(\cdot), \dots, \xi_{nt}(\cdot)]^\top = (\text{Id} - \mathcal{A})^{-1}[X_t\beta_0 + F_0] + (\text{Id} - \mathcal{A})^{-1}[\cdot] : \mathcal{H}_{n,2} \rightarrow \mathcal{H}_{n,2}$$

such that $Y_{it} = \xi_{it}(\mathcal{E}_t)$ holds for each $i \in [n]$.

Define

$$\mathcal{E}_{1,it}^{(\delta)} := \{\varepsilon_{jt}\}_{j:\Delta(i,j) \leq \delta}, \quad \mathcal{E}_{2,it}^{(\delta)} := \{\varepsilon_{jt}\}_{j:\Delta(i,j) > \delta}$$

for some $\delta > 0$. Since $L^2(0, 1)$ is separable, both $\mathcal{E}_{1,it}^{(\delta)}$ and $\mathcal{E}_{2,it}^{(\delta)}$ are Polish space-valued random elements in $(\mathcal{H}_{|\{j:\Delta(i,j) \leq \delta\}|,2}, \|\cdot\|_{\infty,2})$ and $(\mathcal{H}_{|\{j:\Delta(i,j) > \delta\}|,2}, \|\cdot\|_{\infty,2})$, respectively. Then, by Lemma 2.11 of Dudley and Philipp (1983) (see also Lemma A.1 of Jenish (2012)), a function χ exists such that $(\mathcal{E}_{1,it}^{(\delta)}, \chi(U, \mathcal{E}_{1,it}^{(\delta)}))$ has the same law as that of $(\mathcal{E}_{1,it}^{(\delta)}, \mathcal{E}_{2,it}^{(\delta)})$, which is an appropriate permutation of $\mathcal{E}$, where U is a random variable uniformly distributed on $[0, 1]$ and independent of $\mathcal{E}_{1,it}^{(\delta)}$.

Now, with a slight abuse of notation, we write

$$\begin{aligned}Y_{it} &= \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \mathcal{E}_{2,it}^{(\delta)}) \equiv \xi_{it}(\mathcal{E}_t) \\ Y_{it}^{(\delta)} &:= \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \chi(U, \mathcal{E}_{1,it}^{(\delta)})) \equiv \xi_{it}(\mathcal{E}_t^{(\delta)})\end{aligned}$$

where $\mathcal{E}_t^{(\delta)} = (\varepsilon_{1t}^{(\delta)}, \dots, \varepsilon_{nt}^{(\delta)})^\top$. To be specific,

$$Y_{it}^{(\delta)}(s) = \left\{ (\text{Id} - \mathcal{A})^{-1}[X_t\beta_0 + F_0 + \mathcal{E}_t^{(\delta)}](s) \right\}_i$$

$$= \alpha_0(s) \sum_{j=1}^n w_{i,j} A(Y_{jt}^{(\delta)}, s) + X_{it}^\top \beta_0(s) + f_{0i}(s) + \varepsilon_{it}^{(\delta)}(s).$$

By construction, we have

$$\begin{aligned} \mathbb{E}[Y_{it}(s) \mid \mathcal{F}_{it}(\delta)] &= \mathbb{E} \left[\xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \mathcal{E}_{2,it}^{(\delta)})(s) \mid \mathcal{E}_{1,it}^{(\delta)} \right] \\ &= \mathbb{E} \left[\xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \chi(U, \mathcal{E}_{1,it}^{(\delta)}))(s) \mid \mathcal{E}_{1,it}^{(\delta)} \right] = \mathbb{E}[Y_{it}^{(\delta)}(s) \mid \mathcal{F}_{it}(\delta)], \end{aligned}$$

where $\mathcal{F}_{it}(\delta)$ is the σ -field generated by $\mathcal{E}_{1,it}^{(\delta)} = \{\varepsilon_{jt} : \Delta(i, j) \leq \delta\}$. Similarly, we have $\|Y_{it}(u) - Y_{it}^{(\delta)}(u)\|_2^2 \leq 4\|Y_{it}(u)\|_2^2$.

Here, suppose that $0 < \delta < \bar{\Delta}$, where $\bar{\Delta}$ is as provided in Assumption 3.1(ii). Then, because at least i 's own ε_{it} is included in $\mathcal{E}_{1,it}^{(\delta)}$, we have $\varepsilon_{it} \equiv \varepsilon_{it}^{(\delta)}$, and hence

$$Y_{it}(s) - Y_{it}^{(\delta)}(s) = \alpha_0(s) \sum_{j=1}^n w_{i,j} A(Y_{jt} - Y_{jt}^{(\delta)}, s)$$

holds. Thus, by Minkowski's inequality and Assumptions 3.2(ii) and 3.4,

$$\begin{aligned} \|Y_{it}(s) - Y_{it}^{(\delta)}(s)\|_2 &= \left\| \alpha_0(s) \sum_{j=1}^n w_{i,j} A(Y_{jt} - Y_{jt}^{(\delta)}, s) \right\|_2 \\ &\leq |\alpha_0(s)| \sum_{j=1}^n |w_{i,j}| \cdot \|A(Y_{jt} - Y_{jt}^{(\delta)}, s)\|_2 \\ &\leq |\alpha_0(s)| \sum_{j=1}^n |w_{i,j}| \cdot \left(\int_0^1 \|Y_{jt}(u) - Y_{jt}^{(\delta)}(u)\|_2^2 \omega_2(u, s) du \right)^{1/2} \leq C \cdot \varrho, \end{aligned}$$

where $C := 2 \max_{i,t} \text{ess sup}_{u \in [0,1]} \|Y_{it}(u)\|_2$, and $\varrho := \bar{\alpha}_0 \|W_n\|_\infty$. Similarly, when $\bar{\Delta} \leq \delta < 2\bar{\Delta}$ holds, noting now that under Assumption 3.1(ii) we have $\varepsilon_{jt} \equiv \varepsilon_{jt}^{(\delta)}$ for all j 's who are direct

neighbors of i ,

$$\begin{aligned}
\|Y_{it}(s) - Y_{it}^{(\delta)}(s)\|_2 &\leq \bar{\alpha}_0 \sum_{j=1}^n |w_{i,j}| \cdot \|A(Y_{jt} - Y_{jt}^{(\delta)}, s)\|_2 \\
&= \bar{\alpha}_0 \sum_{j=1}^n |w_{i,j}| \cdot \left\| \sum_{k=1}^n w_{j,k} A(\alpha_0(\cdot) A(Y_{kt} - Y_{kt}^{(\delta)}, \cdot), s) \right\|_2 \\
&\leq \bar{\alpha}_0 \sum_{j=1}^n |w_{i,j}| \sum_{k=1}^n |w_{j,k}| \cdot \|A(\alpha_0(\cdot) A(Y_{kt} - Y_{kt}^{(\delta)}, \cdot), s)\|_2 \\
&\leq \bar{\alpha}_0 \sum_{j=1}^n |w_{i,j}| \sum_{k=1}^n |w_{j,k}| \cdot \left(\int_0^1 \|\alpha_0(u) A(Y_{kt} - Y_{kt}^{(\delta)}, u)\|_2^2 \omega_2(u, s) du \right)^{1/2} \\
&\leq \bar{\alpha}_0^2 \sum_{j=1}^n |w_{i,j}| \sum_{k=1}^n |w_{j,k}| \cdot \left(\int_0^1 \|A(Y_{kt} - Y_{kt}^{(\delta)}, u)\|_2^2 \omega_2(u, s) du \right)^{1/2} \\
&\leq \bar{\alpha}_0^2 \sum_{j=1}^n |w_{i,j}| \sum_{k=1}^n |w_{j,k}| \cdot \left(\int_0^1 \int_0^1 \|Y_{kt}(t) - Y_{kt}^{(\delta)}(t)\|_2^2 \omega_2(t, u) \omega_2(u, s) dt du \right)^{1/2} \\
&\leq C \cdot \varrho^2.
\end{aligned}$$

Applying the same argument recursively, for $m\bar{\Delta} \leq \delta < (m+1)\bar{\Delta}$ such that $\varepsilon_{jt} \equiv \varepsilon_{jt}^{(\delta)}$ for all j 's in the m -th order neighborhood of i , we obtain

$$\|Y_{it}(s) - Y_{it}^{(\delta)}(s)\|_2 \leq C \cdot \varrho^{\lfloor \delta/\bar{\Delta} \rfloor + 1}. \quad (\text{B.4})$$

Finally, by Jensen's inequality and (B.4),

$$\begin{aligned}
\|Y_{it}(s) - \mathbb{E}[Y_{it}(s) \mid \mathcal{F}_{it}(\delta)]\|_2 &= \left\| \int_0^1 \left[\xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \mathcal{E}_{2,it}^{(\delta)})(s) - \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \chi(u, \mathcal{E}_{1,it}^{(\delta)}))(s) \right] du \right\|_2 \\
&\leq \left\{ \mathbb{E} \int_0^1 \left| \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \mathcal{E}_{2,it}^{(\delta)})(s) - \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \chi(u, \mathcal{E}_{1,it}^{(\delta)}))(s) \right|^2 du \right\}^{1/2} \\
&= \left\{ \mathbb{E} \left| \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \mathcal{E}_{2,it}^{(\delta)})(s) - \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \chi(U, \mathcal{E}_{1,it}^{(\delta)}))(s) \right|^2 \right\}^{1/2} \\
&= \left\| \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \mathcal{E}_{2,it}^{(\delta)})(s) - \xi_{it}(\mathcal{E}_{1,it}^{(\delta)}, \chi(U, \mathcal{E}_{1,it}^{(\delta)}))(s) \right\|_2 \\
&= \|Y_{it}(s) - Y_{it}^{(\delta)}(s)\|_2 \leq C \cdot \varrho^{\lfloor \delta/\bar{\Delta} \rfloor + 1} \rightarrow 0
\end{aligned}$$

as $\delta \rightarrow \infty$ by Assumption 2.1. This completes the proof. $\square$

Lemma B.4. Suppose that Assumptions 2.1, 3.1, 3.2, and 3.4 hold. Then, for any given $s \in [0, 1]$ and all $t \in [T]$, $\{A(\bar{Y}_{it}, s) : i \in \mathcal{D}_n; n \geq 1\}$ is uniformly and geometrically L^2 -NED

on $\{\varepsilon_{it} : i \in \mathcal{D}_n; n \geq 1\}$.

Proof. Note that $\mathcal{F}_{jt}((\delta - 1)\bar{\Delta}) \subseteq \mathcal{F}_{it}(\delta\bar{\Delta})$ for (i, j) with $\Delta(i, j) \leq \bar{\Delta}$ and $\delta > 1$. Thus, by Lemma B.3,

$$\begin{aligned} \|\bar{Y}_{it}(s) - \mathbb{E}[\bar{Y}_{it}(s) \mid \mathcal{F}_{it}(\delta\bar{\Delta})]\|_2 &\leq \sum_{j=1}^n |w_{i,j}| \|Y_{jt}(s) - \mathbb{E}[Y_{jt}(s) \mid \mathcal{F}_{it}(\delta\bar{\Delta})]\|_2 \\ &\lesssim \sum_{j: \Delta(i,j) \leq \bar{\Delta}} \|Y_{jt}(s) - \mathbb{E}[Y_{jt}(s) \mid \mathcal{F}_{jt}((\delta - 1)\bar{\Delta})]\|_2 \\ &\lesssim \varrho^{[\delta]}, \end{aligned}$$

which implies that $\{\bar{Y}_{it}(s)\}$ is uniformly and geometrically L^2 -NED. By Assumption 3.4,

$$\begin{aligned} \|A(\bar{Y}_{it}, s) - \mathbb{E}[A(\bar{Y}_{it}, s) \mid \mathcal{F}_{it}(\delta)]\|_2 &= \|A(\bar{Y}_{it} - \mathbb{E}[\bar{Y}_{it} \mid \mathcal{F}_{it}(\delta)], s)\|_2 \\ &\leq \left(\int_0^1 \|\bar{Y}_{it}(u) - \mathbb{E}[\bar{Y}_{it}(u) \mid \mathcal{F}_{it}(\delta)]\|_2^2 \omega_2(u, s) du \right)^{1/2} \\ &\lesssim \varrho^{[\delta/\bar{\Delta}]}. \end{aligned}$$

This proves the desired result. $\square$

As a useful consequence from Lemma B.3 and B.4, we have

$$\begin{aligned} \|e_{it}(s; \theta) - \mathbb{E}[e_{it}(s; \theta) \mid \mathcal{F}_{it}(\delta)]\|_2 &\leq \|Y_{it}(s) - \mathbb{E}[Y_{it}(s) \mid \mathcal{F}_{it}(\delta)]\|_2 \\ &\quad + |\alpha(s; \theta)| \cdot \|A(\bar{Y}_{it}, s) - \mathbb{E}[A(\bar{Y}_{it}, s) \mid \mathcal{F}_{it}(\delta)]\|_2 \\ &\lesssim \varrho^{[\delta/\bar{\Delta}]} \end{aligned}$$

uniformly in $s \in [0, 1]$, $\theta \in \Theta_K$, and (i, t) ; that is, $\{e_{it}(s; \theta)\}$ is uniformly and geometrically L^2 -NED.

Lemma B.5. Suppose that Assumption 3.3(i) holds. Let $\{\xi_{it} : i \in \mathcal{D}_n; n \geq 1\}$ be a geometrically L^2 -NED random field on $\{\varepsilon_{it} : i \in \mathcal{D}_n; n \geq 1\}$ for all $t \in [T]$, independent of $\{\varepsilon_{it'} : i \in \mathcal{D}_n; n \geq 1\}$ for $t' \neq t$. Denote $\vec{\xi}_{it} := \xi_{i,t+1} - \xi_{it}$ and $C_\xi := \sup_{i,t} \|\xi_{it}\|_2$. Then,

- (i) $\left| \text{Cov} \left(\vec{\xi}_{it}, \vec{\xi}_{jt} \right) \right| \lesssim C_\xi^2 \rho(\Delta(i, j)/3)$ for all $t \in [T]$ with some geometric NED coefficient ρ ;
- (ii) $\left| \text{Cov} \left(\vec{\xi}_{i,t+1}, \vec{\xi}_{jt} \right) \right| \lesssim C_\xi^2 \rho(\Delta(i, j)/3)$ for all $t \in [T-1]$ with some geometric NED coefficient ρ .

Proof. Since the proofs are similar, we only prove (ii). Decompose $\vec{\xi}_{it} = \vec{\xi}_{1,it}^{(\delta)} + \vec{\xi}_{2,it}^{(\delta)}$, where

$$\vec{\xi}_{1,it}^{(\delta)} := \mathbb{E} \left[\vec{\xi}_{it} \mid \mathcal{F}_{it}^+(\delta) \right], \quad \text{and} \quad \vec{\xi}_{2,it}^{(\delta)} := \vec{\xi}_{it} - \mathbb{E} \left[\vec{\xi}_{it} \mid \mathcal{F}_{it}^+(\delta) \right],$$

where $\mathcal{F}_{it}^+(\delta)$ is the sigma field generated from $\{(\varepsilon_{i't}, \varepsilon_{i',t+1}) : \Delta(i, i') \leq \delta\}$. Note that $\mathcal{F}_{it}^+(\delta) = \mathcal{F}_{it}(\delta) \vee \mathcal{F}_{i,t+1}(\delta)$ holds. Then, for each pair $\vec{\xi}_{i,t+1}$ and $\vec{\xi}_{jt}$, denoting $\delta_{ij} := \Delta(i, j)/3$,

$$\begin{aligned} \left| \text{Cov} \left(\vec{\xi}_{i,t+1}, \vec{\xi}_{jt} \right) \right| &= \left| \text{Cov} \left(\vec{\xi}_{1,i,t+1}^{(\delta_{ij})} + \vec{\xi}_{2,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{1,jt}^{(\delta_{ij})} + \vec{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| \\ &\leq \left| \text{Cov} \left(\vec{\xi}_{1,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{1,jt}^{(\delta_{ij})} \right) \right| + \left| \text{Cov} \left(\vec{\xi}_{1,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| \\ &\quad + \left| \text{Cov} \left(\vec{\xi}_{2,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{1,jt}^{(\delta_{ij})} \right) \right| + \left| \text{Cov} \left(\vec{\xi}_{2,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{2,jt}^{(\delta_{ij})} \right) \right|. \end{aligned}$$

The first term on the right-hand side is zero by Assumption 3.3(i). Note that, by Jensen's and triangle inequalities, $\|\vec{\xi}_{1,it}^{(\delta_{ij})}\|_2 \leq \|\vec{\xi}_{it}\|_2 \leq \|\xi_{i,t+1}\|_2 + \|\xi_{it}\|_2 \leq 2C_\xi$. In addition, $\|\vec{\xi}_{2,it}^{(\delta_{ij})}\|_2 \leq 2\|\vec{\xi}_{it}\|_2 \leq 4C_\xi$. Then, since $\{\xi_{it}\}$ is assumed to be L^2 -NED on $\{\varepsilon_{it}\}$ at each t , it holds that

$$\begin{aligned} \|\vec{\xi}_{2,it}^{(\delta_{ij})}\|_2 &= \left\| \vec{\xi}_{it} - \mathbb{E} \left[\vec{\xi}_{it} \mid \mathcal{F}_{it}^+(\delta_{ij}) \right] \right\|_2 \\ &\leq \|\xi_{i,t+1} - \mathbb{E}[\xi_{i,t+1} \mid \mathcal{F}_{i,t+1}(\delta_{ij})]\|_2 + \|\xi_{it} - \mathbb{E}[\xi_{it} \mid \mathcal{F}_{it}(\delta_{ij})]\|_2 \leq 4C_\xi \rho(\delta_{ij}). \end{aligned}$$

Hence, Cauchy–Schwarz inequality gives

$$\left| \text{Cov} \left(\vec{\xi}_{1,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| \lesssim \left\| \vec{\xi}_{1,i,t+1}^{(\delta_{ij})} \right\|_2 \left\| \vec{\xi}_{2,jt}^{(\delta_{ij})} \right\|_2 \lesssim C_\xi^2 \rho(\delta_{ij}).$$

The same inequality applies to $\left| \text{Cov} \left(\vec{\xi}_{2,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{1,jt}^{(\delta_{ij})} \right) \right|$. Furthermore,

$$\left| \text{Cov} \left(\vec{\xi}_{2,i,t+1}^{(\delta_{ij})}, \vec{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| \lesssim \left\| \vec{\xi}_{2,i,t+1}^{(\delta_{ij})} \right\|_2 \left\| \vec{\xi}_{2,jt}^{(\delta_{ij})} \right\|_2 \lesssim C_\xi^2 \rho(\delta_{ij}).$$

This completes the proof. $\square$

Lemma B.6. Suppose that Assumptions 2.1, 3.1 – 3.5, and 3.7 hold. For all $m \in [M]$ and $\theta \in \Theta_K$,

- (i) $\left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} (\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)]) (\theta_0 - \theta) / (NL) \right\| \lesssim_p \sqrt{K} / \sqrt{nT}$
- (ii) $\left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} (\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)]) / (NL) \right\| \lesssim_p K / \sqrt{nT}$
- (iii) $\left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbf{V}(s_l) / (NL) \right\| \lesssim_p K^{-\pi}, \quad \left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\mathbf{V}(s_l)] / (NL) \right\| \lesssim K^{-\pi}$

- (iv) $\left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) / (NL) \right\| \lesssim_p 1/\sqrt{nT}$
- (v) $\left| \sum_{l=1}^L \left\{ \mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta) - \mathbb{E}[\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta)] \right\} / (NL) \right| \lesssim_p 1/\sqrt{nT}$
- (vi) $\left| \sum_{l=1}^L \mathbf{V}(s_l)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{V}(s_l) / (NL) \right| \lesssim_p K^{-2\pi}$
- (vii) $\left| \sum_{l=1}^L \mathbf{V}(s_l)^\top \mathbf{D}^\top P_m \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) / (NL) \right| \lesssim_p K^{-\pi}$
- (viii) $\left| \sum_{l=1}^L \boldsymbol{\mathcal{E}}(s_l)^\top \mathbf{D}^\top P_m \mathbf{D} \boldsymbol{\mathcal{E}}(s_l) / (NL) \right| \lesssim_p 1/\sqrt{nT}.$

Proof. Below, for a generic variable $\mathbf{x}$ indexed by i and t , we denote $\vec{\mathbf{x}}_{it} = \mathbf{x}_{i,t+1} - \mathbf{x}_{it}$. In addition, we write $a_{it}(s) := A(\bar{Y}_{it}, s)$.

(i) Observe that for each s_l ,

$$\left\| \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} (\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)])(\theta_0 - \theta) \right\| = \left\| \sum_{i=1}^n \sum_{t=1}^{T-1} \vec{B}_{it} \otimes \phi^K(s_l) [\vec{a}_{it}(s_l) - \mathbb{E}(\vec{a}_{it}(s_l))] \alpha(s_l; \theta_0 - \theta) \right\|.$$

As a typical element, the variance of the first element of $\mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} (\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)])(\theta_0 - \theta)$ is given as

$$\begin{aligned} \text{Var}[\{\mathbf{Z}^\top \mathbf{D}^\top \mathbf{D} (\mathbf{H} - \mathbb{E}[\mathbf{H}])(\theta_0 - \theta)\}_1] &= \mathbb{E} \left(\sum_{i=1}^n \sum_{t=1}^{T-1} \vec{Q}_{it}^1 \phi_1(s_l) (\vec{a}_{it} - \mathbb{E}[\vec{a}_{it}]) \alpha(s_l; \theta_0 - \theta) \right)^2 \\ &\lesssim \sum_{t=1}^{T-1} \sum_{t'=1}^{T-1} \sum_{i=1}^n \sum_{i'=1}^n |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't'})| \\ &= \sum_{t=1}^{T-1} \sum_{i=1}^n \text{Var}(\vec{a}_{it}) + \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't})| \\ &\quad + \sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{i=1}^n |\text{Cov}(\vec{a}_{it}, \vec{a}_{it'})| + \sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't'})|. \end{aligned}$$

Here, the dependence on s_l , " (s_l) ", is occasionally omitted for notational simplicity. First, from Assumptions 2.1(i), 3.2(ii), and 3.4, we can easily see that $\sum_{t=1}^{T-1} \sum_{i=1}^n \text{Var}(\vec{a}_{it}) \lesssim nT$.

Second, by Lemma B.4 and B.5(i), there exists a geometric NED coefficient ρ that satisfies

$$\begin{aligned}
\sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't})| &\lesssim \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{i' \neq i} \rho(\Delta(i, i')/3) \\
&= \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{m=1}^{\infty} \sum_{i': \Delta(i, i') \in [m, m+1)} \rho(\Delta(i, i')/3) \\
&\lesssim \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{m=1}^{\infty} m^{d-1} \rho(m) \lesssim nT,
\end{aligned}$$

where the second inequality is from Lemma A.1(iii) of [Jenish and Prucha \(2009\)](#), and the final claim follows from the geometric NED property. Third, since $\vec{a}_{it}$ and $\vec{a}_{i't'}$ are independent if $|t - t'| \geq 2$, it holds that

$$\sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{i=1}^n |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't'})| = \sum_{t' \in \{t-1, t+1\}} \sum_{t=1}^{T-1} \sum_{i=1}^n |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't'})| \lesssim nT.$$

Finally, noting that $\sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't'})| = \sum_{t' \in \{t-1, t+1\}} \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\vec{a}_{it}, \vec{a}_{i't'})|$, Lemma B.5(ii) implies that this term is also of order nT .

Combining the above results suggests that

$$\mathbb{E} \left\| \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} (\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)]) (\theta_0 - \theta) / N \right\|^2 \lesssim K / (nT)$$

for each s_l , which completes the proof by applying Markov's and triangle inequalities.

(ii) Analogous to the proof of (i).

(iii) We prove the first part. Let

$$\mathbf{Z}_L = \begin{pmatrix} \mathbf{Z}(s_1) \\ \vdots \\ \mathbf{Z}(s_L) \end{pmatrix}_{nTL \times (d_q + d_x)K}, \quad \mathbf{V}_L = \begin{pmatrix} \mathbf{V}(s_1) \\ \vdots \\ \mathbf{V}(s_L) \end{pmatrix}_{nTL \times 1}, \quad \mathbf{D}_L = I_L \otimes \mathbf{D}.$$

Then, noting the equivalence $\lambda_{\max} \left(\sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbf{Z}(s_l) / (NL) \right) = \lambda_{\max} \left(\mathbf{D}_L \mathbf{Z}_L \mathbf{Z}_L^\top \mathbf{D}_L^\top / (NL) \right)$, by Assumption 3.5(ii),

$$\left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbf{V}(s_l) / (NL) \right\|^2 = \left\| \mathbf{Z}_L^\top \mathbf{D}_L^\top \mathbf{D}_L \mathbf{V}_L / (NL) \right\|^2$$

$$\begin{aligned}
&= \mathbf{V}_L^\top \mathbf{D}_L^\top \mathbf{D}_L \mathbf{Z}_L \mathbf{Z}_L^\top \mathbf{D}_L^\top \mathbf{D}_L \mathbf{V}_L / (N^2 L^2) \\
&\lesssim \mathbf{V}_L^\top \mathbf{D}_L^\top \mathbf{D}_L \mathbf{V}_L / (NL) \\
&\lesssim \frac{1}{NL} \sum_{l=1}^L \sum_{t=1}^T \sum_{i=1}^n |v_{it}(s_l)|^2
\end{aligned}$$

Then, the result follows from Markov's inequality and (B.3).

The second part can be proved analogously.

(iv) By the triangle inequality,

$$\begin{aligned}
\left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\varepsilon}(s_l) / (NL) \right\| &= \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \vec{B}_{it} \otimes \phi^K(s_l) \vec{\varepsilon}_{it}(s_l) / (NL) \right\| \\
&\leq \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \vec{B}_{it} \otimes \varepsilon_{i,t+1}(s_l) \phi^K(s_l) / (NL) \right\| \\
&\quad + \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \vec{B}_{it} \otimes \varepsilon_{it}(s_l) \phi^K(s_l) / (NL) \right\|.
\end{aligned}$$

Further, by Assumptions 3.3(i) and (iii),

$$\begin{aligned}
&\mathbb{E} \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \vec{B}_{it} \otimes \varepsilon_{it}(s_l) \phi^K(s_l) / (NL) \right\|^2 \\
&= \frac{1}{N^2} \sum_{t=1}^{T-1} \sum_{i=1}^n \text{trace} \left\{ \vec{B}_{it} \vec{B}_{it}^\top \otimes \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_{it}(s_l, s_{l'}) \phi^K(s_l) \phi^K(s_{l'})^\top \right\} \\
&= \frac{1}{N^2} \sum_{t=1}^{T-1} \sum_{i=1}^n \text{trace} \left\{ \vec{B}_{it} \vec{B}_{it}^\top \right\} \text{trace} \left\{ \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_{it}(s_l, s_{l'}) \phi^K(s_l) \phi^K(s_{l'})^\top \right\} \lesssim 1/(nT).
\end{aligned}$$

Repeating the same calculation for the other term, the result follows from Markov's inequality.

(v) Observe that

$$\begin{aligned}
&\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta) - \mathbb{E}[\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta)] \\
&= \sum_{t=1}^{T-1} \sum_{1 \leq i, j \leq n} p_{m,i,j} (\vec{e}_{it}(s_l; \theta) \vec{e}_{jt}(s_l; \theta) - \mathbb{E}[\vec{e}_{it}(s_l; \theta) \vec{e}_{jt}(s_l; \theta)]).
\end{aligned}$$

Here, let $\mathbf{e}_{m,jt}(s_l; \theta) := \sum_{i=1}^n p_{m,i,j} e_{it}(s_l; \theta)$ and recall that there is a constant $\bar{\Delta}_m$ such that $p_{m,i,j} = 0$ if $\Delta(i, j) > \bar{\Delta}_m$. Then, noting that $\mathcal{F}_{it}((\delta - 1)\bar{\Delta}_m) \subseteq \mathcal{F}_{jt}(\delta\bar{\Delta}_m)$ for (i, j) with

$\Delta(i, j) \leq \bar{\Delta}_m$ and $\delta > 1$,

$$\begin{aligned}
\|\mathbf{e}_{m,jt}(s_l; \theta) - \mathbb{E}[\mathbf{e}_{m,jt}(s_l; \theta) \mid \mathcal{F}_{jt}(\delta \bar{\Delta}_m)]\|_2 &\leq \sum_{i=1}^n |p_{m,i,j}| \|e_{it}(s_l; \theta) - \mathbb{E}[e_{it}(s_l; \theta) \mid \mathcal{F}_{jt}(\delta \bar{\Delta}_m)]\|_2 \\
&\lesssim \sum_{i: \Delta(i,j) \leq \bar{\Delta}_m} \|e_{it}(s_l; \theta) - \mathbb{E}[e_{it}(s_l; \theta) \mid \mathcal{F}_{it}((\delta - 1)\bar{\Delta}_m)]\|_2 \\
&\lesssim \varrho^{l(\delta-1)\bar{\Delta}_m/\bar{\Delta}},
\end{aligned}$$

which implies that $\{\mathbf{e}_{m,jt}(s_l; \theta)\}$ is uniformly and geometrically L^2 -NED, for all $l \in [L]$, $m \in [M]$, and $\theta \in \Theta_K$.

Now, suppressing the dependence on both s_l and θ ,

$$\begin{aligned}
\text{Var}[\mathbf{E}^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}] &= \mathbb{E} \left(\sum_{t=1}^{T-1} \sum_{1 \leq i, j \leq n} p_{m,i,j} \{\vec{e}_{it} \vec{e}_{jt} - \mathbb{E}[\vec{e}_{it} \vec{e}_{jt}]\} \right)^2 \\
&= \mathbb{E} \left(\sum_{t=1}^{T-1} \sum_{j=1}^n \{\vec{\mathbf{e}}_{m,jt} \vec{e}_{jt} - \mathbb{E}[\vec{\mathbf{e}}_{m,jt} \vec{e}_{jt}]\} \right)^2 \\
&\leq \sum_{t=1}^{T-1} \sum_{t'=1}^{T-1} \sum_{j=1}^n \sum_{j'=1}^n |\text{Cov}(\vec{\mathbf{e}}_{m,jt} \vec{e}_{jt}, \vec{\mathbf{e}}_{m,j't'} \vec{e}_{j't'})| \\
&= \sum_{t=1}^{T-1} \sum_{j=1}^n \text{Var}(\vec{\mathbf{e}}_{m,jt} \vec{e}_{jt}) + \sum_{t=1}^{T-1} \sum_{j=1}^n \sum_{j' \neq j} |\text{Cov}(\vec{\mathbf{e}}_{m,jt} \vec{e}_{jt}, \vec{\mathbf{e}}_{m,j't} \vec{e}_{j't})| \\
&\quad + \sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{j=1}^n |\text{Cov}(\vec{\mathbf{e}}_{m,jt} \vec{e}_{jt}, \vec{\mathbf{e}}_{m,j't'} \vec{e}_{j't'})| + \sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{j=1}^n \sum_{j' \neq j} |\text{Cov}(\vec{\mathbf{e}}_{m,jt} \vec{e}_{jt}, \vec{\mathbf{e}}_{m,j't'} \vec{e}_{j't'})|.
\end{aligned}$$

As we have seen in (A.1), we have $\|e_{jt}\|_p, \|\mathbf{e}_{m,jt}\|_p < \infty$ for $p > 4$. This allows us to use Lemma A.1 of Xu and Lee (2015) (see also Corollary 4.3(b) of Gallant and White (1988)) to show that $\{\mathbf{e}_{m,jt} e_{jt}\}$ is uniformly and geometrically L^2 -NED. Then, following the same argument as in the proof of (i), we can show that $\text{Var}[\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta)] \lesssim nT$ for each s_l , which gives the desired result by the triangle inequality.

(vi), (vii) These can be proved in a similar manner to the proof of Lemma B.1.

(viii) For each s_l , $|\boldsymbol{\mathcal{E}}(s_l)^\top \mathbf{D}^\top P_m \mathbf{D} \boldsymbol{\mathcal{E}}(s_l)/N| \lesssim_p 1/\sqrt{nT}$ holds under Assumptions 3.3(i) and (ii), as in Lemma 9 in Yu *et al.* (2008) and Lemma 1 in Lee and Yu (2014). Then, the result is straightforward.

□

Lemma B.7. Suppose that Assumptions 2.1(i), 3.2, 3.4, and 3.5(i) hold. Then, $\sup_{\theta \in \Theta_K} \|\mathbb{E}[\bar{g}_{nT}(\theta)]\| \lesssim \sqrt{K}$.

Proof. Observe that

$$\|\mathbb{E}[\bar{g}_{nT}(\theta)]\| \leq \left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\mathbf{E}(s_l; \theta)] / (NL) \right\| + \sum_{m=1}^M \left| \sum_{l=1}^L \mathbb{E}[\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta)] / (NL) \right|.$$

For the first term,

$$\begin{aligned} \left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\mathbf{E}(s_l; \theta)] / (NL) \right\| &= \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \vec{B}_{it} \otimes \mathbb{E}[\vec{e}_{it}(s_l; \theta)] \phi^K(s_l) / (NL) \right\| \\ &\lesssim \frac{1}{NL} \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \|\mathbb{E}[\vec{e}_{it}(s_l; \theta)] \phi^K(s_l)\| \\ &\lesssim \sqrt{K} \end{aligned}$$

uniformly in $\theta \in \Theta_K$, since $|\mathbb{E}[e_{it}(s_l; \theta)]| \leq \mathbb{E}|e_{it}(s_l; \theta)| \lesssim 1$ and $\sup_{s \in [0,1]} \|\phi^K(s)\| \lesssim \sqrt{K}$.

For the second term,

$$\begin{aligned} \left| \sum_{l=1}^L \mathbb{E}[\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta)] / (NL) \right| &= \left| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{1 \leq i, j \leq n} p_{m,i,j} \mathbb{E}[\vec{e}_{it}(s_l; \theta) \vec{e}_{jt}(s_l; \theta)] / (NL) \right| \\ &\leq \frac{1}{NL} \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{1 \leq i, j \leq n} |p_{m,i,j}| \cdot |\mathbb{E}[\vec{e}_{it}(s_l; \theta) \vec{e}_{jt}(s_l; \theta)]| \\ &\lesssim 1 \end{aligned}$$

uniformly in $\theta \in \Theta_K$, since

$$\begin{aligned} |\mathbb{E}[\vec{e}_{it}(s_l; \theta) \vec{e}_{jt}(s_l; \theta)]| &\leq \mathbb{E}[|\vec{e}_{it}(s_l; \theta)| \cdot |\vec{e}_{jt}(s_l; \theta)|] \\ &\leq \|\vec{e}_{it}(s_l; \theta)\|_2 \cdot \|\vec{e}_{jt}(s_l; \theta)\|_2 \\ &\leq \{ \|e_{i,t+1}(s_l; \theta)\|_2 + \|e_{it}(s_l; \theta)\|_2 \} \cdot \{ \|e_{j,t+1}(s_l; \theta)\|_2 + \|e_{jt}(s_l; \theta)\|_2 \} \\ &\lesssim 1. \end{aligned}$$

This completes the proof. □

Lemma B.8. Suppose that Assumptions 2.1, 3.1 – 3.5, and 3.7 hold. In addition, assume that $K/\sqrt{nT} \rightarrow 0$ and $K^{1/2-\pi} \rightarrow 0$ as $nT \rightarrow \infty$. Then, $\|\hat{\theta}_{nT} - \theta_0\| = o_P(1)$.

Proof. Observe that

$$\bar{g}_{nT}(\theta) - \mathbb{E}[\bar{g}_{nT}(\theta)] = \begin{pmatrix} A_0(\theta) \\ A_1(\theta) \\ \vdots \\ A_M(\theta) \end{pmatrix}$$

where

$$\begin{aligned} A_0(\theta) := & \underbrace{\sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} (\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)])(\theta_0 - \theta)/(NL)}_{\lesssim_p \sqrt{K}/\sqrt{nT}: \text{Lemma B.6(i)}} \\ & + \underbrace{\sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbf{V}(s_l)/(NL) - \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbb{E}[\mathbf{V}(s_l)]/(NL)}_{\lesssim_p K^{-\pi}: \text{Lemma B.6(iii)}} + \underbrace{\sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\varepsilon}(s_l)/(NL)}_{\lesssim_p 1/\sqrt{nT}: \text{Lemma B.6(iv)}} \end{aligned}$$

and, for $m = 1, \dots, M$,

$$A_m(s; \theta) := \underbrace{\sum_{l=1}^L \{ \mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta) - \mathbb{E}[\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{E}(s_l; \theta)] \}}_{\lesssim_p 1/\sqrt{nT}: \text{Lemma B.6(v)}} / (NL).$$

Hence,

$$\begin{aligned} \|\bar{g}_{nT}(\theta) - \mathbb{E}[\bar{g}_{nT}(\theta)]\| & \leq \|A_0(\theta)\| + \sum_{m=1}^M |A_m(\theta)| \\ & \lesssim_p \sqrt{K}/\sqrt{nT} + K^{-\pi} \end{aligned}$$

uniformly in $\theta \in \Theta_K$. Further, by Cauchy-Schwarz inequality and Lemma B.7,

$$\begin{aligned} \sup_{\theta \in \Theta_K} |\mathcal{Q}_{nT}(\theta) - \mathcal{Q}_{nT}^*(\theta)| & \leq \sup_{\theta \in \Theta_K} \left| (\bar{g}_{nT}(\theta) - \mathbb{E}[\bar{g}_{nT}(\theta)])^\top \Omega_{nT} (\bar{g}_{nT}(\theta) - \mathbb{E}[\bar{g}_{nT}(\theta)]) \right| \\ & \quad + 2 \sup_{\theta \in \Theta_K} \left| (\bar{g}_{nT}(\theta) - \mathbb{E}[\bar{g}_{nT}(\theta)])^\top \Omega_{nT} \mathbb{E}[\bar{g}_{nT}(\theta)] \right| \end{aligned}$$

$$\begin{aligned}
&\lesssim \sup_{\theta \in \Theta_K} \|\bar{g}_{nT}(\theta) - \mathbb{E}[\bar{g}_{nT}(\theta)]\|^2 + \sup_{\theta \in \Theta_K} \|\mathbb{E}[\bar{g}_{nT}(\theta)]\| \sup_{\theta \in \Theta_K} \|\bar{g}_{nT}(\theta) - \mathbb{E}[\bar{g}_{nT}(\theta)]\| \\
&\lesssim_p K/\sqrt{nT} + K^{1/2-\pi}.
\end{aligned}$$

Combined with the identifiability of θ_0 (Lemma B.2), the above result implies the consistency of $\hat{\theta}_{nT}$ (see, e.g., the proof of Theorem 3.3 in Su and Hoshino (2016)). $\square$

B.2.2 Proof of Theorem 3.1.

(i) Given the consistency result in Lemma B.8, if we can show that for an arbitrary $\epsilon > 0$, there exists a constant C_ϵ such that for all sufficiently large nT ,

$$\Pr \left(\inf_{\|\mathbf{u}\|=C_\epsilon} \mathcal{Q}_{nT}(\theta_0 + \zeta_{nT}\mathbf{u}) > \mathcal{Q}_{nT}(\theta_0) \right) \geq 1 - \epsilon,$$

we can conclude that $\|\hat{\theta}_{nT} - \theta_0\| \lesssim_p \zeta_{nT}$.

Decompose

$$\begin{aligned}
\mathcal{Q}_{nT}(\theta_0 + \zeta_{nT}\mathbf{u}) - \mathcal{Q}_{nT}(\theta_0) &= \underbrace{(\bar{g}_{nT}(\theta_0 + \zeta_{nT}\mathbf{u}) - \bar{g}_{nT}(\theta_0))^\top \Omega_{nT} (\bar{g}_{nT}(\theta_0 + \zeta_{nT}\mathbf{u}) - \bar{g}_{nT}(\theta_0))}_{=:\tilde{A}_{nT}(\theta)} \\
&\quad + 2 (\bar{g}_{nT}(\theta_0 + \zeta_{nT}\mathbf{u}) - \bar{g}_{nT}(\theta_0))^\top \Omega_{nT} \bar{g}_{nT}(\theta_0).
\end{aligned}$$

Lemma B.6(ii) implies that $\|\Pi_{nT} - \hat{\Pi}_{nT}\| = o_P(1)$, where $\hat{\Pi}_{nT} := \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \mathbf{H}(s_l)/(NL)$. Thus, by Assumption 3.6, we have $\lambda_{\min}(\hat{\Pi}_{nT}^\top \hat{\Pi}_{nT}) > 0$ with probability approaching one. Observing that

$$\bar{g}_{nT}(\theta) - \bar{g}_{nT}(\theta_0) = \begin{pmatrix} \hat{\Pi}_{nT} \\ \sum_{l=1}^L [\mathbf{H}(s_l)(\theta_0 - \theta) + 2\mathbf{V}(s_l) + 2\mathbf{E}(s_l)]^\top \mathbf{D}^\top P_1 \mathbf{D} \mathbf{H}(s_l)/(NL) \\ \vdots \\ \sum_{l=1}^L [\mathbf{H}(s_l)(\theta_0 - \theta) + 2\mathbf{V}(s_l) + 2\mathbf{E}(s_l)]^\top \mathbf{D}^\top P_M \mathbf{D} \mathbf{H}(s_l)/(NL) \end{pmatrix} (\theta_0 - \theta),$$

we obtain $\tilde{A}_{nT}(\theta) \geq c_1 \zeta_{nT}^2 C_\epsilon^2$ for some $c_1 > 0$ with probability approaching one.

For the second term, we can find by Cauchy-Schwarz inequality that

$$\left| (\bar{g}_{nT}(\theta_0 + \zeta_{nT}\mathbf{u}) - \bar{g}_{nT}(\theta_0))^\top \Omega_{nT} \bar{g}_{nT}(\theta_0) \right| \leq (\tilde{A}_{nT}(\theta))^{1/2} (\bar{g}_{nT}(\theta_0)^\top \Omega_{nT} \bar{g}_{nT}(\theta_0))^{1/2}$$

$$\leq c_2(\tilde{A}_{nT}(\theta))^{1/2}\|\bar{g}_{nT}(\theta_0)\|.$$

Hence,

$$\begin{aligned}\mathcal{Q}_{nT}(\theta_0 + \zeta_{nT}\mathbf{u}) - \mathcal{Q}_{nT}(\theta_0) &\geq \tilde{A}_{nT}(\theta) - 2c_2(\tilde{A}_{nT}(\theta))^{1/2}\|\bar{g}_{nT}(\theta_0)\| \\ &= (\tilde{A}_{nT}(\theta))^{1/2}((\tilde{A}_{nT}(\theta))^{1/2} - 2c_2\|\bar{g}_{nT}(\theta_0)\|).\end{aligned}$$

Since $(\tilde{A}_{nT}(\theta))^{1/2}$ is bounded below from $\sqrt{c_1}\zeta_{nT}C_\epsilon$ with probability approaching one, if we set ζ_{nT} proportional to $\|\bar{g}_{nT}(\theta_0)\|$, we can obtain the desired inequality by choosing a sufficiently large C_ϵ . From Lemma B.6(iii), (iv), (vi), (vii), and (viii), we have

$$\|\bar{g}_{nT}(\theta_0)\| \lesssim 1/\sqrt{nT} + K^{-\pi}, \quad (\text{B.5})$$

and this completes the proof.

(ii) Note that $\int_0^1 \phi^K(s)\phi^K(s)^\top ds = I_K$ by orthonormality. Then, by result (i) and Assumption 3.7,

$$\begin{aligned}\|\hat{\alpha}_{nT} - \alpha_0\|_{L^2} &\leq \left\| \phi^K(\cdot)^\top (\hat{\theta}_{nT,\alpha} - \theta_{0\alpha}) \right\|_{L^2} + \left\| \phi^K(\cdot)^\top \theta_{0\alpha} - \alpha_0(\cdot) \right\|_{L^2} \\ &\lesssim \left((\hat{\theta}_{nT,\alpha} - \theta_{0\alpha})^\top \left[\int_0^1 \phi^K(s)\phi^K(s)^\top ds \right] (\hat{\theta}_{nT,\alpha} - \theta_{0\alpha}) \right)^{1/2} + K^{-\pi} \\ &\lesssim_p 1/\sqrt{nT} + K^{-\pi}.\end{aligned}$$

It is also straightforward to see that $\sup_{s \in [0,1]} |\hat{\alpha}_{nT}(s) - \alpha_0(s)| \leq \sup_{s \in [0,1]} \|\phi^K(s)\| \cdot \|\hat{\theta}_{nT,\alpha} - \theta_{0\alpha}\| + K^{-\pi} \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi}$.

(iii) Analogous to the proof of (ii). □

B.3 Proof of Theorem 3.2

B.3.1 Lemmas

Lemma B.9. Suppose that Assumptions 2.1, 3.1 – 3.7, and 3.8(i), (ii) hold. In addition, assume that $K/\sqrt{nT} \rightarrow 0$ and $K^{1/2-\pi} \rightarrow 0$ as $nT \rightarrow \infty$. Let $\bar{\theta}_{nT}$ be any vector in between $\hat{\theta}_{nT}$ and θ_0 . Then,

$$(i) \quad \left\| \bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT} \right\| \lesssim_p K/\sqrt{nT} + K^{-\pi}$$

$$\begin{aligned}
\text{(ii)} \quad & \left\| \bar{J}_{nT}(\hat{\theta}_{nT}) \bar{J}_{nT}(\hat{\theta}_{nT})^\top - \bar{J}_{nT} \bar{J}_{nT}^\top \right\| \lesssim_p K/\sqrt{nT} + K^{-\pi} \\
\text{(iii)} \quad & \left\| \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) - \bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right\| \lesssim_p K/\sqrt{nT} + K^{-\pi} \\
\text{(iv)} \quad & \left\| \left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) \right)^\top - \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \right\| \lesssim_p K/\sqrt{nT} + K^{-\pi}
\end{aligned}$$

Proof. (i) Observe that

$$\left\| \bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT} \right\| \leq B_{1,nT} + 2 \sum_{m=1}^M (B_{2,m,nT} + B_{3,m,nT})$$

where

$$\begin{aligned}
B_{1,nT} &:= \left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \{ \mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)] \} / (NL) \right\| \lesssim_p \frac{K/\sqrt{nT}}{\text{Lemma B.6(ii)}} \\
B_{2,m,nT} &:= \left\| \sum_{l=1}^L \left\{ \mathbf{E}(s_l; \hat{\theta}_{nT}) - \mathbf{E}(s_l; \theta_0) \right\}^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{H}(s_l) / (NL) \right\| \\
B_{3,m,nT} &:= \left\| \sum_{l=1}^L \left\{ \mathbf{E}(s_l; \theta_0)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{H}(s_l) - \mathbb{E}[\mathbf{E}(s_l; \theta_0)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{H}(s_l)] \right\} / (NL) \right\|.
\end{aligned}$$

In a similar manner to the proof of Lemma B.6(v), we can show that $\| (NL)^{-1} \sum_{l=1}^L \{ \mathbf{H}(s_l)^\top \mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)^\top \mathbf{H}(s_l)] \} \| \lesssim_p K/\sqrt{nT}$. Then, by Assumption 3.8(i) and Theorem 3.1(i), we have

$$\begin{aligned}
B_{2,m,nT} &= \left\| \left(\frac{1}{NL} \sum_{l=1}^L \mathbf{H}(s_l)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{H}(s_l) \right) (\hat{\theta}_{nT} - \theta_0) \right\| \\
&\leq \lambda_{\max} \left(\frac{1}{NL} \sum_{l=1}^L \mathbf{H}(s_l)^\top \mathbf{D}^\top P_m \mathbf{D} \mathbf{H}(s_l) \right) \cdot \|\hat{\theta}_{nT} - \theta_0\| \\
&\leq \lambda_{\max} (\mathbf{D}^\top P_m \mathbf{D}) \cdot \lambda_{\max} \left(\frac{1}{NL} \sum_{l=1}^L \mathbf{H}(s_l)^\top \mathbf{H}(s_l) \right) \cdot \|\hat{\theta}_{nT} - \theta_0\| \lesssim_p 1/\sqrt{nT} + K^{-\pi}.
\end{aligned}$$

For $B_{3,m,nT}$, by the same argument as in Lemma B.6(v), we can show that $B_{3,m,nT} \lesssim_p \sqrt{K}/\sqrt{nT}$. This completes the proof.

(ii) By the triangle inequality,

$$\begin{aligned}
\left\| \bar{J}_{nT}(\hat{\theta}_{nT}) \bar{J}_{nT}(\hat{\theta}_{nT})^\top - \bar{J}_{nT} \bar{J}_{nT}^\top \right\| &\leq \left\| (\bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT})(\bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT})^\top \right\| \\
&\quad + 2 \left\| \bar{J}_{nT}(\bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT})^\top \right\|
\end{aligned}$$

$$\lesssim_p K/\sqrt{nT} + K^{-\pi}$$

where the last inequality is from result (i) and Assumption 3.8(ii).

(iii) By definition of $\bar{\theta}_{nT}$, we have $\|\bar{\theta}_{nT} - \theta_0\| \lesssim_p 1/\sqrt{nT} + K^{-\pi}$ and thus $\|\bar{J}_{nT}(\bar{\theta}_{nT}) - \bar{J}_{nT}\| \lesssim_p K/\sqrt{nT} + K^{-\pi}$, as in result (i). Then, by the triangle inequality,

$$\begin{aligned} \left\| \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) - \bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right\| &\leq \left\| (\bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT})^\top \Omega_{nT} (\bar{J}_{nT}(\bar{\theta}_{nT}) - \bar{J}_{nT}) \right\| \\ &\quad + \left\| \bar{J}_{nT} \Omega_{nT} (\bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT})^\top \right\| \\ &\quad + \left\| \bar{J}_{nT} \Omega_{nT} (\bar{J}_{nT}(\bar{\theta}_{nT}) - \bar{J}_{nT})^\top \right\| \\ &\lesssim_p K/\sqrt{nT} + K^{-\pi}. \end{aligned}$$

(iv) As a result of (iii), we have $\lambda_{\min} \left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) \right) > 0$ with probability approaching one by Assumptions 3.5(ii) and 3.8(ii). Then, noting the equality

$$\begin{aligned} &\left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) \right)^- - \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \\ &= \left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) \right)^- \left[\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} - \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) \right] \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1}, \end{aligned}$$

the result is straightforward. $\square$

B.3.2 Proof of Theorem 3.2.

Since the proof is similar, we only prove (i). By the first-order condition of minimization and the mean-value expansion, we have

$$\begin{aligned} \mathbf{0}_{(d_x+1)K} &= \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{g}_{nT}(\hat{\theta}_{nT}) \\ &= \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \left[\bar{g}_{nT}(\theta_0) + \bar{J}_{nT}(\bar{\theta}_{nT}) \left(\hat{\theta}_{nT} - \theta_0 \right) \right], \end{aligned}$$

where $\bar{\theta}_{nT} \in [\hat{\theta}_{nT}, \theta_0]$, leading to

$$\begin{aligned} \hat{\theta}_{nT} - \theta_0 &= - \left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) \right)^- \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{g}_{nT}(\theta_0) \\ &= -[G_{1,nT} + G_{2,nT} + G_{3,nT} + G_{4,nT}], \end{aligned}$$

with

$$\begin{aligned}
G_{1,nT} &:= \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \bar{J}_{nT}^\top \Omega_{nT} \bar{g}_{1,nT} \\
G_{2,nT} &:= \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \bar{J}_{nT}^\top \Omega_{nT} \bar{g}_{2,nT} \\
G_{3,nT} &:= \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \left\{ \bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT} \right\}^\top \Omega_{nT} \bar{g}_{nT}(\theta_0) \\
G_{4,nT} &:= \left\{ \left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\bar{\theta}_{nT}) \right)^- - \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \right\} \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{g}_{nT}(\theta_0).
\end{aligned}$$

First, observing that

$$\begin{aligned}
\|G_{2,nT}\|^2 &= \bar{g}_{2,nT}^\top \Omega_{nT} \bar{J}_{nT} \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-2} \bar{J}_{nT}^\top \Omega_{nT} \bar{g}_{2,nT} \\
&\lesssim \bar{g}_{2,nT}^\top \Omega_{nT} \bar{J}_{nT} \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \bar{J}_{nT}^\top \Omega_{nT} \bar{g}_{2,nT} \\
&\lesssim \bar{g}_{2,nT}^\top \Omega_{nT} \bar{g}_{2,nT} \\
&\lesssim \|\bar{g}_{2,nT}\|^2,
\end{aligned}$$

we can find that $\|G_{2,nT}\| \lesssim_p K^{-\pi}$ by Lemma B.6(iii), (vi), and (vii). Next, it is easy to see that

$$\begin{aligned}
\|G_{3,nT}\| &\lesssim \underbrace{\left\| \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1} \left\{ \bar{J}_{nT}(\hat{\theta}_{nT}) - \bar{J}_{nT} \right\}^\top \right\|}_{\lesssim_p K/\sqrt{nT} + K^{-\pi}: \text{Lemma B.9(i)}} \cdot \underbrace{\|\bar{g}_{nT}(\theta_0)\|}_{\lesssim_p 1/\sqrt{nT} + K^{-\pi}: \text{(B.5)}} \\
&\lesssim_p K/(nT)
\end{aligned}$$

from $\sqrt{nT}K^{-\pi} \rightarrow 0$. Further, for $G_{4,nT}$, by Lemma B.9(ii) and (iv) and (B.5) with Assumption 3.8(ii), we can show that $\|G_{4,nT}\| \lesssim_p K/(nT)$.

Combining all these results and noting that

$$[\sigma_{nT,\alpha}(s)]^2 \geq c \phi^K(s)^\top \underbrace{\mathbb{S}_\alpha \mathbb{S}_\alpha^\top}_{=I_K} \phi^K(s) \geq c \|\phi^K(s)\|^2 > 0$$

for sufficiently large nT , we have

$$\begin{aligned}
\frac{\sqrt{N}(\hat{\alpha}_{nT}(s) - \alpha_0(s))}{\sigma_{nT,\alpha}(s)} &= -\frac{\sqrt{N}\phi^K(s)^\top \mathbb{S}_\alpha [G_{1,nT} + O_P(K/(nT)) + O_P(K^{-\pi})]}{\sigma_{nT,\alpha}(s)} + \frac{\sqrt{N}O(K^{-\pi})}{\sigma_{nT,\alpha}(s)} \\
&= -\frac{\sqrt{N}\phi^K(s)^\top \mathbb{S}_\alpha G_{1,nT}}{\sigma_{nT,\alpha}(s)} + o_P(1)
\end{aligned}$$

Here, let $\Lambda_{z,nT}(s)$ and $\Lambda_{m,nT}(s)$ denote the first $(d_q + d_x)K$ elements and $((d_q + d_x)K + m)$ -th element of $\phi^K(s)^\top \mathbb{S}_\alpha \left(\bar{\mathbf{J}}_{nT}^\top \Omega_{nT} \bar{\mathbf{J}}_{nT} \right)^{-1} \bar{\mathbf{J}}_{nT}^\top \Omega_{nT} / \sigma_{nT,\alpha}(s)$, respectively. Then, we can write

$$\begin{aligned} & \frac{\sqrt{N} \phi^K(s)^\top \mathbb{S}_\alpha G_{1,nT}}{\sigma_{nT,\alpha}(s)} \\ &= \frac{\phi^K(s)^\top \mathbb{S}_\alpha \left(\bar{\mathbf{J}}_{nT}^\top \Omega_{nT} \bar{\mathbf{J}}_{nT} \right)^{-1} \bar{\mathbf{J}}_{nT}^\top \Omega_{nT} [\sqrt{N} \bar{g}_{1,nT}]}{\sigma_{nT,\alpha}(s)} \\ &= \frac{1}{L\sqrt{N}} \sum_{l=1}^L \left[\Lambda_{z,nT}(s) \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\varepsilon}(s_l) + \boldsymbol{\varepsilon}(s_l)^\top \mathbf{D}^\top \left(\sum_{m=1}^M \Lambda_{m,nT}(s) P_m \right) \mathbf{D} \boldsymbol{\varepsilon}(s_l) \right]. \end{aligned}$$

Hereinafter in this proof, we use the labelling ($i = 1, \dots, \mathbf{n}$) introduced in Appendix A. Let us denote $\Pi_M(s) := \mathbf{D}^\top \left(\sum_{m=1}^M \Lambda_{m,nT}(s) P_m \right) \mathbf{D} = (\pi_{M,i,j}(s))$. Recalling the block-diagonal structure of P_m and that its diagonals are all zero, we can find that the diagonal elements of $\Pi_M(s)$ are also all zero. Further, note that $\Pi_M(s)$ is symmetric. Now, recalling that $z_i^\dagger(s_l)$ is the i -th column of $\mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{D}$, define

$$\begin{aligned} a_i(s) &:= \frac{1}{L\sqrt{N}} \sum_{l=1}^L \Lambda_{z,nT}(s) z_i^\dagger(s_l) \varepsilon_i(s_l) \\ b_{i,j}(s) &:= \frac{1}{L\sqrt{N}} \sum_{l=1}^L \pi_{M,i,j}(s) \varepsilon_i(s_l) \varepsilon_j(s_l) \\ \gamma_i(s) &:= a_i(s) + 2 \underbrace{\sum_{j=1}^{i-1} b_{i,j}(s)}_{=0 \text{ if } i=1}, \end{aligned}$$

and we further re-write

$$\frac{\sqrt{N} \phi^K(s)^\top \mathbb{S}_\alpha G_{1,nT}}{\sigma_{nT,\alpha}(s)} = \sum_{i=1}^{\mathbf{n}} \gamma_i(s).$$

Here, let $\mathcal{F}_{nT}(i)$ denote the σ -field generated by $\{\varepsilon_j : 1 \leq j \leq i\}$. Under Assumption 3.3(i), we have $\mathbb{E}[\gamma_i(s) \mid \mathcal{F}_{nT}(i-1)] = 0$, implying that $\{\gamma_i(s)\}$ forms a martingale difference sequence. Then, it suffices to check the following two conditions for the central limit theorem of Scott (1973):

$$(1) \quad \sum_{i=1}^{\mathbf{n}} \mathbb{E}[(\gamma_i(s))^2 \mid \mathcal{F}_{nT}(i-1)] \xrightarrow{P} 1$$

$$(2) \sum_{i=1}^n \mathbb{E}[(\gamma_i(s))^2 \mathbf{1}\{|\gamma_i(s)| \geq \eta\} \mid \mathcal{F}_{nT}(i-1)] \xrightarrow{p} 0 \text{ for any } \eta > 0$$

Verification of condition (1) Observe that

$$\begin{aligned} \mathbb{E}[(\gamma_i(s))^2 \mid \mathcal{F}_{nT}(i-1)] &= \mathbb{E}[(a_i(s))^2] + 4 \sum_{j_1=1}^{i-1} \sum_{j_2=1}^{i-1} \mathbb{E}[b_{i,j_1}(s)b_{i,j_2}(s) \mid \mathcal{F}_{nT}(i-1)] \\ &\quad + 4 \sum_{j=1}^{i-1} \mathbb{E}[a_i(s)b_{i,j}(s) \mid \mathcal{F}_{nT}(i-1)]. \end{aligned}$$

Recalling the definition of $\mathcal{V}_{z,nT}$ in (A.4), we can easily see that

$$\sum_{i=1}^n \mathbb{E}[(a_i(s))^2] = \Lambda_{z,nT}(s) \mathcal{V}_{z,nT} \Lambda_{z,nT}(s)^\top.$$

For the second term on the right-hand side, noting that $j_1, j_2 \leq i-1$,

$$\begin{aligned} &4 \sum_{i=1}^n \sum_{j_1=1}^{i-1} \sum_{j_2=1}^{i-1} \mathbb{E}[b_{i,j_1}(s)b_{i,j_2}(s) \mid \mathcal{F}_{nT}(i-1)] \\ &= \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \underbrace{\frac{4}{N} \sum_{i=1}^n \sum_{j_1=1}^{i-1} \sum_{j_2=1}^{i-1} \pi_{M,i,j_1}(s) \pi_{M,i,j_2}(s) \Gamma_i(s_l, s_{l'}) \varepsilon_{j_1}(s_l) \varepsilon_{j_2}(s_{l'})}_{=: D(s, s_l, s_{l'})}. \end{aligned}$$

Since $\Pi_M(s)$ is symmetric and its diagonals are zero, recalling the definition of $\mathcal{V}_{ab,nT}$ in (A.5), direct calculation yields

$$\begin{aligned} &\frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \mathbb{E}[D(s, s_l, s_{l'})] \\ &= \frac{4}{N} \sum_{i=1}^n \sum_{j=1}^{i-1} [\pi_{M,i,j}(s)]^2 \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_j(s_l, s_{l'}) \\ &= \frac{2}{N} \sum_{1 \leq i,j \leq n} [\pi_{M,i,j}(s)]^2 \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_j(s_l, s_{l'}) \\ &= \frac{2}{N} \sum_{1 \leq i,j \leq n} \left[\sum_{m=1}^M \Lambda_{m,nT}(s) \tilde{p}_{m,i,j} \right]^2 \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_j(s_l, s_{l'}) \\ &= \sum_{a=1}^M \sum_{b=1}^M \Lambda_{a,nT}(s) \Lambda_{b,nT}(s) \frac{2}{N} \sum_{1 \leq i,j \leq n} \tilde{p}_{a,i,j} \tilde{p}_{b,i,j} \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_j(s_l, s_{l'}) \end{aligned}$$

$$= \sum_{a=1}^M \sum_{b=1}^M \Lambda_{a,nT}(s) \Lambda_{b,nT}(s) \mathcal{V}_{ab,nT}.$$

Meanwhile,

$$\begin{aligned} \text{Var}(D(s, s_l, s_{l'})) &\lesssim \frac{1}{N^2} \sum_{i=1}^n \sum_{j_1=1}^{i-1} \sum_{j_2=1}^{i-1} \sum_{i'=1}^n \sum_{k_1=1}^{i'-1} \sum_{k_2=1}^{i'-1} |\pi_{M,i,j_1}(s) \pi_{M,i,j_2}(s) \pi_{M,i',k_1}(s) \pi_{M,i',k_2}(s)| \\ &\quad \times |\mathbb{E}\{(\varepsilon_{j_1}(s_l) \varepsilon_{j_2}(s_{l'}) - \mathbb{E}[\varepsilon_{j_1}(s_l) \varepsilon_{j_2}(s_{l'})])(\varepsilon_{k_1}(s_l) \varepsilon_{k_2}(s_{l'}) - \mathbb{E}[\varepsilon_{k_1}(s_l) \varepsilon_{k_2}(s_{l'})])\}| \\ &\lesssim \frac{1}{N^2} \sum_{i=1}^n \sum_{i'=1}^n \sum_{j_1=1}^{i-1} \sum_{j_2=1}^{i-1} |\pi_{M,i,j_1}(s) \pi_{M,i,j_2}(s) \pi_{M,i',j_1}(s) \pi_{M,i',j_2}(s)| \\ &= \frac{1}{N^2} \sum_{i'=1}^n \sum_{j_1=1}^{i-1} \sum_{j_2=1}^{i-1} \sum_{i=1}^n |\pi_{M,i,j_1}(s)| \cdot |\pi_{M,i,j_2}(s)| \cdot |\pi_{M,i',j_1}(s)| \cdot |\pi_{M,i',j_2}(s)| \\ &\leq \frac{1}{N^2} \sum_{i'=1}^n \|\Pi_M(s)\|_1^2 \cdot \|\Pi_M(s)\|_\infty^2 \lesssim 1/(nT). \end{aligned}$$

Consequently, $4 \sum_{i=1}^n \sum_{j_1=1}^{i-1} \sum_{j_2=1}^{i-1} \mathbb{E}[b_{i,j_1}(s) b_{i,j_2}(s) \mid \mathcal{F}_{nT}(i-1)] \xrightarrow{p} \sum_{a=1}^M \sum_{b=1}^M \Lambda_{a,nT}(s) \Lambda_{b,nT}(s) \mathcal{V}_{ab,nT}$ holds from Chebyshev's inequality.

For the third term, for $j \leq i-1$,

$$\begin{aligned} \mathbb{E}[a_i(s) b_{i,j}(s) \mid \mathcal{F}_{nT}(i-1)] &= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \Lambda_{z,nT}(s) z_i^\dagger(s_l) \pi_{M,i,j}(s) \mathbb{E}[\varepsilon_i(s_l) \varepsilon_i(s_{l'})] \varepsilon_j(s_{l'}) \\ &= \frac{1}{LN} \sum_{l=1}^L \pi_{M,i,j}(s) h_i(s, s_l) \varepsilon_j(s_l), \end{aligned}$$

where $h_i(s, s_{l'}) := L^{-1} \sum_{l=1}^L \Lambda_{z,nT}(s) z_i^\dagger(s_l) \mathbb{E}[\varepsilon_i(s_l) \varepsilon_i(s_{l'})]$. Hence, we can write

$$4 \sum_{i=1}^n \sum_{j=1}^{i-1} \mathbb{E}[a_i(s) b_{i,j}(s) \mid \mathcal{F}_{nT}(i-1)] = \frac{4}{L} \sum_{l=1}^L \left(\frac{1}{N} \sum_{j=1}^n \sum_{i=j+1}^n \pi_{M,i,j}(s) h_i(s, s_l) \varepsilon_j(s_l) \right).$$

Noting that $|h_i(s, s_{l'})| \lesssim K/\|\phi^K(s)\|$,

$$\begin{aligned} \mathbb{E} \left| \frac{1}{N} \sum_{j=1}^n \sum_{i=j+1}^n \pi_{M,i,j}(s) h_i(s, s_l) \varepsilon_j(s_l) \right|^2 &\lesssim \frac{K^2}{N^2} \sum_{j=1}^n \sum_{i=j+1}^n \sum_{i'=j+1}^n |\pi_{M,i,j}(s)| \cdot |\pi_{M,i',j}(s)| \\ &\leq \frac{K^2}{N^2} \sum_{j=1}^n \|\Pi_M(s)\|_1^2 \lesssim K^2/(nT) \end{aligned}$$

Then, by Markov's inequality, we obtain $4 \sum_{i=1}^n \sum_{j=1}^{i-1} \mathbb{E}[a_i(s)b_{i,j}(s) \mid \mathcal{F}_{nT}(i-1)] \xrightarrow{p} 0$.

Finally, combining the above results gives

$$\begin{aligned} & \mathbb{E}[(\gamma_i(s))^2 \mid \mathcal{F}_{nT}(i-1)] \xrightarrow{p} \Lambda_{z,nT}(s) \mathcal{V}_{z,nT} \Lambda_{z,nT}(s)^\top + \sum_{a=1}^M \sum_{b=1}^M \Lambda_{a,nT}(s) \Lambda_{b,nT}(s) \mathcal{V}_{ab,nT} \\ &= \begin{pmatrix} \Lambda_{z,nT}(s) & \Lambda_{1,nT}(s) & \cdots & \Lambda_{M,nT}(s) \end{pmatrix} \\ & \quad \times \begin{pmatrix} \mathcal{V}_{z,nT} & \mathbf{0}_{(d_q+d_x)K \times 1} & \cdots & \mathbf{0}_{(d_q+d_x)K \times 1} \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \mathcal{V}_{11,nT} & \cdots & \mathcal{V}_{1M,nT} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \mathcal{V}_{M1,nT} & \cdots & \mathcal{V}_{MM,nT} \end{pmatrix} \begin{pmatrix} \Lambda_{z,nT}(s)^\top \\ \Lambda_{1,nT}(s) \\ \vdots \\ \Lambda_{M,nT}(s) \end{pmatrix} \\ &= \frac{\phi^K(s)^\top \mathbb{S}_\alpha \Sigma_{nT} \mathbb{S}_\alpha^\top \phi^K(s)}{[\sigma_{nT,\alpha}(s)]^2} = 1, \end{aligned}$$

as desired.

Verification of condition (2) To verify condition (2), it is sufficient to show that $\sum_{i=1}^n \mathbb{E}[|\gamma_i(s)|^4 \mid \mathcal{F}_{nT}(i-1)] \xrightarrow{p} 0$. Moreover, by the c_r inequality,

$$\sum_{i=1}^n \mathbb{E}[|\gamma_i(s)|^4 \mid \mathcal{F}_{nT}(i-1)] \leq 8 \sum_{i=1}^n \mathbb{E}[|a_i(s)|^4] + 128 \sum_{i=1}^n \mathbb{E} \left[\left| \sum_{j=1}^{i-1} b_{i,j}(s) \right|^4 \mid \mathcal{F}_{nT}(i-1) \right].$$

For the first term on the right-hand side, noting that Assumption 3.3(ii) implies $\mathbb{E}[\prod_{k=1}^4 \varepsilon_{it}(s_k)] < \infty$ by Hölder's inequality,

$$\begin{aligned} \sum_{i=1}^n \mathbb{E}[|a_i(s)|^4] &= \frac{1}{L^4 N^2} \sum_{i=1}^n \sum_{1 \leq l_1, l_2, l_3, l_4 \leq L} \left| \prod_{j=1}^4 \Lambda_{z,nT}(s) z_i^\dagger(s_{l_j}) \right| \cdot \mathbb{E} \left[\left| \prod_{j=1}^4 \varepsilon_i(s_{l_j}) \right| \right] \\ &\lesssim \frac{1}{N^2} \sum_{i=1}^n \frac{1}{L^4} \sum_{1 \leq l_1, l_2, l_3, l_4 \leq L} \left| \prod_{j=1}^4 \Lambda_{z,nT}(s) z_i^\dagger(s_{l_j}) \right| \lesssim \frac{K^4}{nT \|\phi^K(s)\|^4}. \end{aligned}$$

For the second term, observe that

$$\begin{aligned} & \sum_{i=1}^n \mathbb{E} \left[\left| \sum_{j=1}^{i-1} b_{i,j}(s) \right|^4 \mid \mathcal{F}_{nT}(i-1) \right] \\ &= \frac{1}{L^4 N^2} \sum_{i=1}^n \sum_{1 \leq l_1, l_2, l_3, l_4 \leq L} \left(\sum_{j_1=1}^{i-1} \pi_{M,i,j_1}(s) \varepsilon_{j_1}(s_{l_1}) \right) \cdots \left(\sum_{j_4=1}^{i-1} \pi_{M,i,j_4}(s) \varepsilon_{j_4}(s_{l_4}) \right) \mathbb{E} \left(\prod_{k=1}^4 \varepsilon_i(s_{l_k}) \right). \end{aligned}$$

Further, it is easy to see that $\sum_{j=1}^{i-1} \pi_{M,i,j}(s) \varepsilon_j(s_l) \lesssim_p 1$ by Markov's inequality. Hence, we have

$\sum_{i=1}^n \mathbb{E} \left[\left| \sum_{j=1}^{i-1} b_{i,j}(s) \right|^4 \mid \mathcal{F}_{nT}(i-1) \right] \lesssim_p 1/(nT)$, and combining this with the previous result implies condition (2). $\square$

B.4 Proof of Corollary 3.1.

In view of Proposition C.1(iii) and (iv), the results will follow from Slutsky's theorem. $\square$

B.5 Proof of Proposition 4.1

Since the proofs of (i) and (ii) are almost identical, we only prove (i). By the triangle inequality,

$$\left\| \widehat{M}_{nT}^S(i, j, s) - M(i, j, s) \right\|_{\infty} \leq \left\| \widehat{M}_{nT}^S(i, j, s) - M^S(i, j, s) \right\|_{\infty} + \left\| M^S(i, j, s) - M(i, j, s) \right\|_{\infty},$$

where $M^S(i, j, s) := \sum_{\ell=0}^S W_n^{\ell} \mathbf{e}_i \gamma^{\ell}(\beta_{0j}, s)$. For the first term on the right-hand side, observe that

$$\begin{aligned} \left| \left\{ \widehat{M}_{nT}^S(i, j, s) \right\}_k - \left\{ M^S(i, j, s) \right\}_k \right| &\leq \sum_{\ell=0}^S \left| \{W_n^{\ell} \mathbf{e}_i\}_k \right| \cdot \left| \widehat{\gamma}_{nT}^{\ell}(\widehat{\beta}_{nT,j}, s) - \gamma^{\ell}(\beta_{0j}, s) \right| \\ &\lesssim \sum_{\ell=0}^S \left| \widehat{\gamma}_{nT}^{\ell}(\widehat{\beta}_{nT,j}, s) - \gamma^{\ell}(\beta_{0j}, s) \right| \end{aligned}$$

for all $k \in [n]$. By definition, when $\ell = 0$,

$$\left| \widehat{\gamma}_{nT}^0(\widehat{\beta}_{nT,j}, s) - \gamma^0(\beta_{0j}, s) \right| = \left| \widehat{\beta}_{nT,j}(s) - \beta_{0j}(s) \right| \lesssim_p c_n$$

uniformly in $s \in [0, 1]$, where $c_n := \sqrt{K}/\sqrt{nT} + K^{1/2-\pi}$. When $\ell = 1$, by Assumption 3.4,

$$\begin{aligned} \left| \widehat{\gamma}_{nT}^1(\widehat{\beta}_{nT,j}, s) - \gamma^1(\beta_{0j}, s) \right| &= \left| \widehat{\alpha}_{nT}(s) A(\widehat{\beta}_{nT,j}, s) - \alpha_0(s) A(\beta_{0j}, s) \right| \\ &\leq \left| \widehat{\alpha}_{nT}(s) \right| \cdot \left| A(\widehat{\beta}_{nT,j} - \beta_{0j}, s) \right| + \left| \widehat{\alpha}_{nT}(s) - \alpha_0(s) \right| \cdot \left| A(\beta_{0j}, s) \right| \\ &\lesssim_p (\bar{\alpha}_0 + c_n) \cdot \sup_{s \in [0,1]} \left| \widehat{\beta}_{nT,j}(s) - \beta_{0j}(s) \right| + c_n \\ &\lesssim \bar{\alpha}_0 c_n + c_n \end{aligned}$$

uniformly in $s \in [0, 1]$. Similarly, when $\ell = 2$, we have

$$\begin{aligned}
\left| \hat{\gamma}_{nT}^2(\hat{\beta}_{nT,j}, s) - \gamma^2(\beta_{0j}, s) \right| &= \left| \hat{\alpha}_{nT}(s) A(\hat{\gamma}_{nT}^1(\hat{\beta}_{nT,j}, \cdot), s) - \alpha_0(s) A(\gamma^1(\beta_{0j}, \cdot), s) \right| \\
&\leq \left| \hat{\alpha}_{nT}(s) \right| \cdot \left| A(\hat{\gamma}_{nT}^1(\hat{\beta}_{nT,j}, \cdot) - \gamma^1(\beta_{0j}, \cdot), s) \right| \\
&\quad + \left| \hat{\alpha}_{nT}(s) - \alpha_0(s) \right| \cdot \left| A(\gamma^1(\beta_{0j}, \cdot), s) \right| \\
&\lesssim_p (\bar{\alpha}_0 + c_n) \cdot (\bar{\alpha}_0 c_n + c_n) + \bar{\alpha}_0 c_n \\
&\lesssim \bar{\alpha}_0^2 c_n + \bar{\alpha}_0 c_n.
\end{aligned}$$

Thus, repeating the same computation recursively, we can obtain $|\hat{\gamma}_{nT}^\ell(\hat{\beta}_{nT,j}, s) - \gamma^\ell(\beta_{0j}, s)| \lesssim_p \bar{\alpha}_0^{\ell-1} c_n$ for general $\ell \geq 1$ under $\bar{\alpha}_0 < 1$. From a straightforward calculation, we have $\sum_{\ell=1}^S \bar{\alpha}_0^{\ell-1} c_n = c_n(1 - \bar{\alpha}_0^S)/(1 - \bar{\alpha}_0)$, which leads to $\|\widehat{M}_{nT}^S(i, j, s) - M^S(i, j, s)\|_\infty \lesssim_p c_n$.

Next, observe that $M^S(i, j, s) - M(i, j, s) = \sum_{\ell=S+1}^\infty W_n^\ell e_i \gamma^\ell(\beta_{0j}, s)$ and that

$$|\gamma^0(\beta_{0j}, s)| \leq \bar{\beta}_{0j}, \quad |\gamma^1(\beta_{0j}, s)| \leq \bar{\alpha}_0 \bar{\beta}_{0j}, \quad \dots, \quad |\gamma^\ell(\beta_{0j}, s)| \leq \bar{\alpha}_0^\ell \bar{\beta}_{0j}$$

by repeatedly applying Assumption 3.4, where $\bar{\beta}_{0j} := \sup_{s \in [0,1]} |\beta_{0j}(s)|$. Hence, we have

$$\|M^S(i, j, s) - M(i, j, s)\|_\infty \lesssim \sum_{\ell=S+1}^\infty |\gamma^\ell(\beta_{0j}, s)| \leq \frac{\bar{\beta}_{0j}}{1 - \bar{\alpha}_0} \cdot \bar{\alpha}_0^{S+1}.$$

Combining these results completes the proof. □

C Consistent variance estimation

First, observe the following alternative representations of $\mathcal{V}_{z,nT}$ and $\mathcal{V}_{ab,nT}$:

$$\begin{aligned}
\mathcal{V}_{z,nT} &= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top \mathbb{E}[\tilde{\varepsilon}_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'})] \\
\mathcal{V}_{ab,nT} &= \frac{2}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{1 \leq i, j \leq n} p_{a,i,j} p_{b,i,j} \mathbb{E}[\tilde{\varepsilon}_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'})] \mathbb{E}[\tilde{\varepsilon}_{jt}(s_l) \tilde{\varepsilon}_{jt'}(s_{l'})].
\end{aligned}$$

Define $\vec{e}_{it}(s) := e_{i,t+1}(s; \hat{\theta}_{nT}) - e_{it}(s; \hat{\theta}_{nT})$,

$$\begin{aligned}\hat{\mathcal{V}}_{z,nT} &:= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \vec{z}_{it}(s_l) \vec{z}_{it'}(s_{l'})^\top \vec{e}_{it}(s_l) \vec{e}_{it'}(s_{l'}) \\ \hat{\mathcal{V}}_{ab,nT} &:= \frac{2}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{1 \leq i,j \leq n} p_{a,i,j} p_{b,i,j} \vec{e}_{it}(s_l) \vec{e}_{it'}(s_{l'}) \vec{e}_{jt}(s_l) \vec{e}_{jt'}(s_{l'}) \\ \hat{\mathcal{V}}_{nT} &:= \begin{pmatrix} \hat{\mathcal{V}}_{z,nT} & \mathbf{0}_{(d_q+d_x)K \times 1} & \cdots & \mathbf{0}_{(d_q+d_x)K \times 1} \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \hat{\mathcal{V}}_{11,nT} & \cdots & \hat{\mathcal{V}}_{1M,nT} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \hat{\mathcal{V}}_{M1,nT} & \cdots & \hat{\mathcal{V}}_{MM,nT} \end{pmatrix} \\ \hat{\Sigma}_{nT} &:= \left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\hat{\theta}_{nT}) \right)^{-1} \bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \hat{\mathcal{V}}_{nT} \Omega_{nT} \bar{J}_{nT}(\hat{\theta}_{nT}) \left(\bar{J}_{nT}(\hat{\theta}_{nT})^\top \Omega_{nT} \bar{J}_{nT}(\hat{\theta}_{nT}) \right)^{-1}.\end{aligned}$$

Then, our standard error estimators for $\sigma_{nT,\alpha}(s)$ and $\sigma_{nT,j}(s)$ are given as

$$\begin{aligned}\hat{\sigma}_{nT,\alpha}(s) &:= \sqrt{\phi^K(s)^\top \mathbb{S}_\alpha \hat{\Sigma}_{nT} \mathbb{S}_\alpha^\top \phi^K(s)} \\ \hat{\sigma}_{nT,j}(s) &:= \sqrt{\phi^K(s)^\top \mathbb{S}_j \hat{\Sigma}_{nT} \mathbb{S}_j^\top \phi^K(s)},\end{aligned}\tag{C.1}$$

respectively.

Proposition C.1 (Consistent variance estimation). Suppose that the assumptions in Theorem 3.2 are satisfied. In addition, assume that $K^3/(nT) \rightarrow 0$ and $K^{3/2-\pi} \rightarrow 0$ as $nT \rightarrow \infty$. Then,

- (i) $\left\| \hat{\mathcal{V}}_{z,nT} - \mathcal{V}_{z,nT} \right\| \lesssim_p K^{3/2}/\sqrt{nT} + K^{3/2-\pi}$
- (ii) $\left\| \hat{\mathcal{V}}_{ab,nT} - \mathcal{V}_{ab,nT} \right\| \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi}$ for all $1 \leq a, b \leq M$
- (iii) $\hat{\sigma}_{nT,\alpha}(s)/\sigma_{nT,\alpha}(s) \xrightarrow{p} 1$
- (iv) $\hat{\sigma}_{nT,j}(s)/\sigma_{nT,j}(s) \xrightarrow{p} 1$ for all $j \in [d_x]$.

Proof. (i) Decompose

$$\hat{\mathcal{V}}_{z,nT} - \mathcal{V}_{z,nT} = \left(\hat{\mathcal{V}}_{z,nT} - \tilde{\mathcal{V}}_{z,nT} \right) + \left(\tilde{\mathcal{V}}_{z,nT} - \mathcal{V}_{z,nT} \right),$$

where

$$\tilde{\mathcal{V}}_{z,nT} = \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top \tilde{\varepsilon}_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'}).$$

Since

$$\tilde{\varepsilon}_{it}(s) = \underbrace{\vec{a}_{it}(s)[\alpha_0(s) - \hat{\alpha}_{nT}(s)] + \sum_{j=1}^{d_x} \vec{X}_{it}^j [\beta_{0j}(s) - \hat{\beta}_{nT,j}(s)]}_{=: b_{it}(s)} + \tilde{\varepsilon}_{it}(s),$$

we have

$$\tilde{\varepsilon}_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'}) = \tilde{\varepsilon}_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'}) + b_{it}(s_l) b_{it'}(s_{l'}) + b_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'}) + \tilde{\varepsilon}_{it}(s_l) b_{it'}(s_{l'}).$$

Then, we can write

$$\begin{aligned} \hat{\mathcal{V}}_{z,nT} - \tilde{\mathcal{V}}_{z,nT} &= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top [\tilde{\varepsilon}_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'}) - \tilde{\varepsilon}_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'})] \\ &= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top b_{it}(s_l) b_{it'}(s_{l'}) \\ &\quad + \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top b_{it}(s_l) \tilde{\varepsilon}_{it'}(s_{l'}) \\ &\quad + \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top \tilde{\varepsilon}_{it}(s_l) b_{it'}(s_{l'}). \end{aligned}$$

In view of Theorem 3.1(ii) and (iii), we can easily find that $|b_{it}(s)| \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi}$ uniformly in s and (i, t) . In addition, it is straightforward to see that $\|\tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top\| \lesssim K$, and $\|\tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top \tilde{\varepsilon}_{it}(s_l)\|, \|\tilde{z}_{it}(s_l) \tilde{z}_{it'}(s_{l'})^\top \tilde{\varepsilon}_{it'}(s_{l'})\| \lesssim_p K$ by Markov's inequality. Hence, we have $\|\hat{\mathcal{V}}_{z,nT} - \tilde{\mathcal{V}}_{z,nT}\| \lesssim_p K^{3/2}/\sqrt{nT} + K^{3/2-\pi}$.

Meanwhile, it is not difficult to see that $\|\tilde{\mathcal{V}}_{z,nT} - \mathcal{V}_{z,nT}\| \lesssim_p K/\sqrt{nT}$ by Markov's inequality. Then, the result follows from the triangle inequality.

(ii) Similar to the above, we decompose

$$\hat{\mathcal{V}}_{ab,nT} - \mathcal{V}_{ab,nT} = \left(\hat{\mathcal{V}}_{ab,nT} - \tilde{\mathcal{V}}_{ab,nT} \right) + \left(\tilde{\mathcal{V}}_{ab,nT} - \mathcal{V}_{ab,nT} \right),$$

where

$$\tilde{\mathcal{V}}_{ab,nT} = \frac{2}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{1 \leq i, j \leq n} p_{a,i,j} p_{b,i,j} \vec{\varepsilon}_{it}(s_l) \vec{\varepsilon}_{it'}(s_{l'}) \vec{\varepsilon}_{jt}(s_l) \vec{\varepsilon}_{jt'}(s_{l'}).$$

For the first term on the right-hand side, noting that

$$\begin{aligned} \vec{\varepsilon}_{it}(s_l) \vec{\varepsilon}_{it'}(s_{l'}) \vec{\varepsilon}_{jt}(s_l) \vec{\varepsilon}_{jt'}(s_{l'}) &= \{ \vec{\varepsilon}_{it}(s_l) \vec{\varepsilon}_{it'}(s_{l'}) + b_{it}(s_l) b_{it'}(s_{l'}) + b_{it}(s_l) \vec{\varepsilon}_{it'}(s_{l'}) + \vec{\varepsilon}_{it}(s_l) b_{it'}(s_{l'}) \} \\ &\quad \times \{ \vec{\varepsilon}_{jt}(s_l) \vec{\varepsilon}_{jt'}(s_{l'}) + b_{jt}(s_l) b_{jt'}(s_{l'}) + b_{jt}(s_l) \vec{\varepsilon}_{jt'}(s_{l'}) + \vec{\varepsilon}_{jt}(s_l) b_{jt'}(s_{l'}) \} \\ &= \vec{\varepsilon}_{it}(s_l) \vec{\varepsilon}_{it'}(s_{l'}) \vec{\varepsilon}_{jt}(s_l) \vec{\varepsilon}_{jt'}(s_{l'}) \\ &\quad + \vec{\varepsilon}_{it}(s_l) \vec{\varepsilon}_{it'}(s_{l'}) \vec{\varepsilon}_{jt}(s_l) b_{jt'}(s_{l'}) + \cdots (\text{three } \vec{\varepsilon}\text{'s * one } b) \\ &\quad + \vec{\varepsilon}_{it}(s_l) \vec{\varepsilon}_{it'}(s_{l'}) b_{jt}(s_l) b_{jt'}(s_{l'}) + \cdots (\text{two } \vec{\varepsilon}\text{'s * two } b\text{'s}) \\ &\quad + \vec{\varepsilon}_{it'}(s_{l'}) b_{it}(s_l) b_{jt}(s_l) b_{jt'}(s_{l'}) + \cdots (\text{one } \vec{\varepsilon} * \text{three } b\text{'s}) \\ &\quad + b_{it}(s_l) b_{it'}(s_{l'}) b_{jt}(s_l) b_{jt'}(s_{l'}), \end{aligned}$$

it is not difficult to show that $|\hat{\mathcal{V}}_{ab,nT} - \tilde{\mathcal{V}}_{ab,nT}| \lesssim_p \sqrt{K}/\sqrt{nT} + K^{1/2-\pi}$.

For the second term, following the analogous argument as in the proof of Proposition 2 [Lin and Lee \(2010\)](#), $|\tilde{\mathcal{V}}_{ab,nT} - \mathcal{V}_{ab,nT}| \lesssim_p 1/\sqrt{nT}$ holds. Hence, we obtain the desired result.

(iii) To prove the result, it suffices to show that

$$\left| \frac{[\hat{\sigma}_{nT,\alpha}(s)]^2 - [\sigma_{nT,\alpha}(s)]^2}{[\sigma_{nT,\alpha}(s)]^2} \right| \xrightarrow{p} 0.$$

As shown in the proof of Theorem 3.2, $[\sigma_{nT,\alpha}(s)]^2$ is bounded below from $c\|\phi^K(s)\|^2$. On the other hand, writing $R_{nT} := \Omega_{nT} \bar{J}_{nT} \left(\bar{J}_{nT}^\top \Omega_{nT} \bar{J}_{nT} \right)^{-1}$ and its estimator counterpart as $\hat{R}_{nT}$, by the triangle inequality,

$$\begin{aligned} |[\hat{\sigma}_{nT,\alpha}(s)]^2 - [\sigma_{nT,\alpha}(s)]^2| &= \left| \phi^K(s)^\top \mathbb{S}_\alpha \left[\hat{\Sigma}_{nT} - \Sigma_{nT} \right] \mathbb{S}_\alpha^\top \phi^K(s) \right| \\ &\leq \left| \phi^K(s)^\top \mathbb{S}_\alpha \left[\hat{R}_{nT}^\top \hat{\mathcal{V}}_{nT} \hat{R}_{nT} - \hat{R}_{nT}^\top \mathcal{V}_{nT} \hat{R}_{nT} \right] \mathbb{S}_\alpha^\top \phi^K(s) \right| \\ &\quad + \left| \phi^K(s)^\top \mathbb{S}_\alpha \left[\hat{R}_{nT}^\top \mathcal{V}_{nT} \hat{R}_{nT} - R_{nT}^\top \mathcal{V}_{nT} R_{nT} \right] \mathbb{S}_\alpha^\top \phi^K(s) \right| \\ &\leq \left\| \hat{\mathcal{V}}_{nT} - \mathcal{V}_{nT} \right\| \cdot \left\| \hat{R}_{nT} \mathbb{S}_\alpha^\top \phi^K(s) \right\|^2 + \lambda_{\max}(\mathcal{V}_{nT}) \cdot \left\| \left\{ \hat{R}_{nT} - R_{nT} \right\} \mathbb{S}_\alpha^\top \phi^K(s) \right\|^2 \\ &\quad + 2 \left| \phi^K(s)^\top \mathbb{S}_\alpha R_{nT}^\top \mathcal{V}_{nT} \left\{ \hat{R}_{nT} - R_{nT} \right\} \mathbb{S}_\alpha^\top \phi^K(s) \right| \end{aligned}$$

$$\lesssim_p \left(K^{3/2}/\sqrt{nT} + K^{2-\pi} \right) \|\phi^K(s)\|^2,$$

where the last inequality is from $\|\hat{R}_{nT} - R_{nT}\| \lesssim_p K/\sqrt{nT} + K^{-\pi}$ by Lemma B.9(i) and (iv) and $\|\hat{\mathcal{V}}_{nT} - \mathcal{V}_{nT}\| \lesssim_p K^{3/2}/\sqrt{nT} + K^{3/2-\pi}$ by results (i) and (ii). Thus, we have

$$\left| \frac{[\hat{\sigma}_{nT,\alpha}(s)]^2 - [\sigma_{nT,\alpha}(s)]^2}{[\sigma_{nT,\alpha}(s)]^2} \right| \lesssim_p K^{3/2}/\sqrt{nT} + K^{3/2-\pi} \rightarrow 0.$$

(iv) Analogous to the proof of (iii). □

D Details of the Monte Carlo experiments

D.1 Setup

Throughout the experiments, we consider the following data-generating process (DGP):

$$Y_{it}(s) = \alpha_0(s) \int_0^1 \bar{Y}_{it}(u) \nu(u, s) du + X_{it} \beta_0(s) + f_{0i}(s) + \varepsilon_{it}(s),$$

where $X_{it} \sim \mathcal{N}(0, 1)$, $\alpha_0(s) = \phi(s; 0.4, 0.5^2) + 0.2s - 0.4s^2$, $\phi(\cdot; \mu, \sigma^2)$ denotes the standard normal density function with mean μ and variance σ^2 , $\nu(u, s) = 0.75(1 - (u - s)^2)$ (i.e., the Epanechnikov kernel function), and the individual fixed effects are given by $f_{0i}(s) = 1 + \cos(is)$. The coefficient function is given by

$$\beta_0(s) = r(0.5 + s + 0.5 * \sin(\pi s)),$$

where r is chosen from $r \in \{0.4, 1\}$. We use $Q_{it} = (\bar{X}_{it}, \bar{\bar{X}}_{it})$ as the IVs for $\bar{Y}_{it}$, and, thus, the magnitude of r determines the strength of these IVs. For the error term, we generate $\varepsilon_{it}(s) = \sqrt{1 + \deg_i}(e_{1,it}, e_{2,it}, e_{3,it})^\top (1, s, s^2)$, where $\deg_i$ denotes the number of units connected to i (i.e., i 's degree), and $e_{j,it} \sim \mathcal{N}(0, 0.4^2)$ for $j = 1, 2, 3$. The weight matrix W_n is a row-normalized adjacency matrix, which is constructed by randomly placing n units on a $[\sqrt{2n}] \times [\sqrt{2n}]$ lattice, where $[a]$ denotes the nearest integer to a . Any two units are connected if the Euclidean distance between them is exactly one. The sample size n is chosen from $n \in \{40, 80\}$, and T is from $T \in \{6, 12\}$.

Since our DGP satisfies the conditions in Assumption 2.1, we can generate the outcome functions using the Neumann series approximation: $Y_t \approx Y_t^{(S)} := \sum_{\ell=0}^S \mathcal{A}^\ell [X_t \beta_0 + F_0 + \mathcal{E}_t]$, where S is increased iteratively until $\max_{i \in [n]} |Y_{it}^{(S)}(s) - Y_{it}^{(S-1)}(s)| < 0.001$ is satisfied for all t

and s . Throughout the simulations, integrals over $[0, 1]$ are approximated by finite summations over 99 equally-spaced grid points.

For the choice of basis functions, we use the cubic B-splines orthonormalized via the Gram–Schmidt procedure. The number of inner knots for the B-splines is selected using cross-validation. Specifically, we use the first T_{train} periods as the training data and the remaining $T - T_{\text{train}}$ periods as the validation data. When $T = 6$, we set $T_{\text{train}} = 4$, and when $T = 12$, we set $T_{\text{train}} = 8$. Once the parameters are estimated, we compute the predicted outcome difference as

$$\hat{Y}_{t+1}^{(S)} - \hat{Y}_t^{(S)} := \sum_{\ell=0}^S \hat{\mathcal{A}}_{nT_{\text{train}}}^{\ell} (X_{t+1} - X_t)^{\top} \hat{\beta}_{nT_{\text{train}}},$$

where $\hat{\mathcal{A}}_{nT_{\text{train}}}$ is obtained by replacing α_0 with $\hat{\alpha}_{nT_{\text{train}}}$. We then compute the average mean squared prediction error (AMSPE) as

$$\text{AMSPE} := \frac{1}{99} \sum_{l=1}^{99} \left(\frac{1}{n(T - T_{\text{train}} - 1)} \sum_{i=1}^n \sum_{t=T_{\text{train}}+1}^{T-1} \left\{ [\hat{Y}_{i,t+1}(s_l) - \hat{Y}_{it}(s_l)] - [Y_{i,t+1}(s_l) - Y_{it}(s_l)] \right\}^2 \right),$$

where $\hat{Y}_{i,t+1}(s_l) - \hat{Y}_{it}(s_l)$ denotes the i -th element of $\hat{Y}_{t+1}^{(S)} - \hat{Y}_t^{(S)}$ evaluated at s_l . In view of Remark 3.2, we set the number of inner knots to $\lfloor c_K(nT)^{1/5} \rfloor$, where c_K is a constant chosen to minimize the AMSPE. In this numerical study, considering computational costs, we conduct a grid search over $c_K \in \{0.4, 0.8, 1.2, 1.6\}$. Moreover, instead of performing the cross-validation in every simulation run, we compute the optimal c_K based on 10 pilot simulations under each design. We then average the selected values of c_K across these pilot runs and use the resulting value throughout the entire simulation analysis. As a result, we find that $c_K \approx 1.1$ is approximately optimal in our setting.

The number of grid points used to evaluate the moment function is chosen from $L = c_L K$, where $c_L \in \{2, 3, 4\}$, with the points $0 < s_1 < \dots < s_L < 1$ evenly spaced over $[0, 1]$. For the quadratic moments, we use two weight matrices ($M = 2$): $P_{1,1} = W_n$ and $P_{2,1} = W_n^{\top} W_n - \text{diag}(W_n^{\top} W_n)$. We then compare the performance of three different estimators: **GMM 1**: the integrated-GMM estimator using the weight matrix given in (3.1), **GMM 2**: the integrated-GMM estimator using the identity weight matrix, and (integrated) **2SLS**: GMM 1 estimator without utilizing the quadratic moment conditions.

D.2 Results

For each setup, we generate the dataset 500 times. The performance of the estimators is evaluated based on the average bias (BIAS) and the average root-mean-squared error (RMSE). Specifically, the BIAS and RMSE of estimating α_0 are defined as

$$\begin{aligned}\text{BIAS}(\alpha_0) &:= \frac{1}{500} \sum_{b=1}^{500} \left[\frac{1}{99} \sum_{l=1}^{99} \left(\hat{\alpha}_{nT}^{(b)}(s_l) - \alpha_0(s_l) \right) \right] \\ \text{RMSE}(\alpha_0) &:= \frac{1}{500} \sum_{b=1}^{500} \sqrt{\frac{1}{99} \sum_{l=1}^{99} \left(\hat{\alpha}_{nT}^{(b)}(s_l) - \alpha_0(s_l) \right)^2}\end{aligned}$$

respectively. Here, $\hat{\alpha}_{nT}^{(b)}$ denotes the estimator of α_0 obtained from the b -th replicated dataset. The BIAS and RMSE for β_0 are defined analogously.

In computing the integrated-GMM estimators, we use the `nloptr` function in **R**, with the 2SLS estimator used to obtain the initial parameter value for optimization. Under our computational environment (Intel Core i7-9700, 32GB RAM), it takes on average about 20 minutes to complete 500 simulation iterations in our minimal setup with $(n, T) = (40, 6)$, and about 45 minutes in the largest setup with $(n, T) = (80, 12)$. The programming code used in this analysis is available as online supplementary material.

The simulation results for the estimation of α_0 are summarized in Table D.1. Table D.2 presents the simulation results for estimating β_0 .

Table D.1: Simulation result: α_0

n	T	c_L	r	GMM 1		GMM 2		2SLS		
				BIAS	RMSE	BIAS	RMSE	BIAS	RMSE	
40	6	2	0.4	-0.0276	0.1628	-0.0377	0.1902	0.0113	0.1949	
			1	-0.0093	0.0692	-0.0140	0.1008	-0.0016	0.0760	
			3	0.4	-0.0273	0.1632	-0.0325	0.1882	0.0113	0.1949
				1	-0.0089	0.0698	-0.0121	0.0994	-0.0016	0.0759
			4	0.4	-0.0273	0.1632	-0.0317	0.1865	0.0113	0.1949
				1	-0.0092	0.0701	-0.0098	0.0958	-0.0016	0.0759
		12	2	0.4	-0.0016	0.1174	-0.0043	0.1330	0.0235	0.1353
			1	-0.0027	0.0474	-0.0047	0.0637	0.0064	0.0532	
			3	0.4	0.0002	0.1185	-0.0023	0.1322	0.0235	0.1352
				1	-0.0018	0.0482	-0.0030	0.0629	0.0064	0.0532
			4	0.4	0.0015	0.1195	-0.0017	0.1331	0.0235	0.1352
				1	-0.0020	0.0485	-0.0026	0.0620	0.0064	0.0532
80	6	2	0.4	-0.0027	0.1229	-0.0032	0.1382	0.0116	0.1327	
			1	-0.0047	0.0491	-0.0032	0.0619	0.0022	0.0530	
			3	0.4	-0.0015	0.1235	-0.0017	0.1414	0.0116	0.1327
				1	-0.0047	0.0496	-0.0029	0.0608	0.0022	0.0530
			4	0.4	-0.0001	0.1243	-0.0015	0.1396	0.0116	0.1327
				1	-0.0045	0.0499	-0.0022	0.0602	0.0022	0.0530
		12	2	0.4	0.0026	0.0892	0.0028	0.0962	0.0072	0.0919
			1	-0.0036	0.0333	-0.0014	0.0411	0.0013	0.0364	
			3	0.4	0.0029	0.0895	0.0028	0.0958	0.0072	0.0919
				1	-0.0031	0.0339	-0.0011	0.0413	0.0013	0.0364
			4	0.4	0.0032	0.0893	0.0031	0.0943	0.0072	0.0919
				1	-0.0031	0.0339	-0.0010	0.0409	0.0013	0.0364

GMM 1: integrated-GMM estimator with the weight matrix in (3.1). GMM 2: integrated-GMM estimator with the identity weight matrix.

Table D.2: Simulation result: β_0

n	T	c_L	r	GMM 1		GMM 2		2SLS	
				BIAS	RMSE	BIAS	RMSE	BIAS	RMSE
40	6	2	0.4	0.0073	0.0604	0.0046	0.0675	0.0061	0.0606
			1	0.0062	0.0599	0.0045	0.0701	0.0028	0.0596
		3	0.4	0.0072	0.0604	0.0047	0.0668	0.0061	0.0606
			1	0.0059	0.0601	0.0041	0.0692	0.0028	0.0596
		4	0.4	0.0072	0.0604	0.0048	0.0665	0.0061	0.0606
			1	0.0059	0.0601	0.0034	0.0685	0.0028	0.0596
	12	2	0.4	0.0078	0.0408	0.0080	0.0430	0.0065	0.0407
			1	0.0078	0.0406	0.0082	0.0442	0.0048	0.0402
		3	0.4	0.0077	0.0407	0.0078	0.0431	0.0065	0.0407
			1	0.0075	0.0406	0.0078	0.0435	0.0048	0.0402
		4	0.4	0.0077	0.0406	0.0077	0.0431	0.0065	0.0407
			1	0.0075	0.0406	0.0076	0.0431	0.0048	0.0402
80	6	2	0.4	0.0012	0.0412	0.0010	0.0428	0.0004	0.0414
			1	0.0013	0.0413	0.0009	0.0439	-0.0012	0.0413
		3	0.4	0.0011	0.0413	0.0013	0.0428	0.0004	0.0414
			1	0.0012	0.0414	0.0008	0.0435	-0.0012	0.0413
		4	0.4	0.0010	0.0413	0.0016	0.0428	0.0004	0.0414
			1	0.0011	0.0413	0.0006	0.0431	-0.0012	0.0413
	12	2	0.4	0.0011	0.0284	0.0010	0.0291	0.0009	0.0282
			1	0.0014	0.0286	0.0009	0.0293	0.0001	0.0281
		3	0.4	0.0011	0.0284	0.0010	0.0291	0.0009	0.0282
			1	0.0013	0.0286	0.0009	0.0293	0.0001	0.0281
		4	0.4	0.0011	0.0284	0.0010	0.0290	0.0009	0.0282
			1	0.0013	0.0286	0.0008	0.0292	0.0001	0.0281

GMM 1: integrated-GMM estimator with the weight matrix in (3.1). GMM 2: integrated-GMM estimator with the identity weight matrix.

E Supplementary Material for the Empirical Analysis

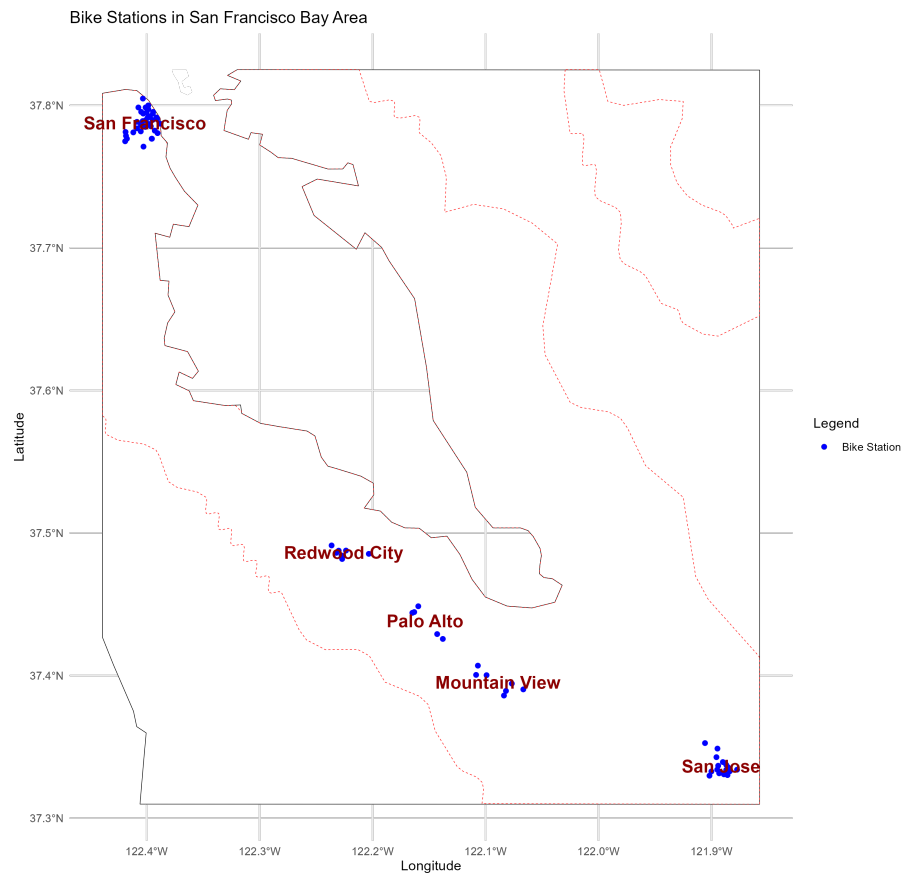


Figure E.1: Locations of bike stations

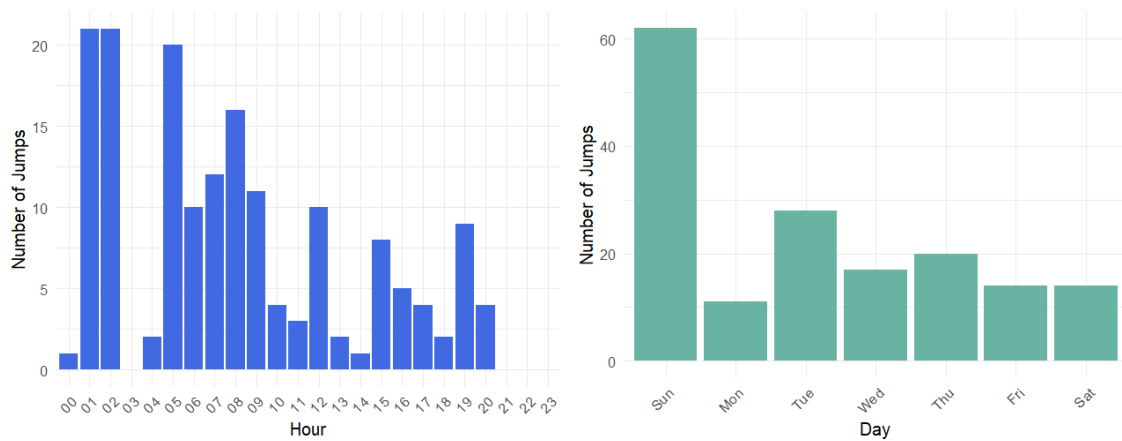
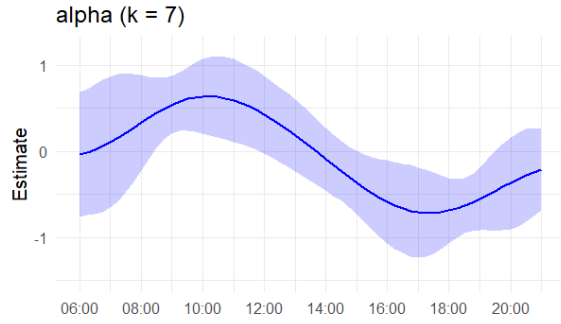
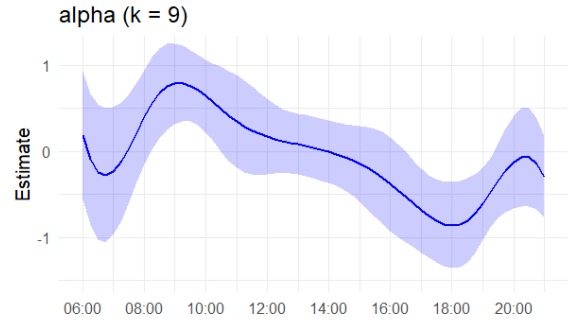


Figure E.2: Distribution of potential bike relocation events

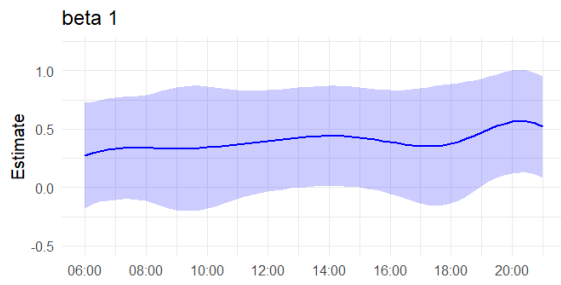


(a) $K = 7$

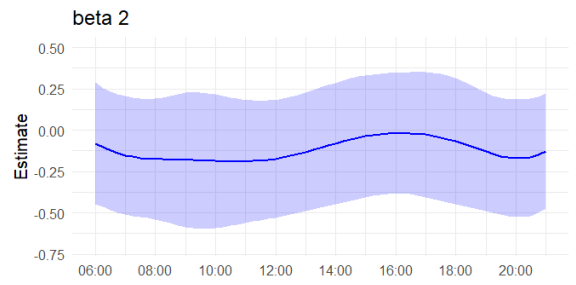


(b) $K = 9$

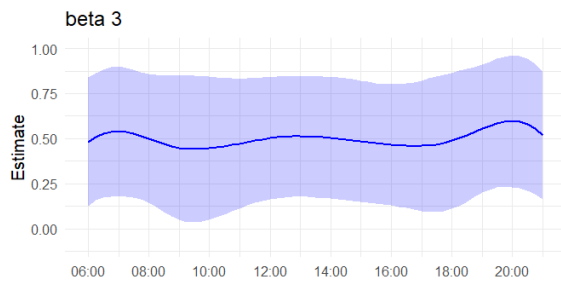
Figure E.3: Estimated $\alpha_0(s)$: robustness check



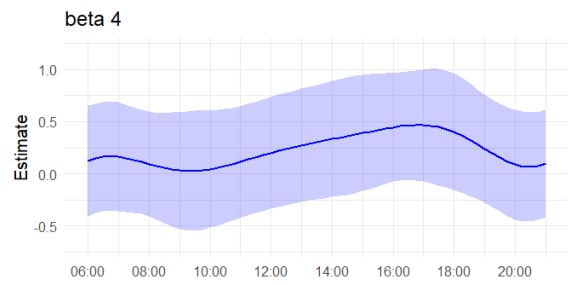
(a) X_{it}^1 : ratio of round trips



(b) X_{it}^2 : ratio of departing subscribers



(c) X_{it}^3 : ratio of arriving subscribers



(d) X_{it}^4 : rainy day

Figure E.4: Estimated $\beta_0(s)$

Table E.1: Estimation result: non-functional panel SAR model with fixed effects

	Estimate	Standard error	<i>t</i> -value
Spatial autoregressive parameter	-0.0790	0.0199	-3.9605
Ratio of round trips	0.6476	0.2509	2.5811
Ratio of departing subscribers	-0.5605	0.1880	-2.9807
Ratio of arriving subscribers	0.8745	0.1684	5.1917
Rainy day	0.1005	0.2187	0.4596
Feb	-0.1989	0.1524	-1.3048
Mar	-0.3019	0.1455	-2.0748
Apr	-0.1772	0.1529	-1.1594
May	-0.0288	0.1332	-0.2161
Jun	-0.0381	0.1282	-0.2969
Jul	-0.0433	0.1325	-0.3266
Aug	-0.0592	0.1324	-0.4472
Sep	0.2561	0.1455	1.7599
Oct	0.1325	0.1533	0.8641
Nov	0.3263	0.1570	2.0785
Dec	0.4202	0.1755	2.3948

Estimation is performed using the `SDPDm` function from the `SDPDmod` package.

F Proof of Theorem 6.1

This appendix outlines the proof of asymptotic normality of the integrated-GMM estimator for the model in (6.1). For simplicity of presentation and ease of comparison, we reuse the same notation as in the main body. As discussed in Subsection 6.1, our estimator can be defined as follows:

$$\hat{\theta}_{nT} = (\hat{\theta}_{nT,\alpha}^\top, \hat{\theta}_{nT,1}^\top, \dots, \hat{\theta}_{nT,d_x}^\top)^\top := \underset{\theta \in \Theta_K}{\operatorname{argmin}} \mathcal{Q}_{nT}(\theta),$$

where $\mathcal{Q}_{nT}(\theta) := \bar{g}_{nT}(\theta)^\top \Omega_{nT} \bar{g}_{nT}(\theta)$,

$$g_{nT}(s; \theta) := \frac{1}{N} \begin{pmatrix} \mathbf{Z}(s)^\top \mathbf{D}^\top \mathcal{R}^\top \mathcal{R} \mathbf{D} \mathbf{E}(s; \theta) \\ \mathbf{E}(s; \theta)^\top \mathbf{D}^\top \mathcal{R}^\top P_1 \mathcal{R} \mathbf{D} \mathbf{E}(s; \theta) \\ \vdots \\ \mathbf{E}(s; \theta)^\top \mathbf{D}^\top \mathcal{R}^\top P_M \mathcal{R} \mathbf{D} \mathbf{E}(s; \theta) \end{pmatrix},$$

and $\bar{g}_{nT}(\theta) := L^{-1} \sum_{l=1}^L g_{nT}(s_l; \theta)$. The estimators of α_0 and β_0 can be defined as $\hat{\alpha}_{nT}(s) := \phi^K(s)^\top \hat{\theta}_{nT,\alpha}$ and $\hat{\beta}_{nT,j}(s) := \phi^K(s)^\top \hat{\theta}_{nT,j}$, $j \in [d_x]$, respectively.

Then, under the assumptions in Theorem 3.2 and Assumption 6.1, we first obtain that, for any given $s \in [0, 1]$,

$$(i) \quad \frac{\sqrt{N} (\hat{\alpha}_{nT}(s) - \alpha_0(s))}{\sigma_{nT,\alpha}(s)} \xrightarrow{d} \mathcal{N}(0, 1), \quad (ii) \quad \frac{\sqrt{N} (\hat{\beta}_{nT,j}(s) - \beta_{0j}(s))}{\sigma_{nT,j}(s)} \xrightarrow{d} \mathcal{N}(0, 1), \quad (\text{F.1})$$

where $[\sigma_{nT,\alpha}(s)]^2 := \phi^K(s)^\top \mathbb{S}_\alpha \Sigma_{nT} \mathbb{S}_\alpha^\top \phi^K(s)$, $[\sigma_{nT,j}(s)]^2 := \phi^K(s)^\top \mathbb{S}_j \Sigma_{nT} \mathbb{S}_j^\top \phi^K(s)$,

$$\begin{aligned} \Sigma_{nT} &:= \left(\bar{\mathbf{J}}_{nT}^\top \Omega_{nT} \bar{\mathbf{J}}_{nT} \right)^{-1} \bar{\mathbf{J}}_{nT}^\top \Omega_{nT} \mathcal{V}_{nT} \Omega_{nT} \bar{\mathbf{J}}_{nT} \left(\bar{\mathbf{J}}_{nT}^\top \Omega_{nT} \bar{\mathbf{J}}_{nT} \right)^{-1}, \\ \mathcal{V}_{nT} &:= \begin{pmatrix} \mathcal{V}_{z,nT} & \mathbf{0}_{(d_q+d_x)K \times 1} & \cdots & \mathbf{0}_{(d_q+d_x)K \times 1} \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \mathcal{V}_{11,nT} & \cdots & \mathcal{V}_{1M,nT} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{1 \times (d_q+d_x)K} & \mathcal{V}_{M1,nT} & \cdots & \mathcal{V}_{MM,nT} \end{pmatrix}, \\ \mathcal{V}_{z,nT} &:= \frac{N}{(NL)^2} \sum_{l=1}^L \sum_{l'=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathcal{R}^\top \mathcal{R} \mathbf{D} \mathbb{E}[\mathcal{E}(s_l) \mathcal{E}(s_{l'})^\top] \mathbf{D}^\top \mathcal{R}^\top \mathcal{R} \mathbf{D} \mathbf{Z}(s_{l'}) \\ &= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{i=1}^n z_i^*(s_l) z_i^*(s_{l'})^\top \Gamma_i(s_l, s_{l'}), \end{aligned}$$

$$\begin{aligned}
\mathcal{V}_{ab,nT} &:= \frac{N}{(NL)^2} \sum_{l=1}^L \sum_{l'=1}^L \mathbb{E}[\boldsymbol{\mathcal{E}}(s_l)^\top \boldsymbol{D}^\top \boldsymbol{\mathcal{R}}^\top P_a \boldsymbol{\mathcal{R}} \boldsymbol{D} \boldsymbol{\mathcal{E}}(s_l) \boldsymbol{\mathcal{E}}(s_{l'})^\top \boldsymbol{D}^\top \boldsymbol{\mathcal{R}}^\top P_b \boldsymbol{\mathcal{R}} \boldsymbol{D} \boldsymbol{\mathcal{E}}(s_{l'})] \\
&= \frac{2}{N} \sum_{1 \leq i, k \leq n} \bar{p}_{a,i,k} \bar{p}_{b,i,k} \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_i(s_l, s_{l'}) \Gamma_k(s_l, s_{l'}),
\end{aligned}$$

$z_i^\star(s)$ denotes the i -th column of $\boldsymbol{Z}(s)^\top \boldsymbol{D}^\top \boldsymbol{\mathcal{R}}^\top \boldsymbol{\mathcal{R}} \boldsymbol{D}$, and $\bar{p}_{m,i,j}$ is the (i, j) -th element of $\boldsymbol{D}^\top \boldsymbol{\mathcal{R}}^\top P_m \boldsymbol{\mathcal{R}} \boldsymbol{D}$. The definition of $\bar{\mathcal{J}}_{nT}$ should be clear from the context.

Here, Assumption 3.6 is to be understood with Π_{nT} re-defined as

$$\Pi_{nT} := (NL)^{-1} \sum_{l=1}^L \boldsymbol{Z}(s_l)^\top \boldsymbol{D}^\top \boldsymbol{\mathcal{R}}^\top \boldsymbol{\mathcal{R}} \boldsymbol{D} \mathbb{E}[\boldsymbol{H}(s_l)],$$

which serves as the main identification condition in this model. Similarly, Assumption 3.8(ii) is to be understood in terms of $\bar{\mathcal{J}}_{nT}$ and $\mathcal{V}_{nT}$ re-defined here.

The proof of (F.1) is almost analogous to that of Theorem 3.2. For example, by treating the matrix $\boldsymbol{\mathcal{R}}^\top P_m \boldsymbol{\mathcal{R}}$ in the same way as the matrix P_m in the original model, the stochastic order evaluations regarding the quadratic moments can be reused without any changes. That said, several lemmas require additional analysis or adjustment. In what follows, rather than reproducing the entire proof of (F.1), we focus only on the parts that require revision.

Lemma B.5' : For all $i, j \in [n]$ such that $\Delta(i, j) > 2\bar{\Delta}$,

- (i) $\left| \text{Cov} \left(\ddot{\xi}_{it}, \ddot{\xi}_{jt} \right) \right| \lesssim C_\xi^2 \rho(\Delta(i, j)/3)$ for all $t \in [T]$ with some geometric NED coefficient ρ ;
- (ii) $\left| \text{Cov} \left(\ddot{\xi}_{i,t+1}, \ddot{\xi}_{jt} \right) \right| \lesssim C_\xi^2 \rho(\Delta(i, j)/3)$ for all $t \in [T-1]$ with some geometric NED coefficient ρ ,

where $\ddot{\xi}_{it} := \xi_{i,t+1} - \bar{\xi}_{i,t+1} - (\xi_{it} - \bar{\xi}_{it})$, and $\bar{\xi}_{it} = \sum_{j=1}^n w_{i,j} \xi_{jt}$.

Proof. We only prove (i). Decompose $\ddot{\xi}_{it} = \ddot{\xi}_{1,it}^{(\delta)} + \ddot{\xi}_{2,it}^{(\delta)}$, where

$$\ddot{\xi}_{1,it}^{(\delta)} := \mathbb{E} \left[\ddot{\xi}_{it} \mid \mathcal{F}_{it}^+(\delta) \right], \quad \text{and} \quad \ddot{\xi}_{2,it}^{(\delta)} := \ddot{\xi}_{it} - \mathbb{E} \left[\ddot{\xi}_{it} \mid \mathcal{F}_{it}^+(\delta) \right].$$

Denoting $\delta_{ij} := (\Delta(i, j) + \bar{\Delta})/3$,

$$\begin{aligned}
\left| \text{Cov} \left(\ddot{\xi}_{it}, \ddot{\xi}_{jt} \right) \right| &= \left| \text{Cov} \left(\ddot{\xi}_{1,it}^{(\delta_{ij})} + \ddot{\xi}_{2,it}^{(\delta_{ij})}, \ddot{\xi}_{1,jt}^{(\delta_{ij})} + \ddot{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| \\
&\leq \left| \text{Cov} \left(\ddot{\xi}_{1,it}^{(\delta_{ij})}, \ddot{\xi}_{1,jt}^{(\delta_{ij})} \right) \right| + \left| \text{Cov} \left(\ddot{\xi}_{1,it}^{(\delta_{ij})}, \ddot{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| \\
&\quad + \left| \text{Cov} \left(\ddot{\xi}_{2,it}^{(\delta_{ij})}, \ddot{\xi}_{1,jt}^{(\delta_{ij})} \right) \right| + \left| \text{Cov} \left(\ddot{\xi}_{2,it}^{(\delta_{ij})}, \ddot{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| \\
&= \left| \text{Cov} \left(\ddot{\xi}_{1,it}^{(\delta_{ij})}, \ddot{\xi}_{2,jt}^{(\delta_{ij})} \right) \right| + \left| \text{Cov} \left(\ddot{\xi}_{2,it}^{(\delta_{ij})}, \ddot{\xi}_{1,jt}^{(\delta_{ij})} \right) \right| + \left| \text{Cov} \left(\ddot{\xi}_{2,it}^{(\delta_{ij})}, \ddot{\xi}_{2,jt}^{(\delta_{ij})} \right) \right|.
\end{aligned}$$

By Jensen's and triangle inequalities,

$$\begin{aligned}
\left\| \ddot{\xi}_{1,it}^{(\delta_{ij})} \right\|_2 &\leq \left\| \ddot{\xi}_{it} \right\|_2 \\
&\leq \left\| \xi_{i,t+1} \right\|_2 + \sum_{k=1}^n |w_{i,k}| \left\| \xi_{k,t+1} \right\|_2 + \left\| \xi_{it} \right\|_2 + \sum_{k=1}^n |w_{i,k}| \left\| \xi_{kt} \right\|_2 \leq 4C_\xi.
\end{aligned}$$

In addition, $\left\| \ddot{\xi}_{2,it}^{(\delta_{ij})} \right\|_2 \leq 2 \left\| \ddot{\xi}_{it} \right\|_2 \leq 8C_\xi$. Furthermore,

$$\begin{aligned}
\left\| \ddot{\xi}_{2,it}^{(\delta_{ij})} \right\|_2 &= \left\| \ddot{\xi}_{it} - \mathbb{E} \left[\ddot{\xi}_{it} \mid \mathcal{F}_{it}^+(\delta_{ij}) \right] \right\|_2 \\
&\leq \left\| \xi_{i,t+1} - \mathbb{E} \left[\xi_{i,t+1} \mid \mathcal{F}_{it}^+(\delta_{ij}) \right] \right\|_2 + \left\| \xi_{it} - \mathbb{E} \left[\xi_{it} \mid \mathcal{F}_{it}^+(\delta_{ij}) \right] \right\|_2 \\
&\quad + \sum_{k=1}^n |w_{i,k}| \left\| \xi_{k,t+1} - \mathbb{E} \left[\xi_{k,t+1} \mid \mathcal{F}_{it}^+(\delta_{ij}) \right] \right\|_2 + \sum_{k=1}^n |w_{i,k}| \left\| \xi_{kt} - \mathbb{E} \left[\xi_{kt} \mid \mathcal{F}_{it}^+(\delta_{ij}) \right] \right\|_2 \\
&\leq \left\| \xi_{i,t+1} - \mathbb{E} \left[\xi_{i,t+1} \mid \mathcal{F}_{i,t+1}(\delta_{ij}) \right] \right\|_2 + \left\| \xi_{it} - \mathbb{E} \left[\xi_{it} \mid \mathcal{F}_{it}(\delta_{ij}) \right] \right\|_2 \\
&\quad + \sum_{k=1}^n |w_{i,k}| \left\| \xi_{k,t+1} - \mathbb{E} \left[\xi_{k,t+1} \mid \mathcal{F}_{k,t+1}(\delta_{ij} - \bar{\Delta}) \right] \right\|_2 \\
&\quad + \sum_{k=1}^n |w_{i,k}| \left\| \xi_{kt} - \mathbb{E} \left[\xi_{kt} \mid \mathcal{F}_{kt}(\delta_{ij} - \bar{\Delta}) \right] \right\|_2 \\
&\lesssim C_\xi \rho(\delta_{ij}).
\end{aligned}$$

Then, the rest of the proof is analogous to Lemma B.5. □

Lemma B.6' :

$$(i') \quad \left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{R}^\top \mathbf{R} \mathbf{D} (\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)])(\theta_0 - \theta) / (NL) \right\| \lesssim_p \sqrt{K} / \sqrt{nT}$$

$$(iv') \quad \left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{R}^\top \mathbf{R} \mathbf{D} \mathbf{E}(s_l) / (NL) \right\| \lesssim_p 1 / \sqrt{nT}$$

$$(v') \quad \left| \sum_{l=1}^L \{ \mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top \mathbf{R}^\top P_m \mathbf{R} \mathbf{D} \mathbf{E}(s_l; \theta) - \mathbb{E}[\mathbf{E}(s_l; \theta)^\top \mathbf{D}^\top \mathbf{R}^\top P_m \mathbf{R} \mathbf{D} \mathbf{E}(s_l; \theta)] \} / (NL) \right| \lesssim_p 1 / \sqrt{nT}.$$

Proof. (i') Observe that for each s_l ,

$$\|\mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{R}^\top \mathbf{R} \mathbf{D}(\mathbf{H}(s_l) - \mathbb{E}[\mathbf{H}(s_l)])(\theta_0 - \theta)\| = \left\| \sum_{i=1}^n \sum_{t=1}^{T-1} \ddot{B}_{it} \otimes \phi^K(s_l) [\ddot{a}_{it}(s_l) - \mathbb{E}[\ddot{a}_{it}(s_l)]] \alpha(s_l; \theta_0 - \theta) \right\|.$$

Hence, for example, the variance of the first element is given as

$$\begin{aligned} \text{Var}[\{\mathbf{Z}^\top \mathbf{D}^\top \mathbf{R}^\top \mathbf{R} \mathbf{D}(\mathbf{H} - \mathbb{E}[\mathbf{H}])(\theta_0 - \theta)\}_1] &= \mathbb{E} \left(\sum_{i=1}^n \sum_{t=1}^{T-1} \ddot{Q}_{it}^1 \phi_1(s_l) (\ddot{a}_{it} - \mathbb{E}[\ddot{a}_{it}]) \alpha(s_l; \theta_0 - \theta) \right)^2 \\ &\lesssim \sum_{t=1}^{T-1} \sum_{t'=1}^{T-1} \sum_{i=1}^n \sum_{i'=1}^n |\text{Cov}(\ddot{a}_{it}, \ddot{a}_{i't'})| \\ &= \sum_{t=1}^{T-1} \sum_{i=1}^n \text{Var}(\ddot{a}_{it}) + \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\ddot{a}_{it}, \ddot{a}_{i't})| \\ &\quad + \sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{i=1}^n |\text{Cov}(\ddot{a}_{it}, \ddot{a}_{i't'})| + \sum_{t=1}^{T-1} \sum_{t' \neq t} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\ddot{a}_{it}, \ddot{a}_{i't'})|. \end{aligned}$$

We can easily see that $\sum_{t=1}^{T-1} \sum_{i=1}^n \text{Var}(\ddot{a}_{it}) \lesssim nT$. Second, by Lemma B.5'(i), there exists a geometric NED coefficient ρ that satisfies

$$\begin{aligned} \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{i' \neq i} |\text{Cov}(\ddot{a}_{it}, \ddot{a}_{i't})| &\lesssim nT + \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{\Delta(i, i') > 2\overline{\Delta}} \rho(\Delta(i, i')/3) \\ &\leq nT + \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{m=\lfloor 2\overline{\Delta} \rfloor}^{\infty} \sum_{i': \Delta(i, i') \in [m, m+1)} \rho(\Delta(i, i')/3) \\ &\lesssim nT + \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{m=\lfloor 2\overline{\Delta} \rfloor}^{\infty} m^{d-1} \rho(m) \lesssim nT. \end{aligned}$$

Similarly, we can verify that the third and fourth terms are also of order nT . The desired result now follows from Markov's inequality.

(iv') By the triangle inequality,

$$\begin{aligned} \left\| \sum_{l=1}^L \mathbf{Z}(s_l)^\top \mathbf{D}^\top \mathbf{R}^\top \mathbf{R} \mathbf{D} \boldsymbol{\varepsilon}(s_l) / (NL) \right\| &= \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \ddot{B}_{it} \otimes \phi^K(s_l) \ddot{\varepsilon}_{it}(s_l) / (NL) \right\| \\ &\leq \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \ddot{B}_{it} \otimes \varepsilon_{i, t+1}(s_l) \phi^K(s_l) / (NL) \right\| \end{aligned}$$

$$\begin{aligned}
& + \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \ddot{B}_{it} \otimes \varepsilon_{it}(s_l) \phi^K(s_l) / (NL) \right\| \\
& + \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{j=1}^n \ddot{B}_{it} w_{i,j} \otimes \varepsilon_{j,t+1}(s_l) \phi^K(s_l) / (NL) \right\| \\
& + \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{j=1}^n \ddot{B}_{it} w_{i,j} \otimes \varepsilon_{jt}(s_l) \phi^K(s_l) / (NL) \right\|.
\end{aligned}$$

The same argument as in the proof of Lemma B.6(iv) shows that the first two terms on the right hand side are stochastically of order $1/\sqrt{nT}$. For the third and fourth terms, observe that

$$\begin{aligned}
& \mathbb{E} \left\| \sum_{l=1}^L \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{j=1}^n \ddot{B}_{it} w_{i,j} \otimes \varepsilon_{jt}(s_l) \phi^K(s_l) / (NL) \right\|^2 \\
& = \frac{1}{N^2} \sum_{t=1}^{T-1} \sum_{i=1}^n \sum_{i'=1}^n \sum_{j=1}^n \text{trace} \left\{ (\ddot{B}_{it} w_{i,j}) (\ddot{B}_{i't} w_{i',j})^\top \otimes \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_{jt}(s_l, s_{l'}) \phi^K(s_l) \phi^K(s_{l'})^\top \right\} \\
& = \frac{1}{N^2} \sum_{t=1}^{T-1} \sum_{j=1}^n \text{trace} \left\{ \left(\sum_{i=1}^n \ddot{B}_{it} w_{i,j} \right) \left(\sum_{i'=1}^n \ddot{B}_{i't} w_{i',j} \right)^\top \right\} \text{trace} \left\{ \frac{1}{L^2} \sum_{l=1}^L \sum_{l'=1}^L \Gamma_{jt}(s_l, s_{l'}) \phi^K(s_l) \phi^K(s_{l'})^\top \right\} \\
& \lesssim 1/(nT).
\end{aligned}$$

Then, the result follows from Markov's inequality.

(v') Let $\check{P}_{m,1} := \mathcal{R}_n^\top P_{m,1} \mathcal{R}_n$ and $\check{P}_m := I_{T-1} \otimes \check{P}_{m,1}$. Then, each (i, j) -th element of $\check{P}_{m,1}$ can be written as

$$\check{p}_{m,i,j} = \sum_{a=1}^n \sum_{b=1}^n r_{a,i} p_{m,a,b} r_{b,j}.$$

By Assumptions 3.1(ii) and 3.5(i), $\check{p}_{m,i,j} \neq 0$ requires the existence of at least one pair (a, b) satisfying $\Delta(i, a) \leq \bar{\Delta}$, $\Delta(a, b) \leq \bar{\Delta}_m$, and $\Delta(j, b) \leq \bar{\Delta}$, simultaneously. In other words, no such pairs exist if $\Delta(i, j) > 2\bar{\Delta} + \bar{\Delta}_m$. Hence, $\check{P}_{m,1} = (\check{p}_{m,i,j})$ satisfies the same threshold condition as in Assumption 3.5(i), and thus (v') can be proved analogously to Lemma B.6(v). $\square$

Now, the rest of the proof of (F.1) proceeds in the same way as that of Theorem 3.2.

Consistent variance estimation can be performed as in Corollary 3.1. Specifically, with an

abuse of notation, define

$$\begin{aligned}\hat{\mathcal{V}}_{z,nT} &:= \frac{1}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{i=1}^n \check{z}_{it}(s_l) \check{z}_{it'}(s_{l'})^\top \vec{e}_{it}(s_l) \vec{e}_{it'}(s_{l'}) \\ \hat{\mathcal{V}}_{ab,nT} &:= \frac{2}{L^2 N} \sum_{l=1}^L \sum_{l'=1}^L \sum_{t=1}^{T-1} \sum_{t': |t'-t| \leq 1} \sum_{1 \leq i,j \leq n} \check{p}_{a,i,j} \check{p}_{b,i,j} \vec{e}_{it}(s_l) \vec{e}_{it'}(s_{l'}) \vec{e}_{jt}(s_l) \vec{e}_{jt'}(s_{l'}),\end{aligned}$$

where $\check{z}_{it}(s)$ denotes the corresponding (i, t) -th element of $\mathbf{R}^\top \mathbf{R} \mathbf{D} \mathbf{Z}(s)$. Similarly, update the definitions of $\hat{\sigma}_{nT,\alpha}(s)$ and $\hat{\sigma}_{nT,j}(s)$ using the re-defined $\hat{\mathcal{V}}_{z,nT}$ and $\hat{\mathcal{V}}_{ab,nT}$. Then, following the same argument as in Proposition C.1, we can show the consistency of $\hat{\mathcal{V}}_{z,nT}$ and $\hat{\mathcal{V}}_{ab,nT}$. Finally, Theorem 6.1 follows from Slutsky's theorem.

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